Pharmodry
Precision in Every Batch.

PHARMACEUTICAL OPERATIONS, QUALITY & COMPLIANCE PLATFORM

Production Manager User Manual

Intended audience: Head of production responsible for plans and batch execution.

Product	Pharmodry 0.1.0 (MVP release) — Precision in Every Batch.
Manual	04 — Production Manager User Manual
Document version	1.0
Generated	2026-10-25 from the running application (screenshots captured 2026-10-25)
Role code	PRODUCTION_MANAGER

Screenshots show fictional demonstration data (NovaCure Pharmaceuticals). Pharmodry is not a validated system; procedures in this manual describe the software and must be used within your company's approved SOPs.

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List of figures: 61 screenshots captured from the running application.

1 Purpose and scope

Owns production plans and batches: creates and approves plans, creates batches, reserves materials, starts, executes, verifies and completes batches, raises deviations and uses the AI assistants. Cannot release batches or issue materials from the warehouse.

This manual covers every Pharmodry screen available to the Production Manager role in release 0.1.0. Your exact access is set by your company's administrator; if your account has additional roles or a site restriction, you may see more or fewer options than described. Configuration-dependent behaviour (for example tolerances, aging thresholds, calibration policy, AI availability) is marked as such.

> **Not available in this release:** Pharmacovigilance, sales/distribution and accounting modules; customer-level distribution in recalls; e-mail notifications; serial-number management; MFA/single sign-on. These are not described as working anywhere in this manual.

2 Role responsibilities

- Create and approve production plans with the planner.
- Create batches and ensure materials are reserved and issued on time.
- Ensure eBMR steps are executed and verified by different people.
- Complete manufacturing and justify or investigate yields outside tolerance.
- Raise deviations promptly.

Dependencies on other departments

- Warehouse — material issue.
- QC/QA — testing and release.
- Engineering — equipment calibration.

3 Signing in and signing out

1. Open the Pharmodry address provided by your administrator in a current browser (Microsoft Edge, Chrome or Firefox).
2. Enter your company e-mail address and password and select Sign in.
3. After five consecutive wrong passwords the account is locked for 15 minutes; an administrator can unlock it.
4. To sign out, open the user menu at the top right and choose Sign out. Always sign out on shared computers.
5. Change your password under Profile & security → Change password (at least 10 characters with upper-case, lower-case and a digit). Your password is also your electronic-signature credential — never share it.
6. Profile → Active sessions lists devices signed in to your account; revoke any session you do not recognise.

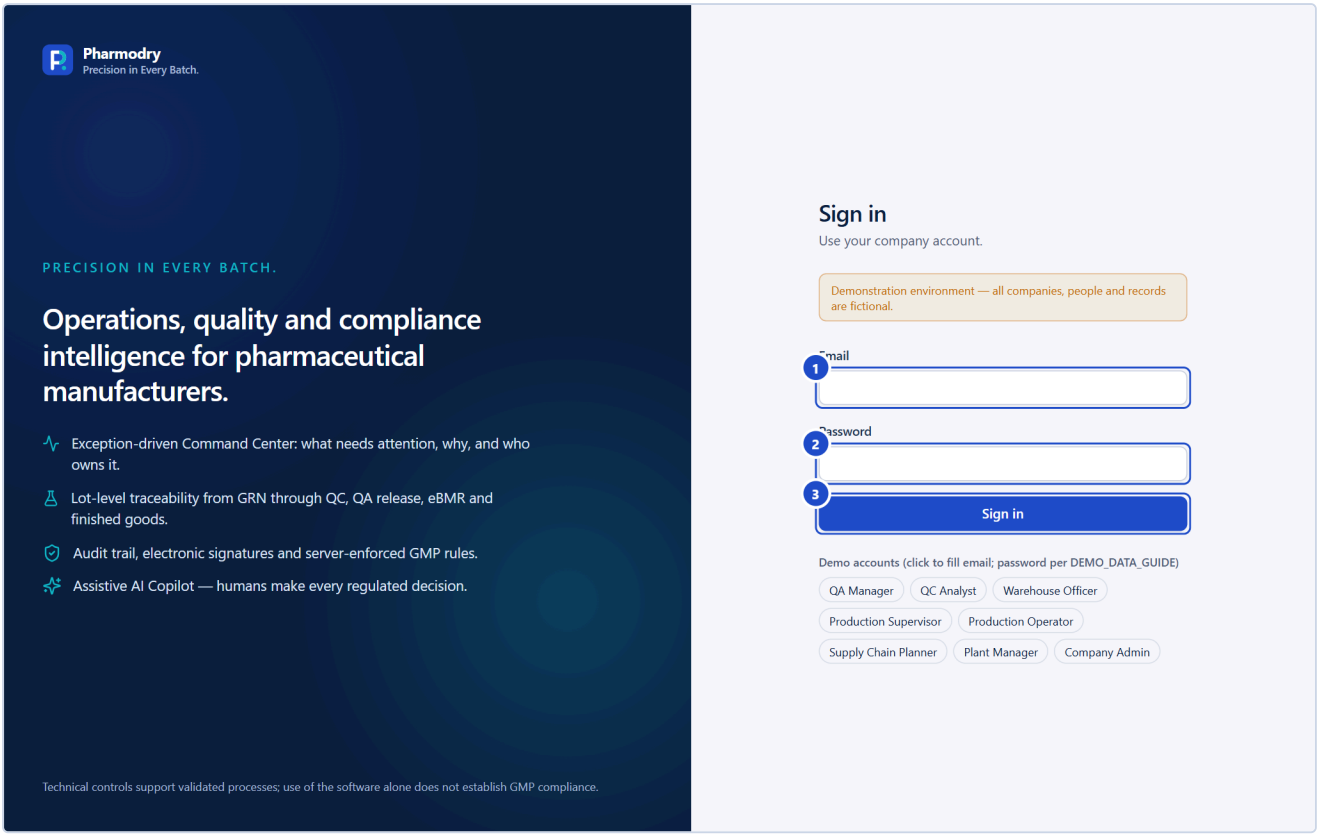


Figure 1 — Sign-in page. Sign-in page. Enter your company e-mail and password, then select Sign in.

1. Company e-mail address
2. Password (also your e-signature credential)
3. Sign in

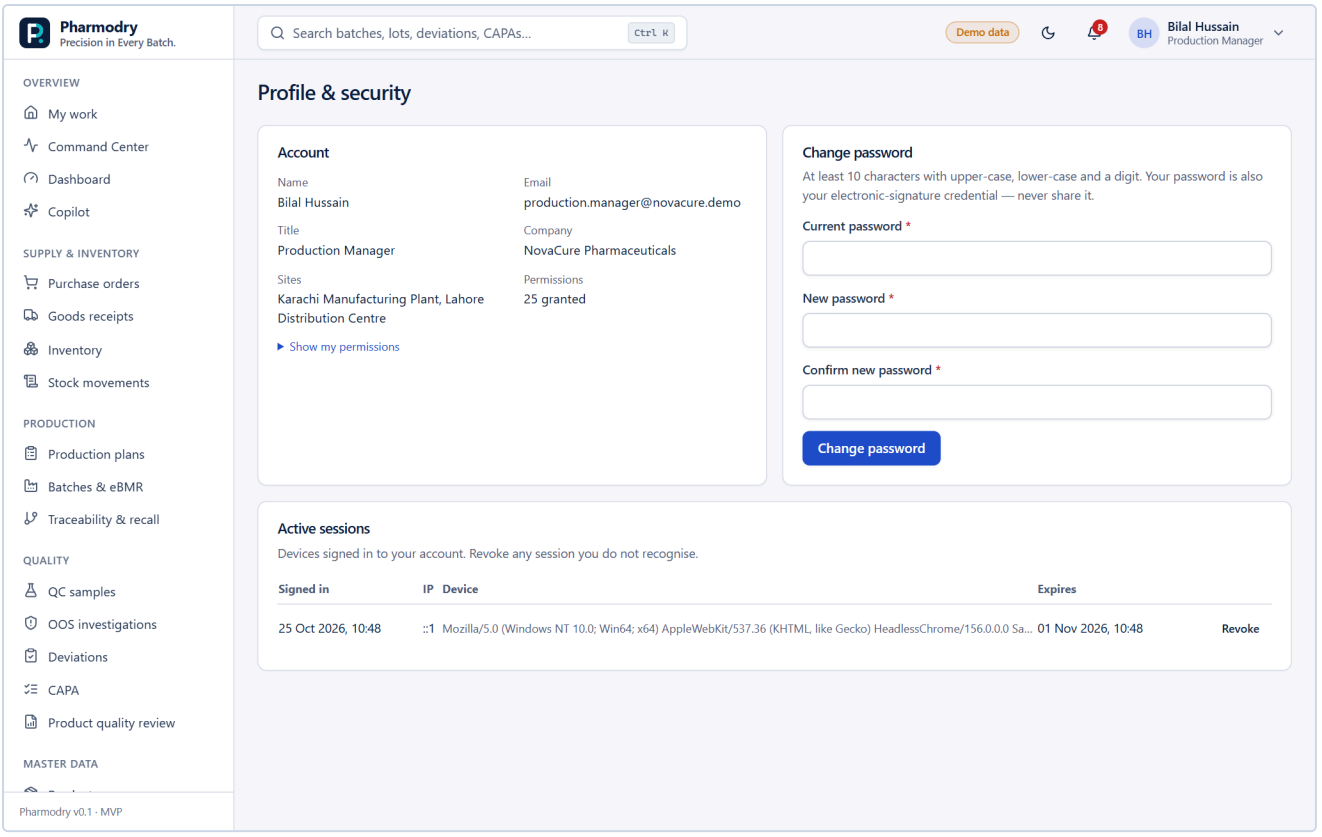


Figure 2 — Profile & security. Profile: password change and active sessions.

4 Home page and dashboard overview

After sign-in you land on My work. It greets you by name and lists the work queues that match your permissions — for example batches awaiting QA release, material lots awaiting disposition, open QC samples, deviations or CAPAs you own, batches on the shop floor, planned batches to prepare and purchase orders awaiting receipt. Each queue links straight to the records. Queues you have no permission for are not shown.

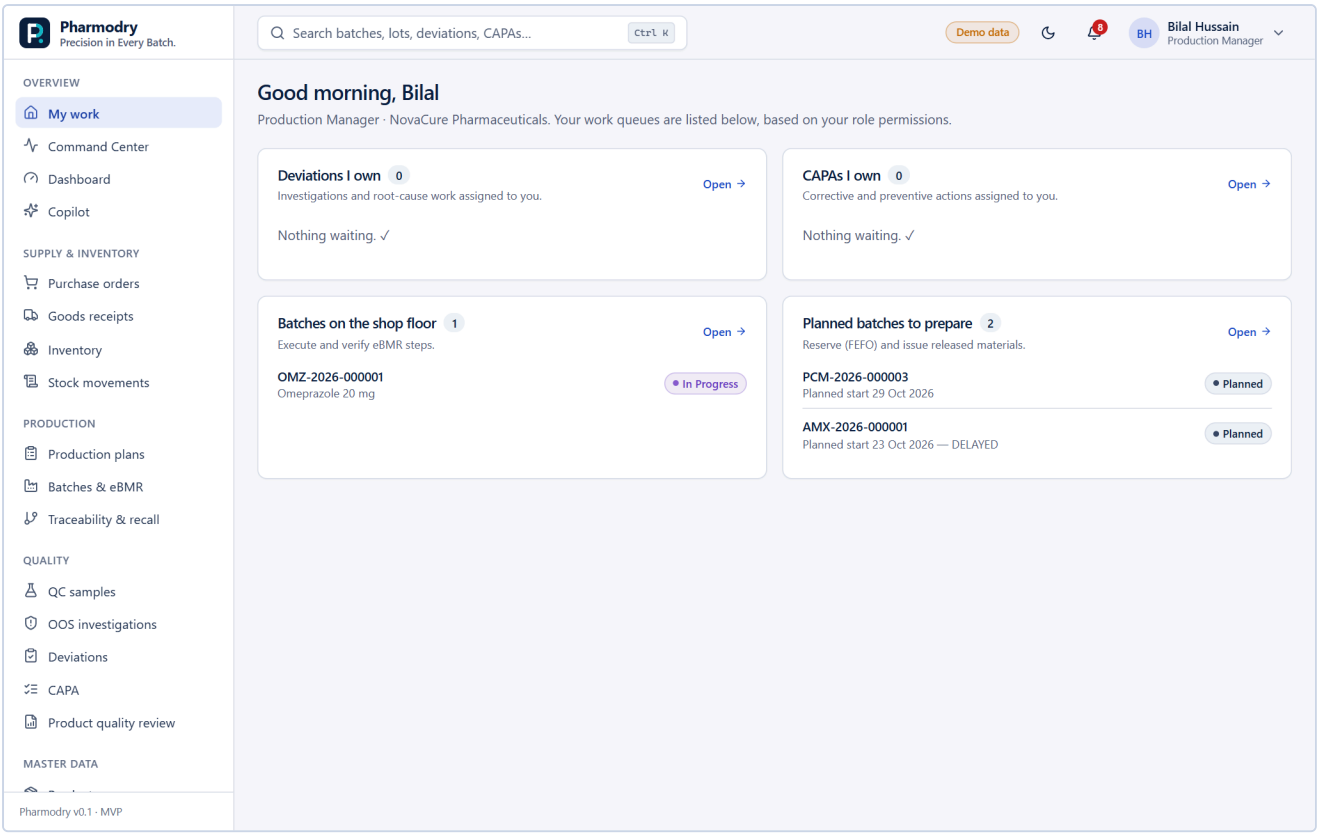


Figure 3 — My work (home). Home page with the work queues available to this role.

Pharma Command Center

The Command Center answers “what needs attention right now, why, which process it affects, who owns it and how long it has been pending”. Items are grouped into Batches at risk, Quality issues, Materials at risk, Inventory risks, Compliance deadlines and Supplier risks, and each carries a severity (Critical, Warning, Info), owner, age and a link to the record. It also lists serialization-readiness gaps such as active products without a GTIN.

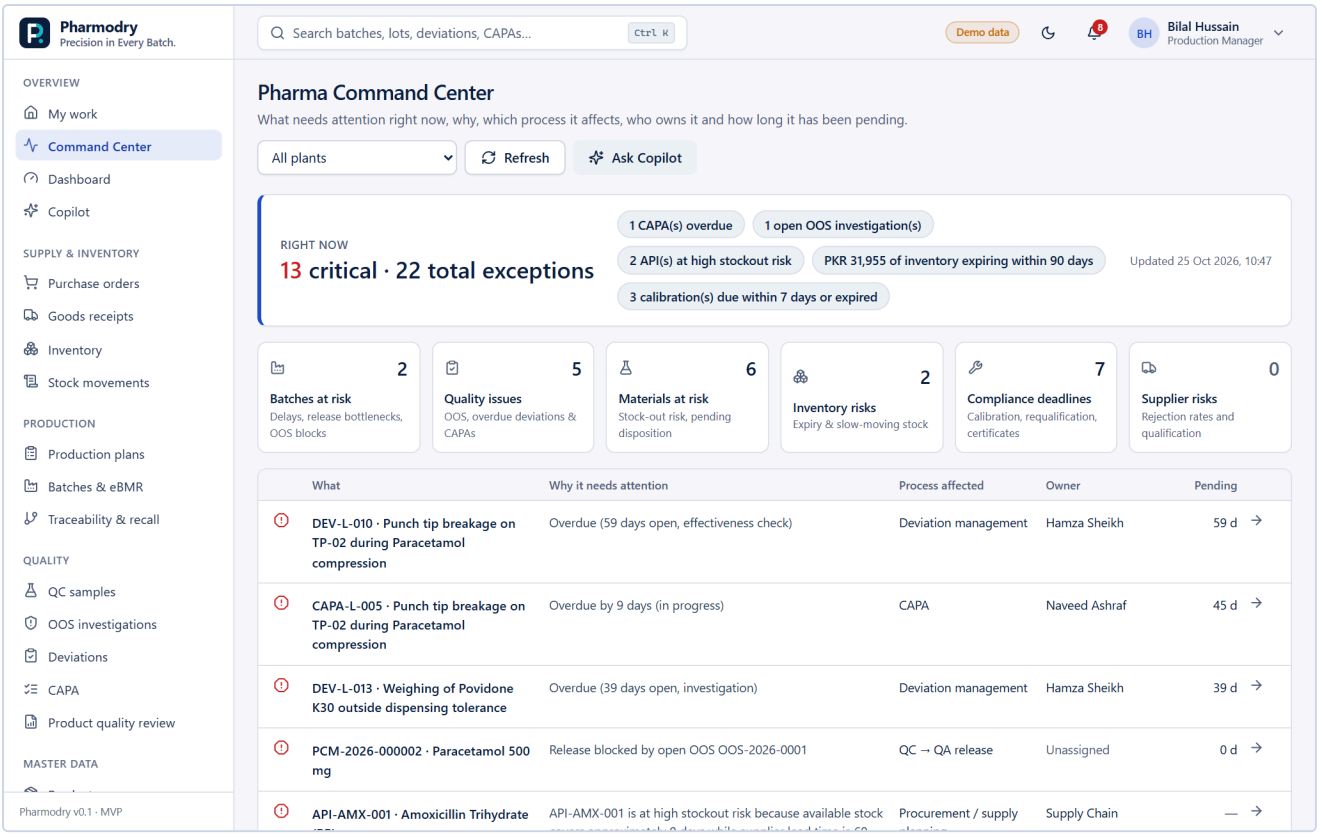


Figure 4 — Command Center. Exception-driven Command Center: what needs attention, why, owner and age.

Management dashboard

KPI cards (Active batches, Awaiting QA release, Open deviations, Overdue CAPAs, Open OOS, Materials at high risk, Near-expiry inventory, Plan achievement, Avg batch release time, Supplier quality score) and charts: Production plan vs actual, Batch yield trend (shaded band = configured tolerance), Deviation trend, CAPA aging, OOS trend, Inventory value by status, Near-expiry inventory, Supplier QC rejection rate and Batch release cycle time. All figures are computed from live regulated records.

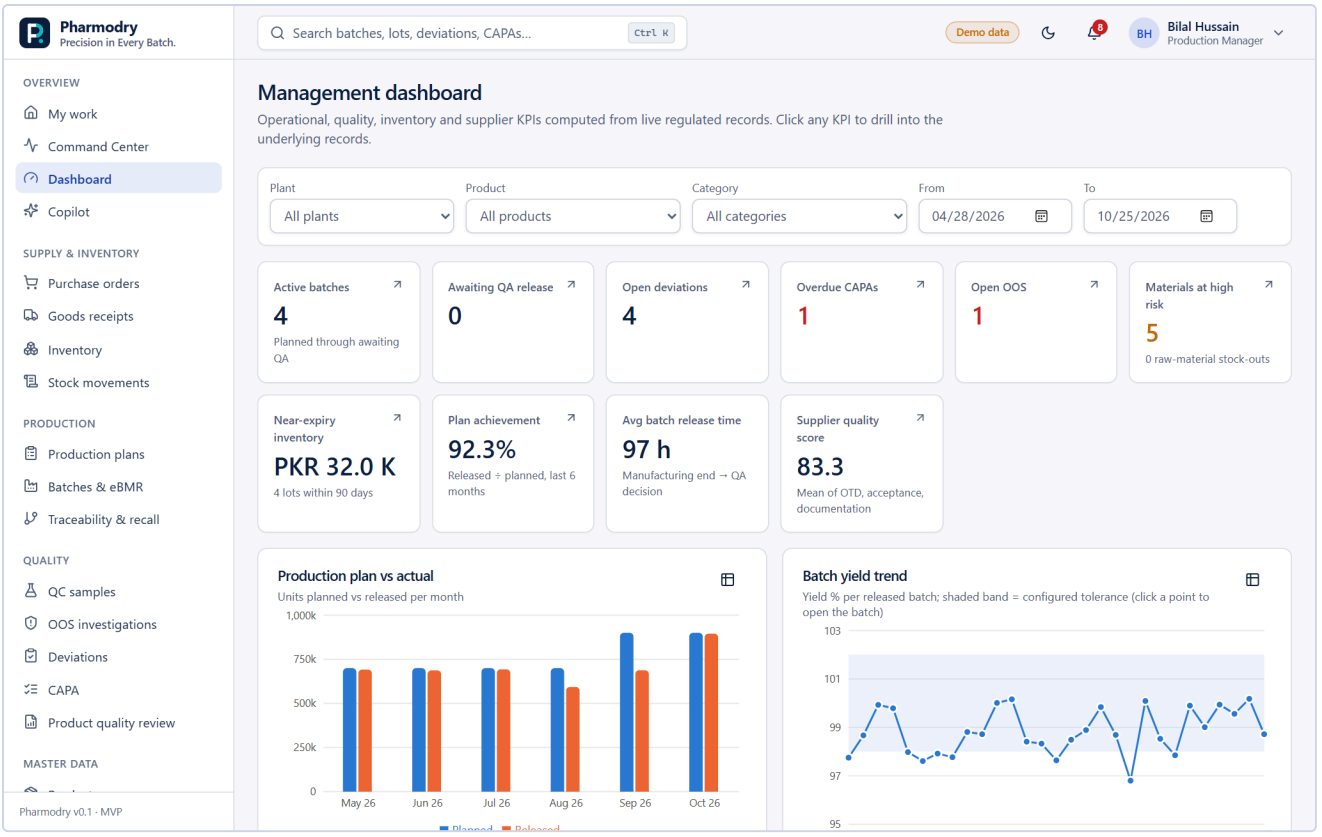


Figure 5 — Management dashboard. KPI cards and charts computed from live records; click a KPI to drill down.

5 Navigation and module overview

- The left sidebar lists only the modules your role can open, grouped into Overview, Supply & inventory, Production, Quality, Master data and Governance.
- The top bar has global search (“Search batches, lots, deviations, CAPAs…”, shortcut Ctrl K), a “Demo data” badge in demonstration environments, the light/dark theme toggle, the notification bell and your user menu (name and job title).
- Record numbers are links; detail pages show a status badge, key facts, actions you may perform and tabs with history.

The sidebar below was recorded from the application while signed in as the demo Production Manager (production.manager@novacure.demo).

Sidebar group	Menu entry	Address
Overview	My work	/
Overview	Command Center	/command-center
Overview	Dashboard	/dashboard
Overview	Copilot	/copilot
Supply & inventory	Purchase orders	/procurement/purchase-orders
Supply & inventory	Goods receipts	/inventory/receipts
Supply & inventory	Inventory	/inventory
Supply & inventory	Stock movements	/inventory/movements
Production	Production plans	/production/plans
Production	Batches & eBMR	/production/batches
Production	Traceability & recall	/traceability
Quality	QC samples	/quality/samples
Quality	OOS investigations	/quality/oos
Quality	Deviations	/quality/deviations
Quality	CAPA	/quality/capa
Quality	Product quality review	/quality/pqr
Master data	Products	/master-data/products
Master data	Materials	/master-data/materials
Master data	Suppliers	/master-data/suppliers
Master data	Equipment	/master-data/equipment
Master data	Specifications	/master-data/specifications
Governance	Audit trail	/audit-trail
Governance	Reports	/reports

6 Features and step-by-step instructions

Each section explains a module you can open, the procedures you are allowed to perform (numbered steps) and screenshots of the real screens. Procedures that need permissions your role does not hold are listed under “Not available to your role”.

Copilot (AI assistant)

Ask questions in plain language about batches, quality events, inventory and suppliers — for example “Which CAPAs are overdue?”, “Which batches are awaiting QA release?”, “Which materials are at risk of stock-out?”. Pharmodry computes the figures from live records you are allowed to see; the AI model only explains them and lists citations (links to the records). The Copilot cannot approve, release or change any record.

> **Note:** If the banner “AI provider not configured” appears, the server has no Mistral API key: you still receive the computed facts and links, but no prose explanation. This is configuration-dependent.

> **Note:** If you ask about data you have no permission for, the Copilot tells you which permission is required instead of answering.

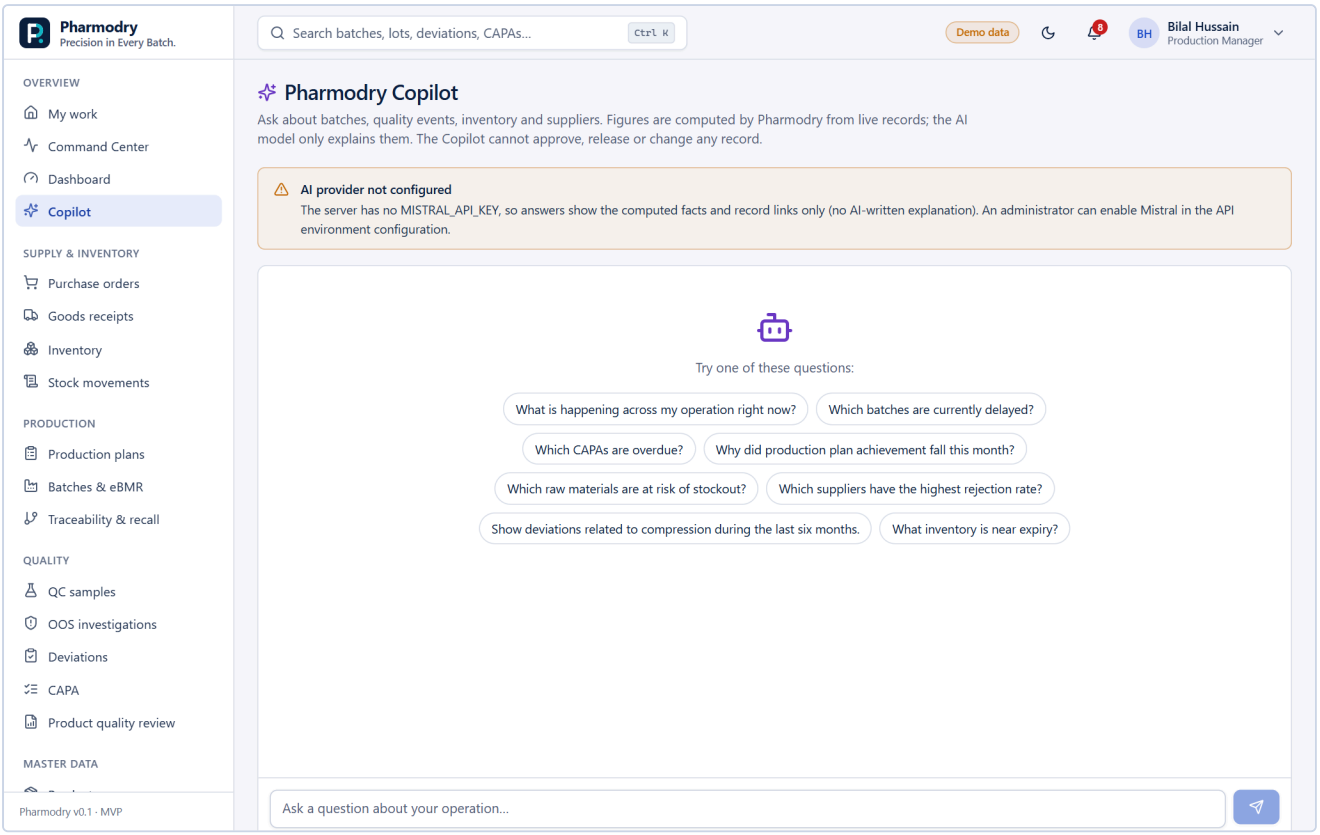


Figure 6 — Copilot. Copilot question box. Answers are computed by Pharmodry; the AI only explains them.

Ask the Copilot

1. Open Copilot from the sidebar.
2. Type your question in “Ask a question about your operation…” and select Send.
3. Read the answer and open the cited records to verify.
4. Rate the answer with Thumbs up / Thumbs down to help administrators monitor quality.

Purchase orders

Purchase orders may only be raised with qualified suppliers (APPROVED or CONDITIONALLY APPROVED) for materials on that supplier's approved list. A new order is a DRAFT until a different user approves it.

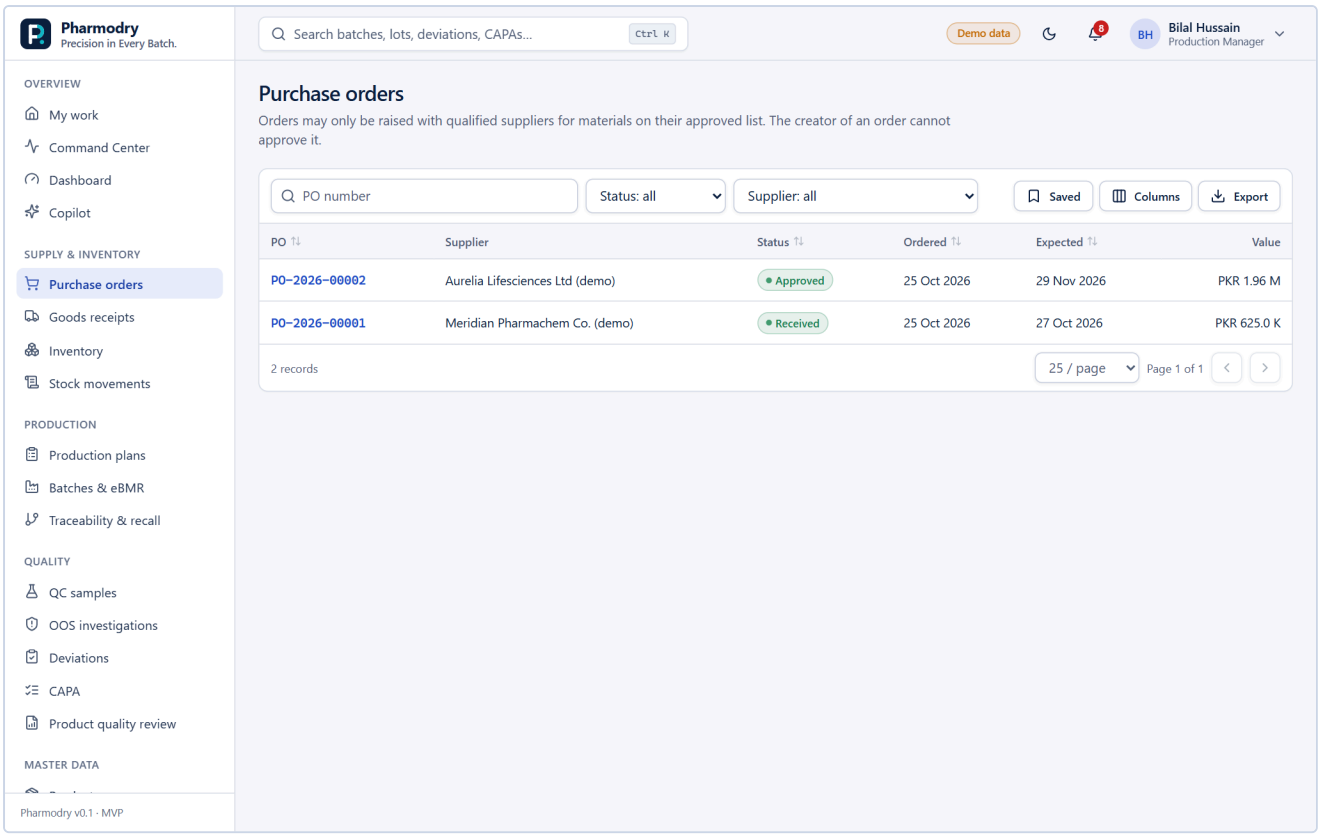


Figure 7 — Purchase orders. Purchase order list with status, supplier and value.

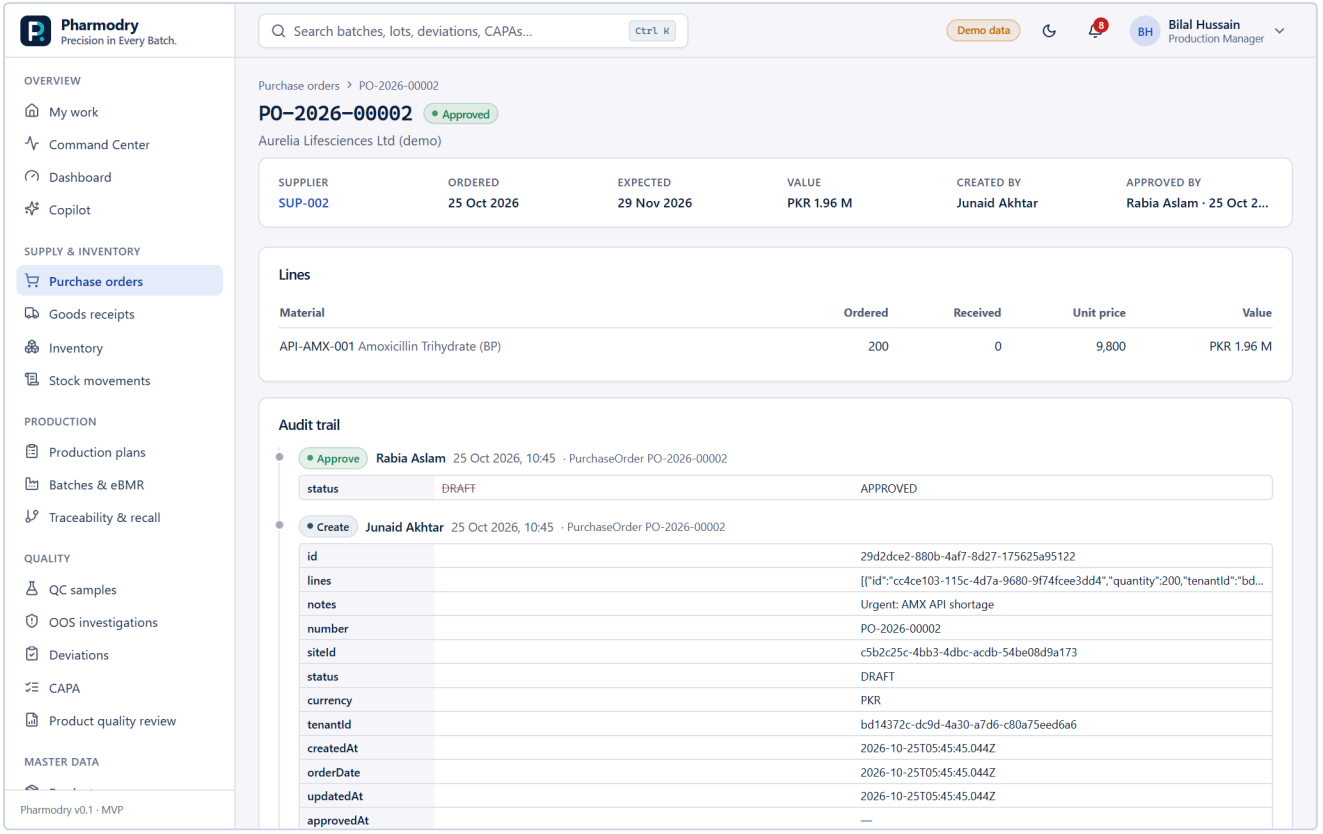


Figure 8 — Purchase order detail. Purchase order detail with lines, approval and audit trail.

> **Not available to your role:** Create a purchase order — requires procurement.po.create; Approve a purchase order — requires procurement.po.approve; Cancel a purchase order — requires procurement.po.create or procurement.po.approve.

Goods receipts (GRN)

Every receipt line creates a separate lot with its own internal lot number in QUARANTINE and automatically requests a QC sample. Material only becomes usable after QC approval and QA release. Receipts are refused for expired material, for suppliers that are not qualified, for materials not approved for the supplier, and when the expiry date is before the manufacturing date.

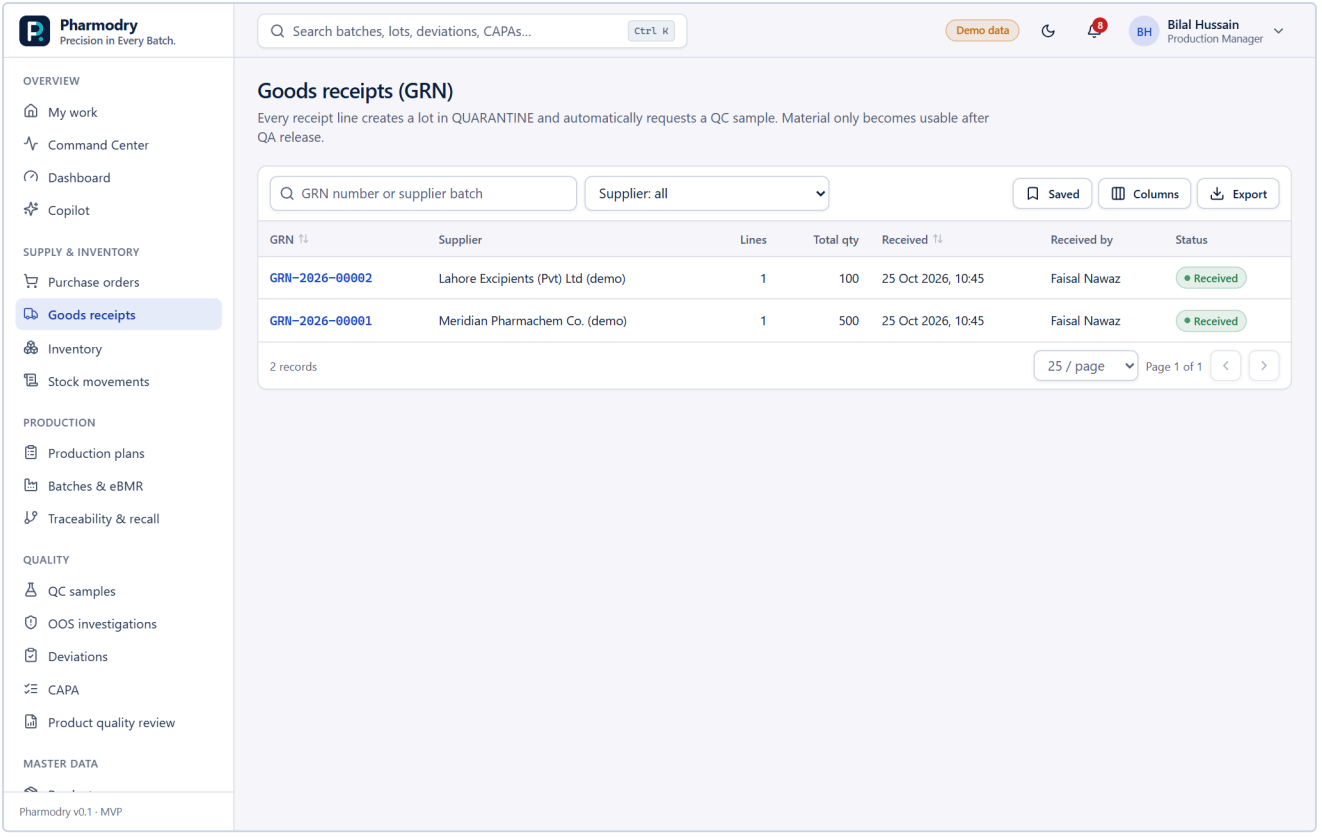


Figure 9 — Goods receipts. Goods receipts (GRN). Every line creates a lot in QUARANTINE.

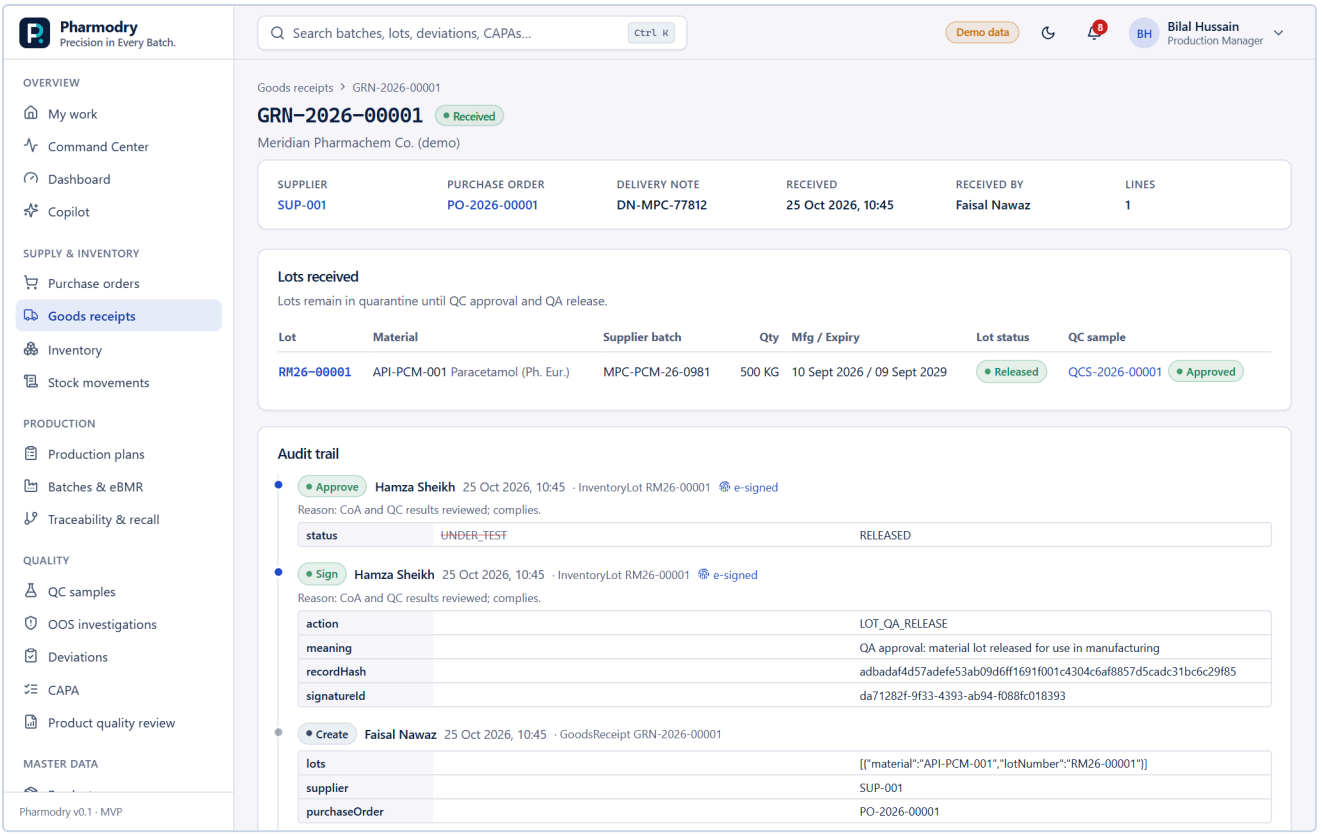


Figure 10 — Goods receipt detail. Received lots and their status after a goods receipt.

> **Not available to your role:** Receive material — requires inventory.receive.

Inventory, lots and stock movements

Inventory is held per lot and location with a status. Only RELEASED, unexpired lots that are not past retest can be reserved or consumed; quarantined, under-test, rejected, blocked or expired lots never can. Tabs: Lots, Stock by material, Stock-out risk and Expiry intelligence. Every quantity change is written to an append-only ledger (Stock movements).

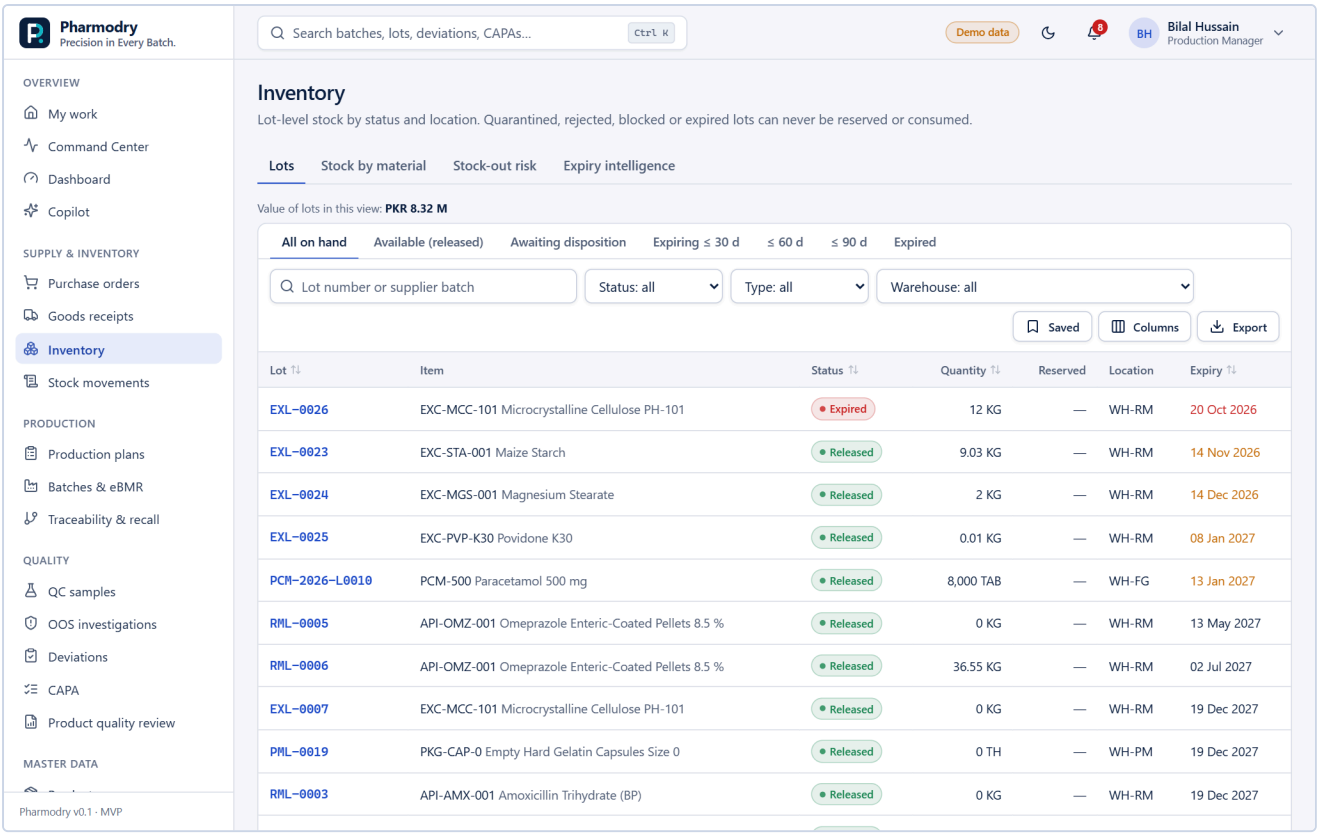


Figure 11 — Inventory. Lot-level inventory with status, location and expiry filters.

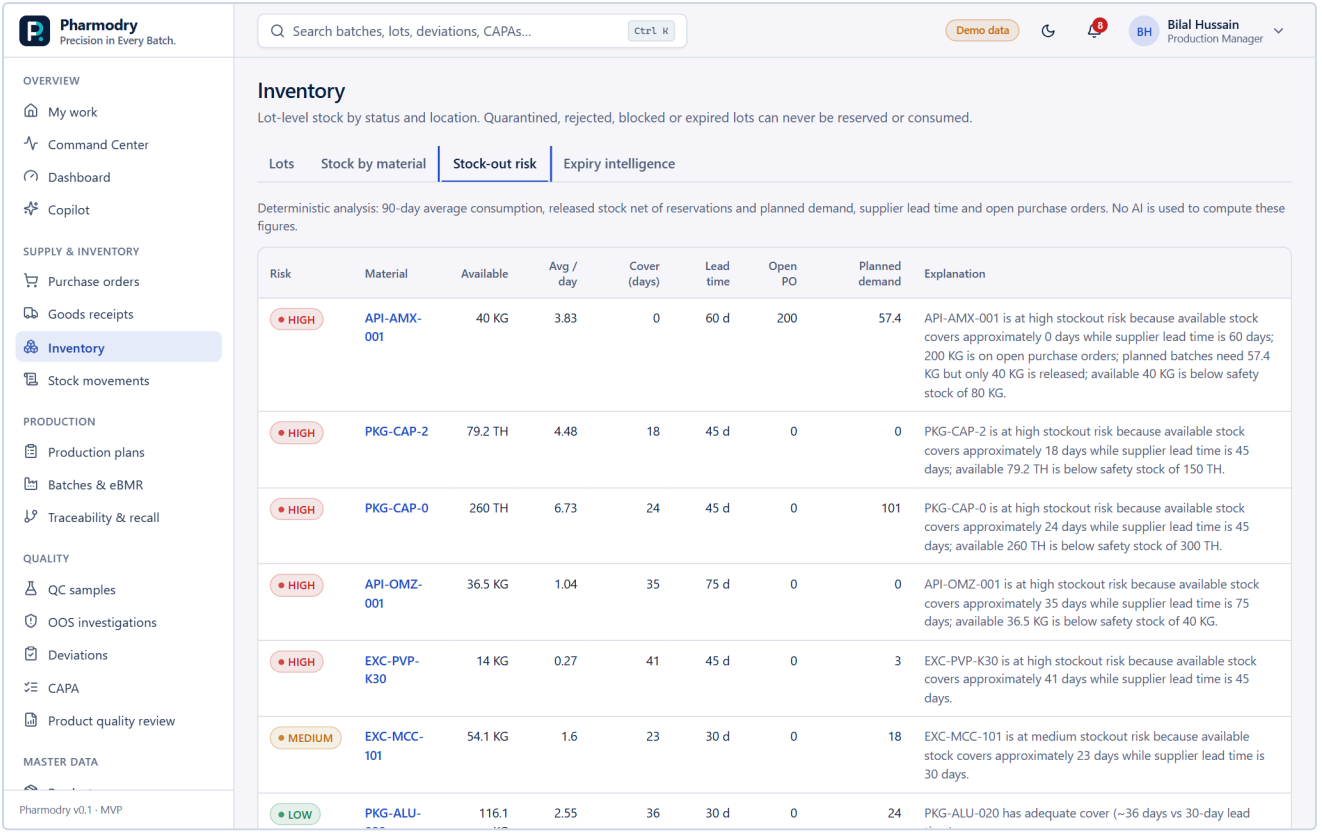


Figure 12 — Stock-out risk. Stock-out risk by material (released stock vs planned demand and lead time).

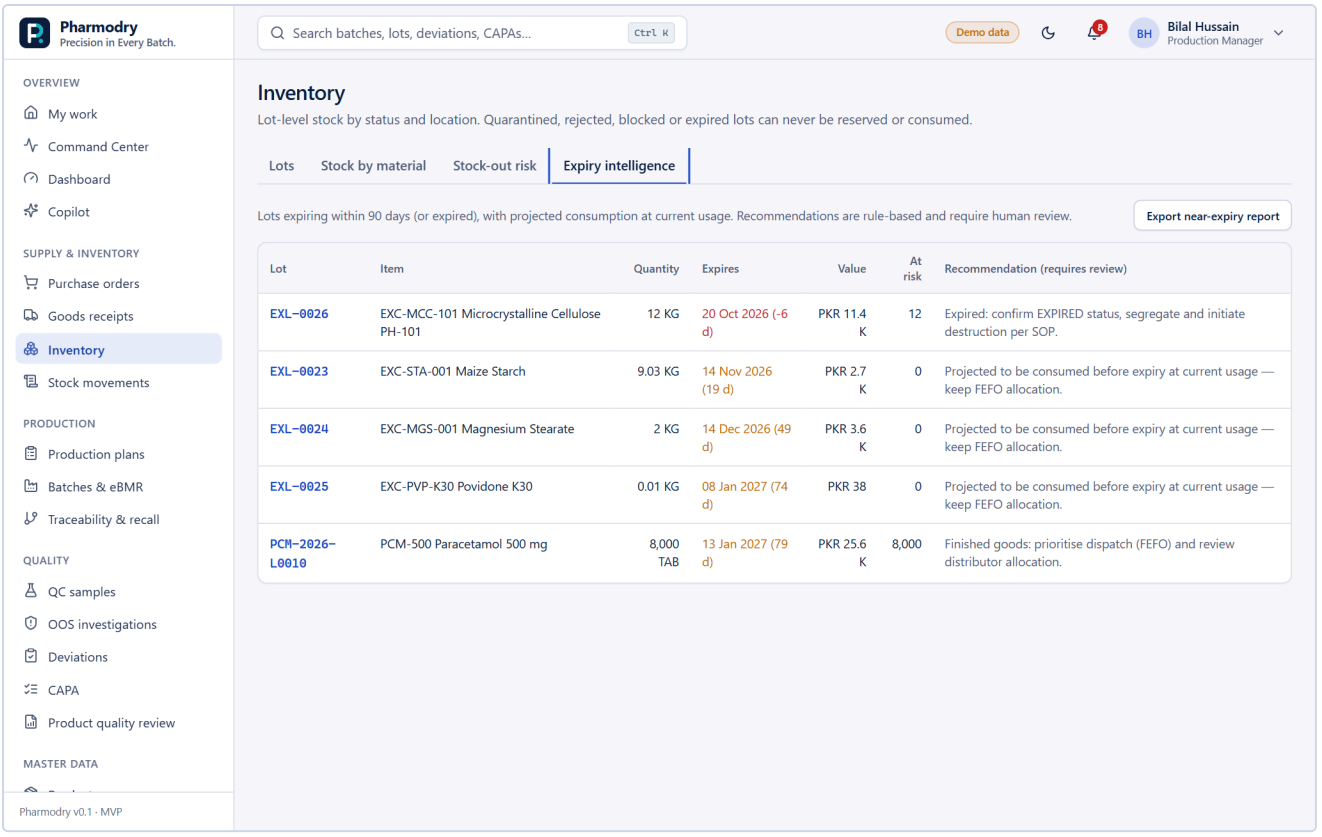


Figure 13 — Expiry intelligence. Expiry intelligence buckets with recommended actions.

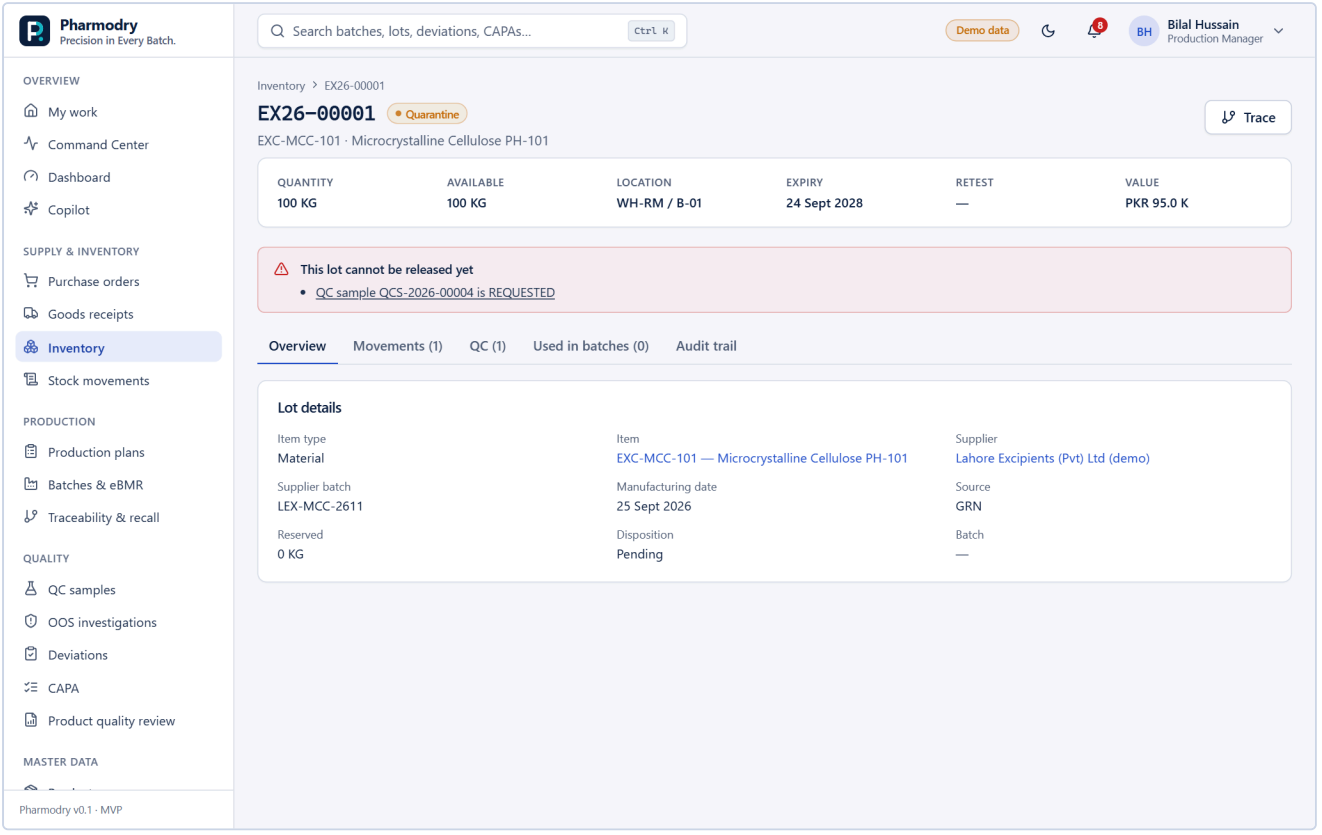


Figure 14 — Lot in quarantine. A quarantined lot: release is blocked until QC approval; blockers are listed.

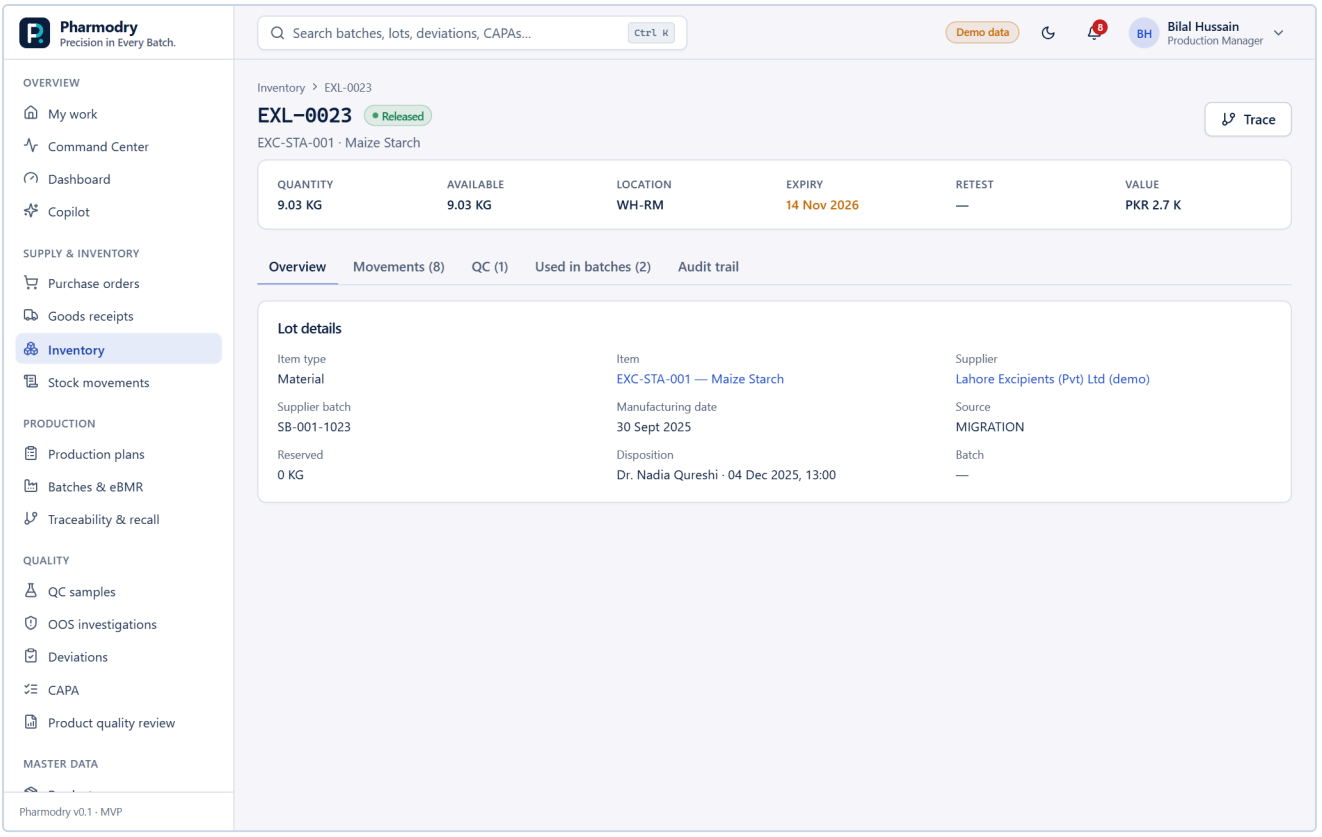


Figure 15 — Released lot. A released material lot with available quantity, location and expiry.

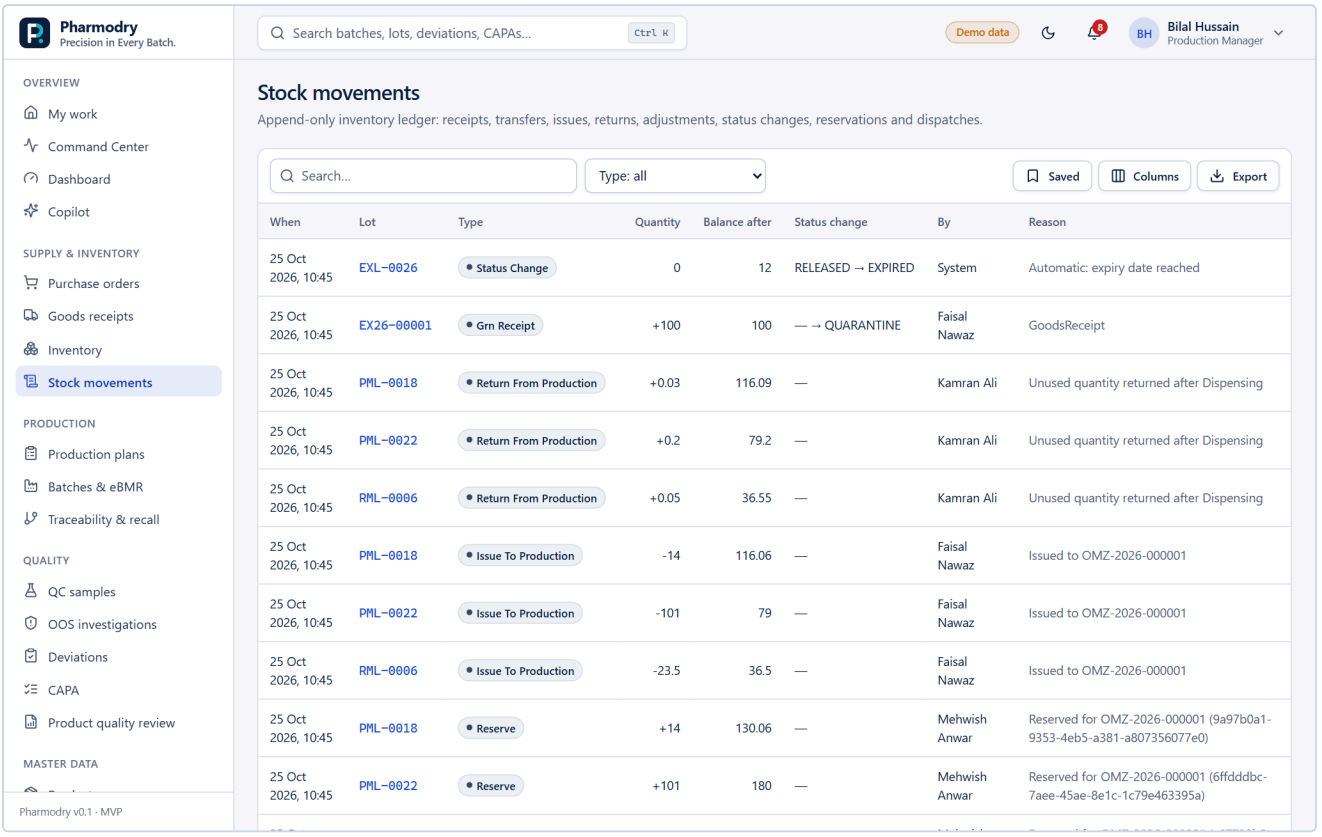


Figure 16 — Stock movements. Append-only inventory ledger of every movement.

Find a lot

1. Open Inventory.

2. Use the quick views (All on hand, Available (released), Awaiting disposition, Expiring ≤ 30 d / ≤ 60 d / ≤ 90 d, Expired) and the Status, Type and Warehouse filters.
3. Type part of a lot or item name in the table search.
4. Select a lot number to open the lot page (Overview, Movements, QC, Used in batches, Audit trail).

Export inventory and expiry lists

1. Use Export CSV on the lot table, or the download on Expiry intelligence.
2. The export is recorded in the audit trail.

> **Not available to your role:** Release or reject a material lot (QA disposition) — requires inventory.release; Transfer stock — requires inventory.transfer; Post a stock adjustment — requires inventory.adjust; Block or unblock a released lot — requires inventory.release.

Production plans

A production plan holds, per product, the planned quantity and number of batches for a period. Achievement = released batch output ÷ planned quantity. Plan detail shows feasibility (Feasible / Shortage) per line based on released stock.

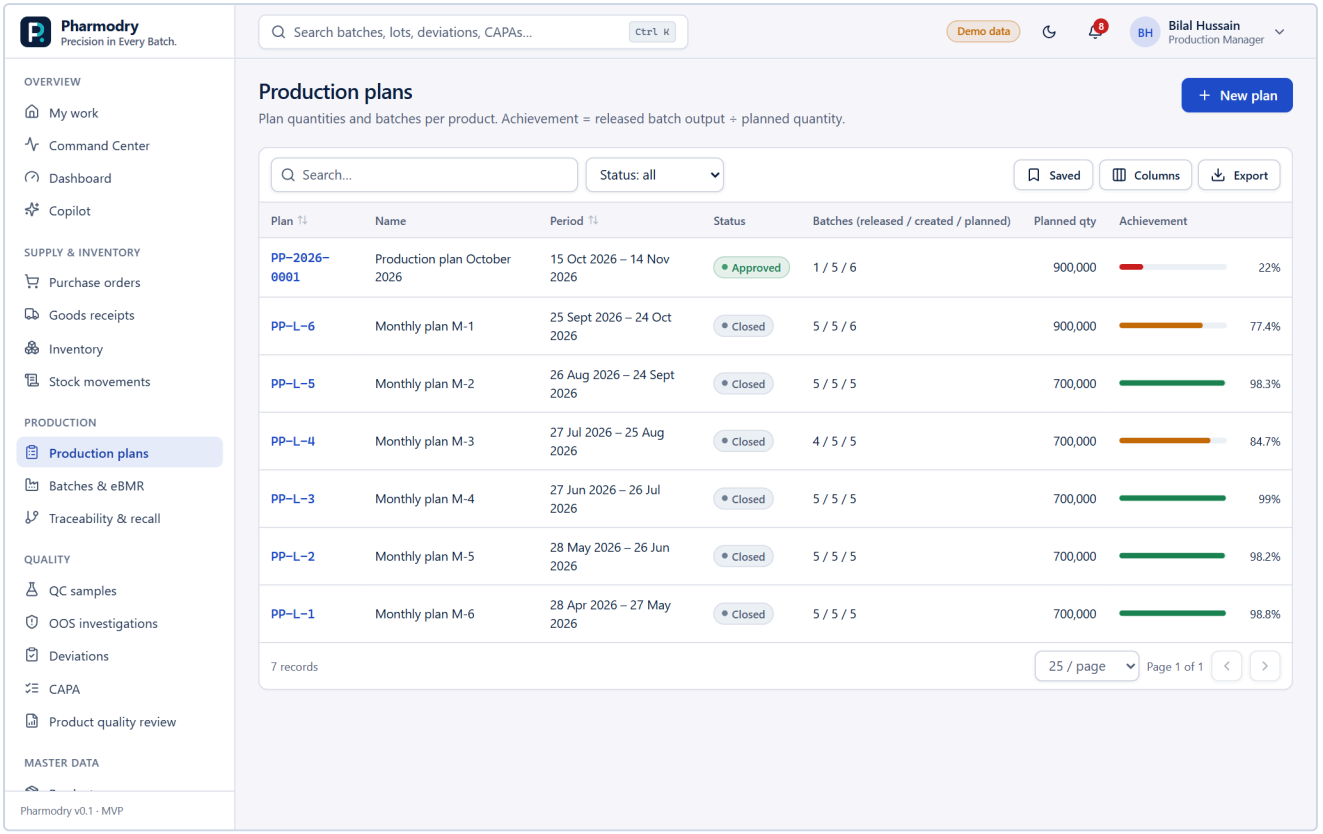


Figure 17 — Production plans. Production plans with achievement against released output.

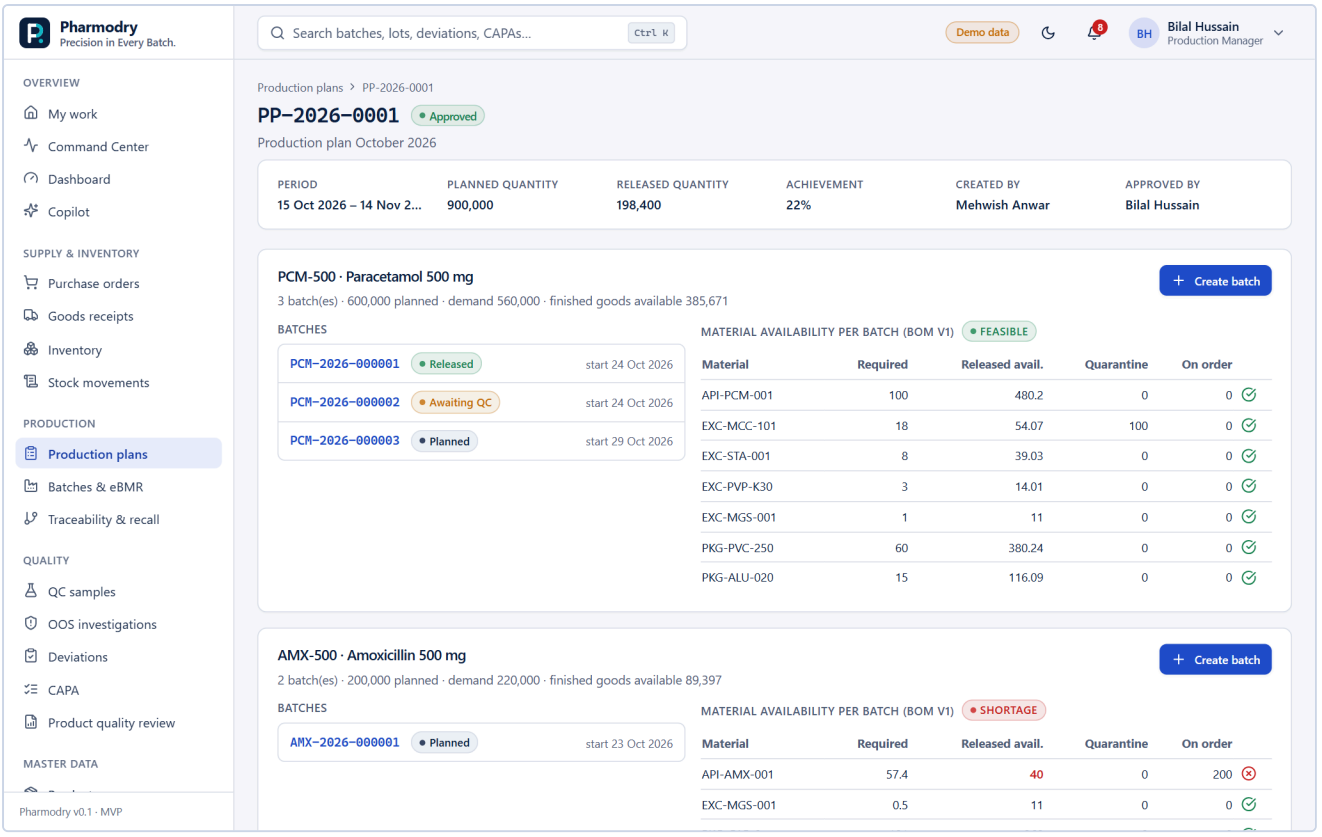


Figure 18 — Production plan detail. Plan lines with material feasibility and batch creation.

Create a production plan

1. Open Production plans and select New production plan.
2. Choose Site, enter Name, Period start and Period end.
3. Add product lines: Product, Batches and Planned quantity (defaults to batches × standard batch size).
4. Select Create plan. The plan is created as DRAFT.

Approve a plan

1. Open the draft plan.
2. Review lines and feasibility.
3. Select Approve plan.

Create a batch from a plan line

1. Open an approved plan.
2. Select Create batch on the line. The batch number is generated by the configured rule and the eBMR is copied from the active manufacturing template.

Batches and electronic batch records (eBMR)

Batch lifecycle: Planned → Material reserved → Ready → In progress → Awaiting QC → Awaiting QA → Released / Rejected → Closed (or Cancelled before start). The batch page has tabs Overview, Manufacturing (eBMR), Materials, QC, Deviations, Approvals and Audit trail, and shows GS1 data (GTIN, expiry, batch) for the 2D DataMatrix when the product has a GTIN.

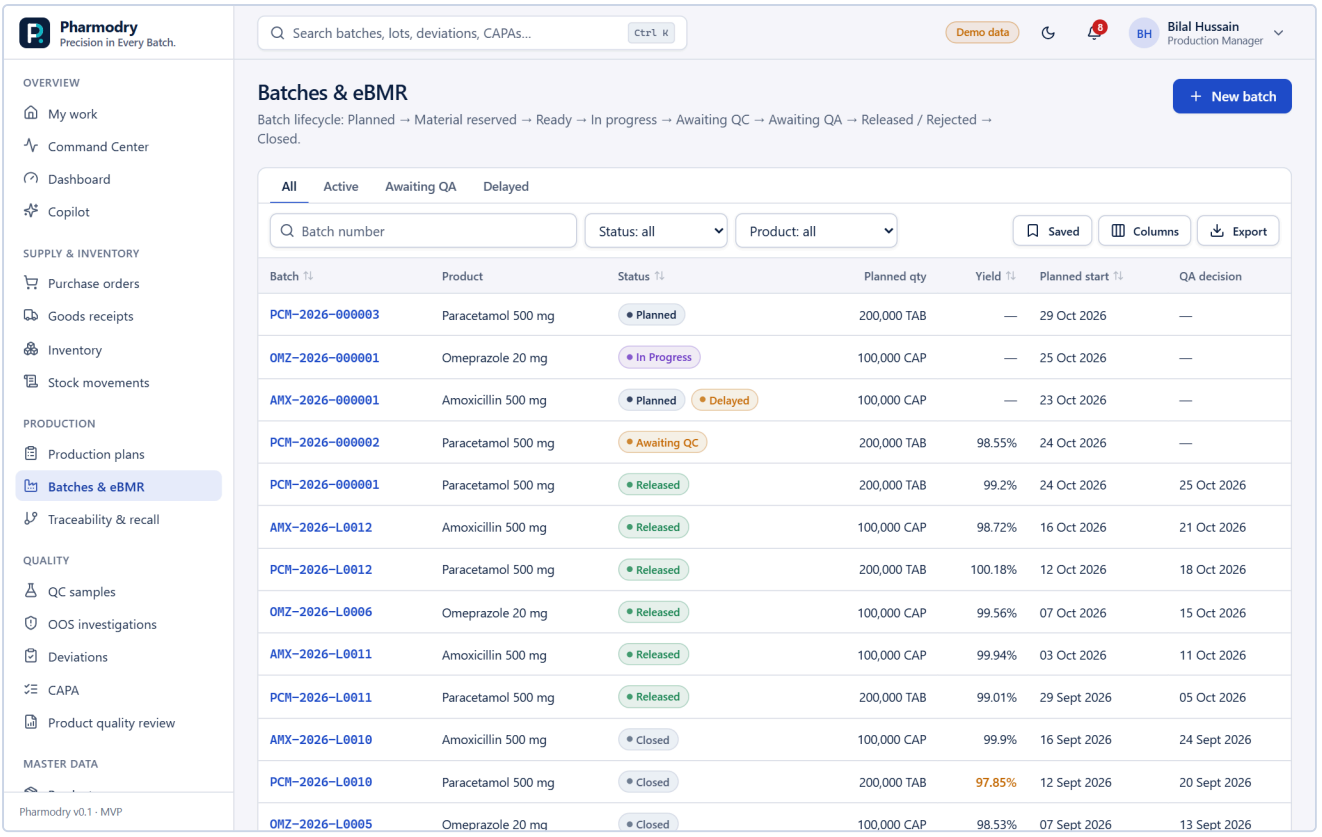


Figure 19 — Batches & eBMR. Batch list with lifecycle status, yield and QA decision.

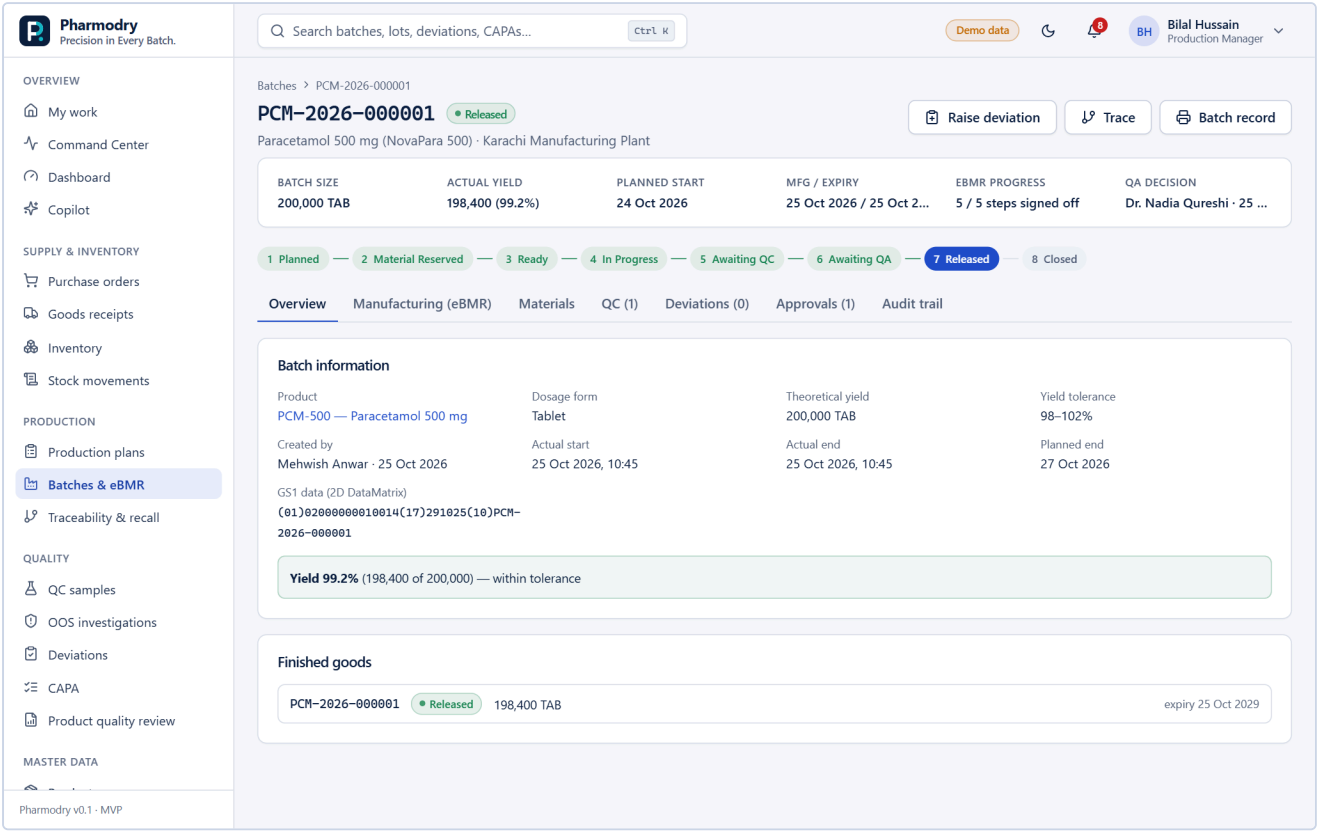


Figure 20 — Released batch — overview. Overview of a released batch: yield, finished goods, GS1 data and QA decision.

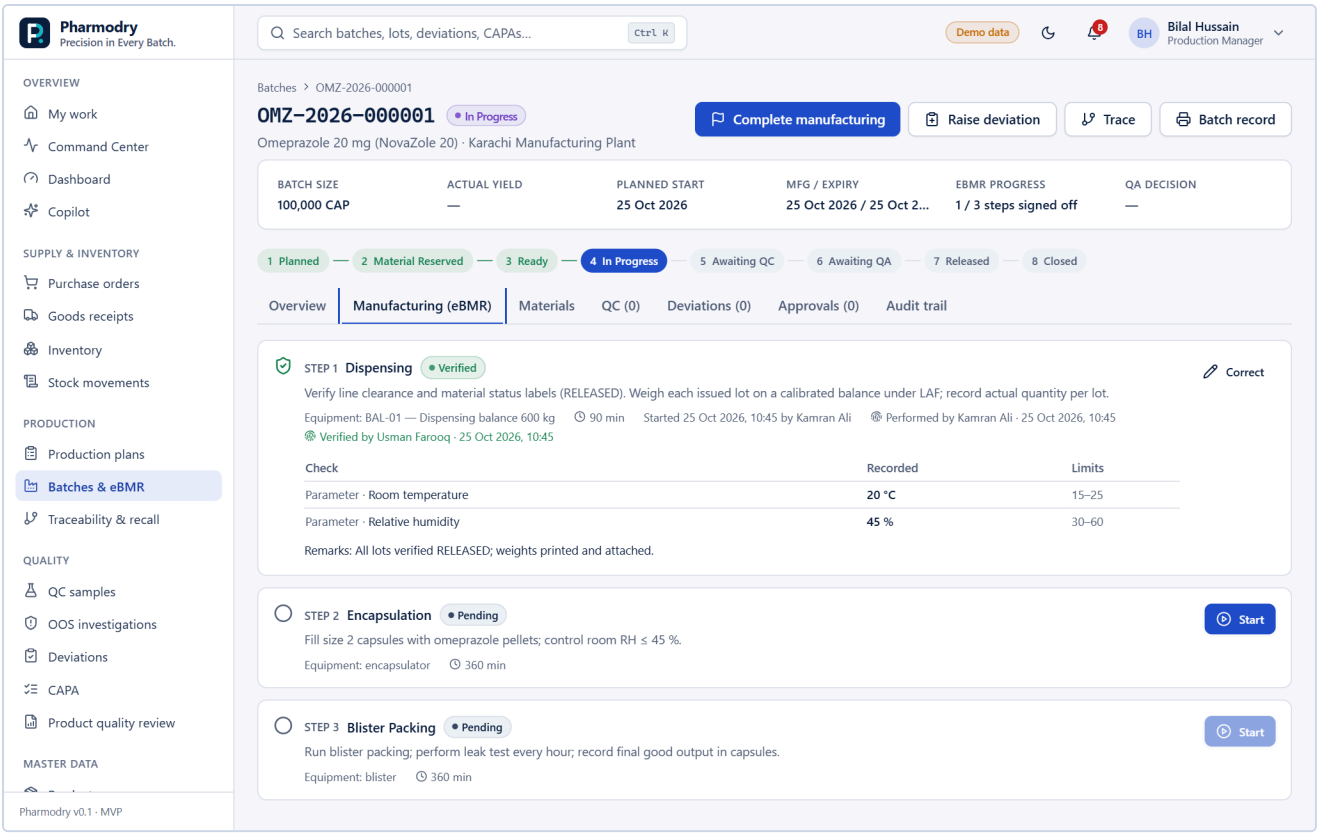


Figure 21 — Batch in progress — eBMR. Electronic batch record of a batch in progress: steps, parameters, signatures.

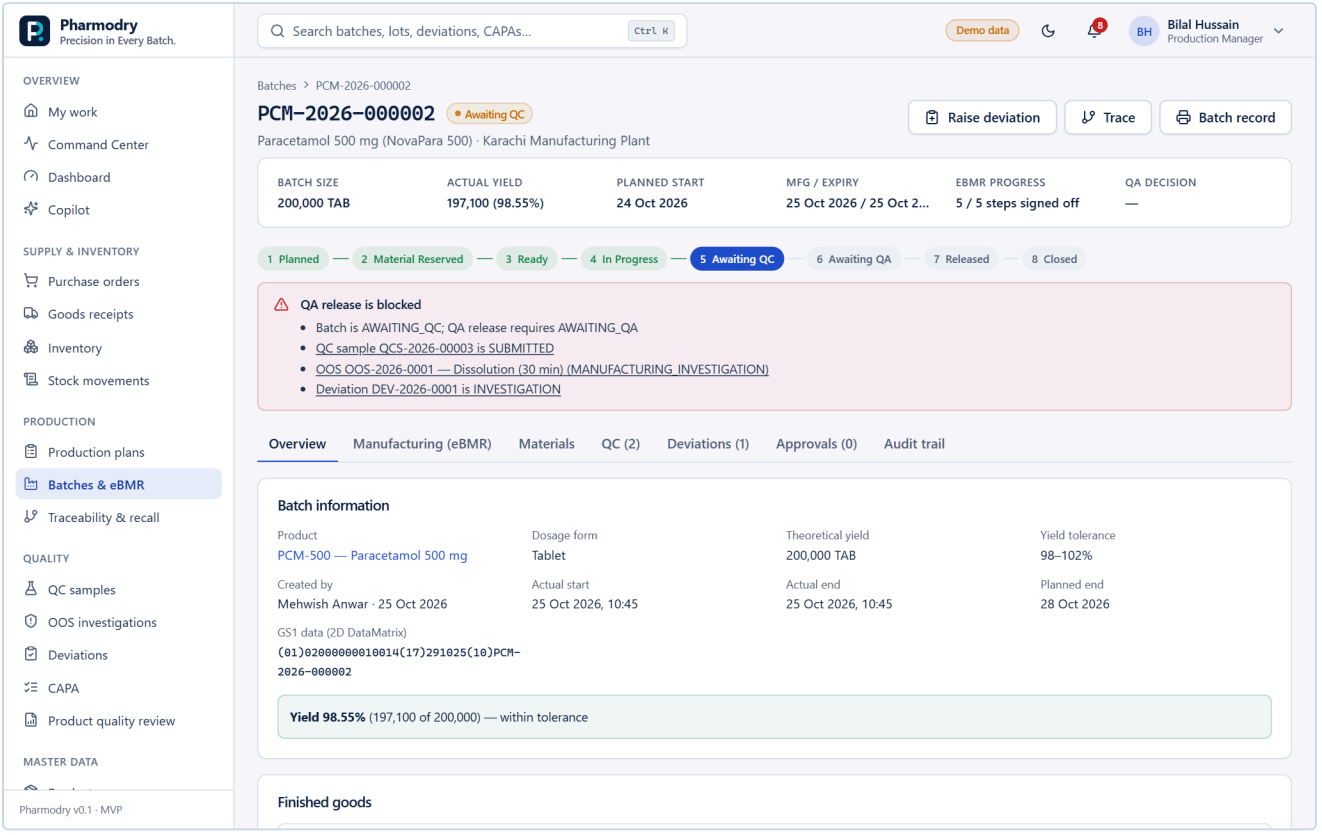


Figure 22 — Batch blocked by OOS. A batch whose release is blocked: failed QC and an open OOS investigation.

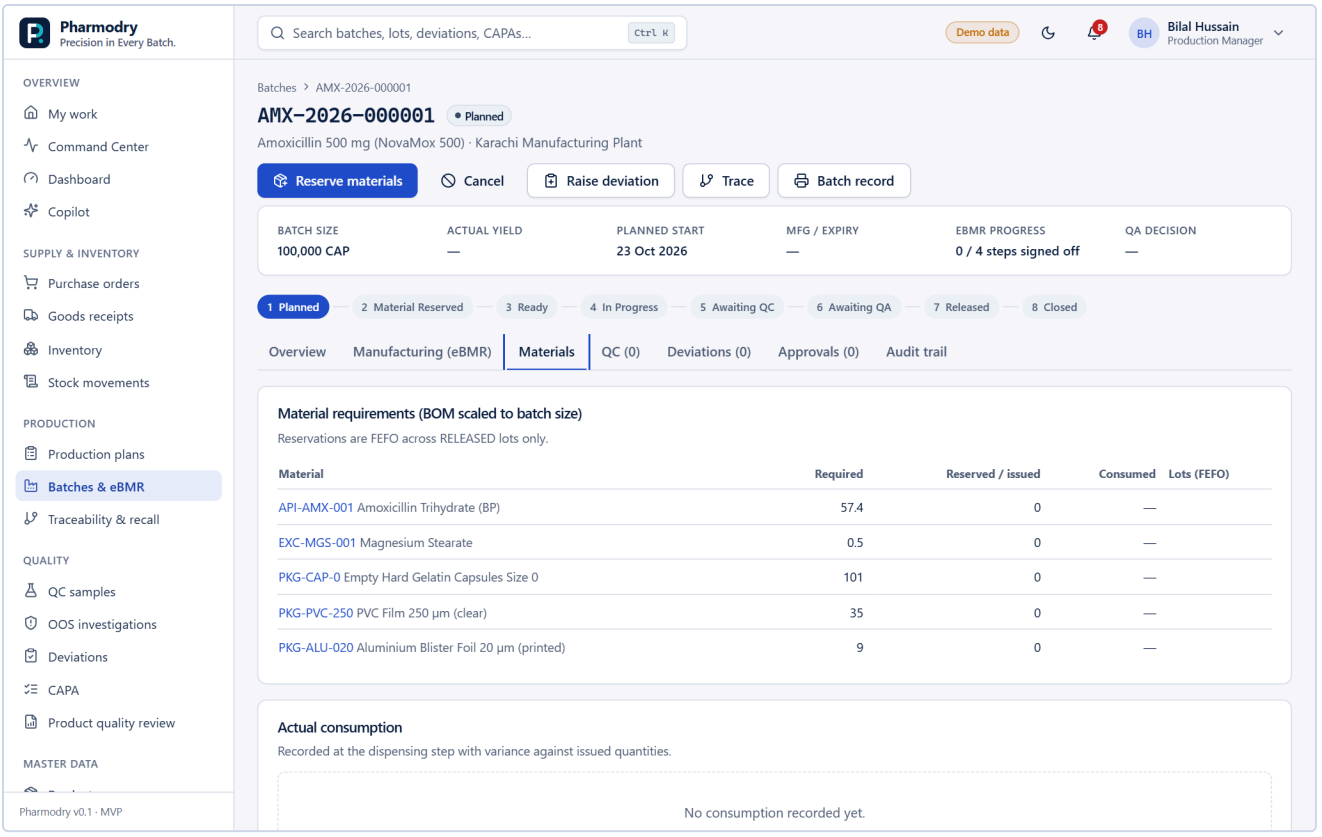


Figure 23 — Batch with material shortage. A planned batch that cannot reserve materials because released stock is insufficient.

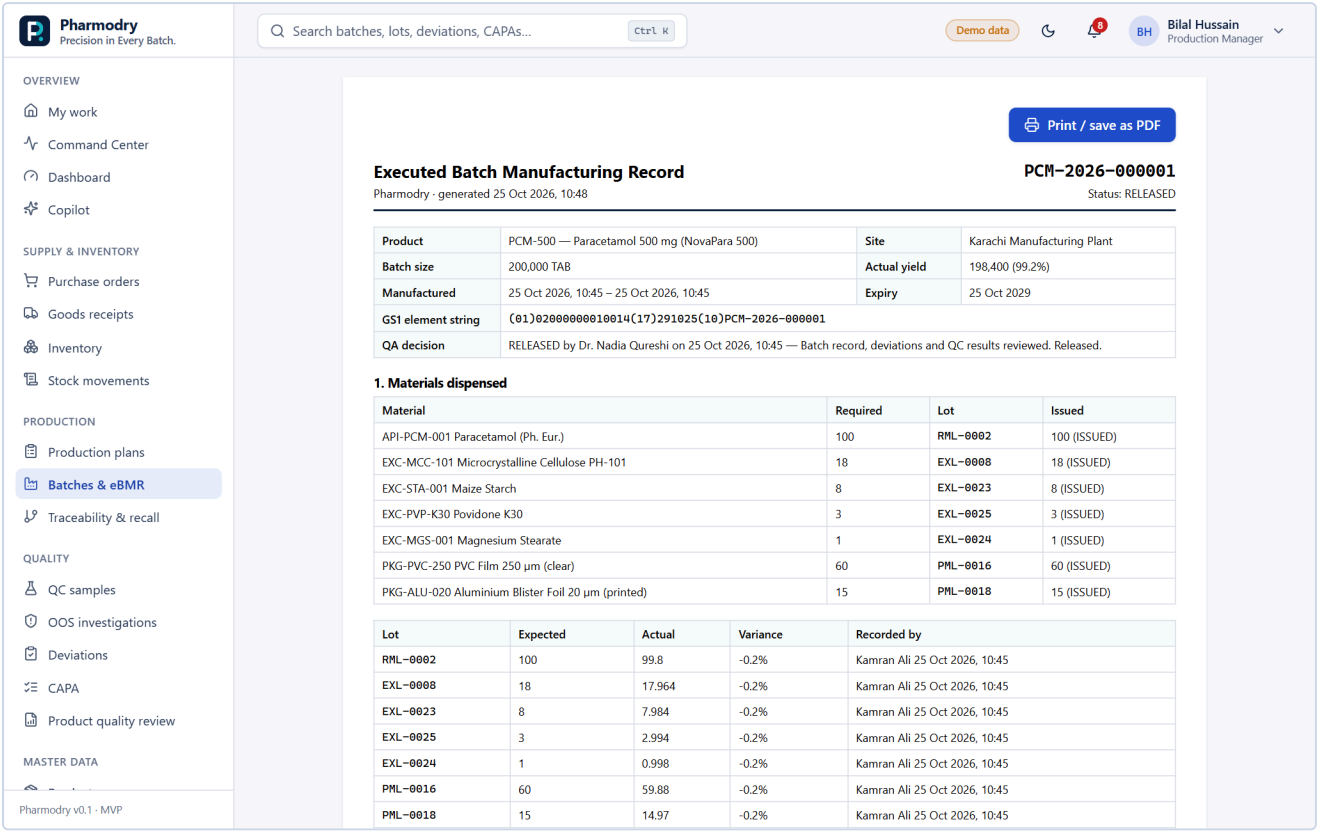


Figure 24 — Printable batch record. Print-ready batch manufacturing record (use the browser's Save as PDF).

Create a batch

1. Open Batches & eBMR and select New (or Create batch on a plan line).
2. Choose Site, Product, optional Production plan line, Batch size and Planned start.
3. Check Material availability: “All materials available” or “Shortage — reservation will be blocked”. Only released, unexpired stock net of reservations counts.
4. Select Create batch. The number follows the configured pattern (e.g. PCM-2026-000123).

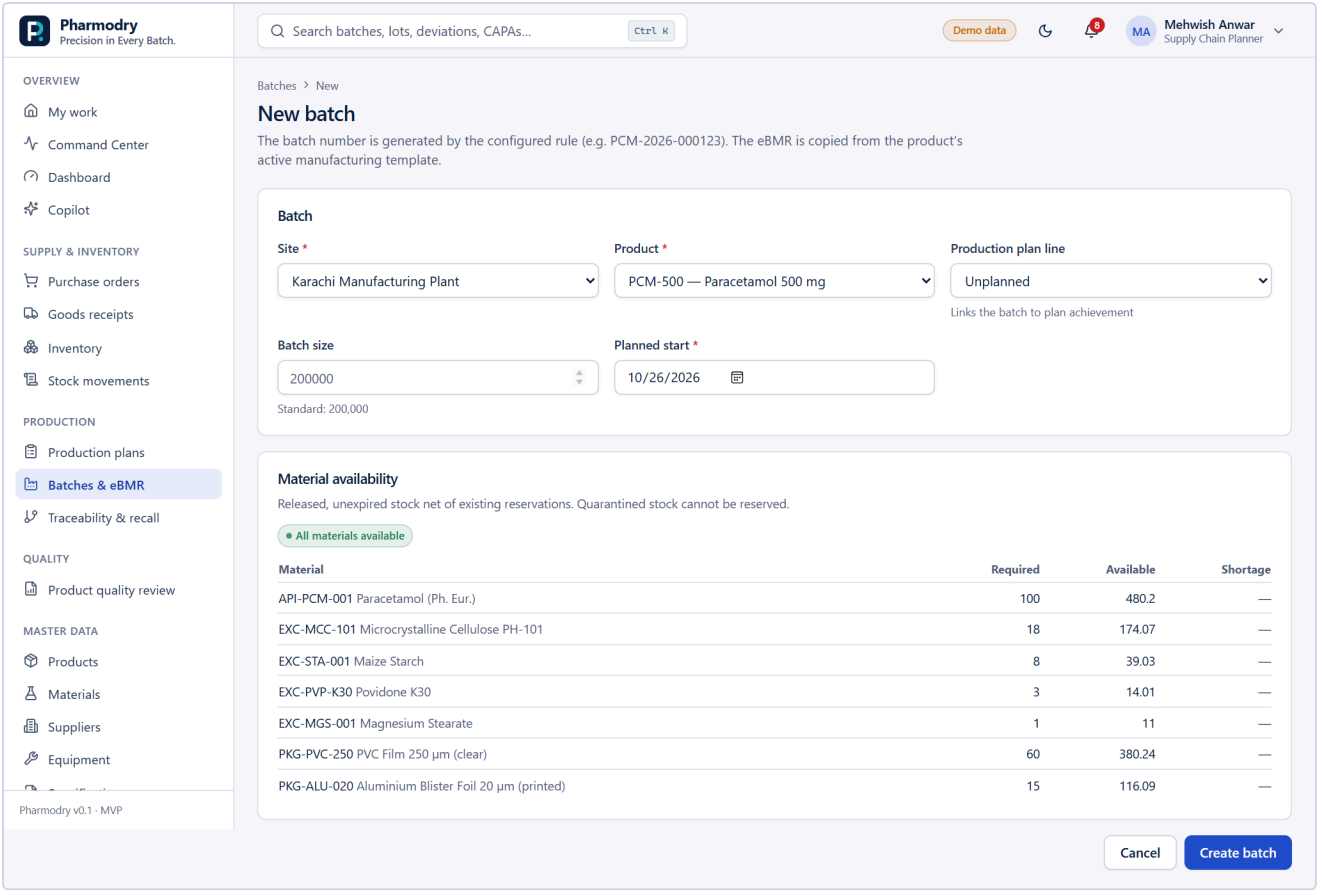


Figure 25 — Create a batch. New batch form with live material availability (released stock only). (screen captured while signed in as Supply Chain Planner)

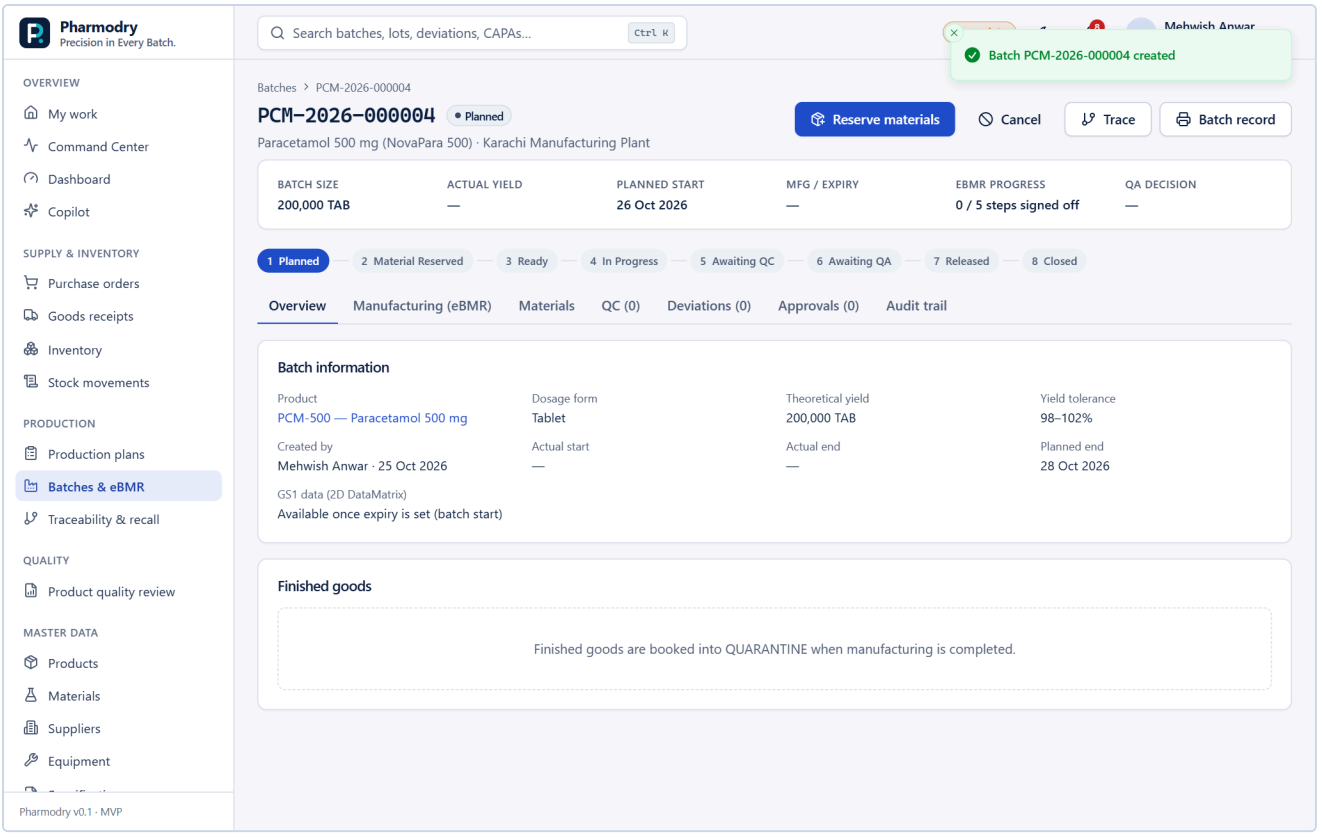


Figure 26 — Batch planned. A planned batch; the next step is FEFO material reservation. (screen captured while signed in as Supply Chain Planner)

Reserve materials (FEFO)

1. Open the PLANNED batch and select Reserve materials.
2. Pharmodry reserves RELEASED lots in first-expiry-first-out order. If stock is insufficient the action is refused.
3. Release reservation returns reserved quantities (with a reason) if the batch will not proceed.

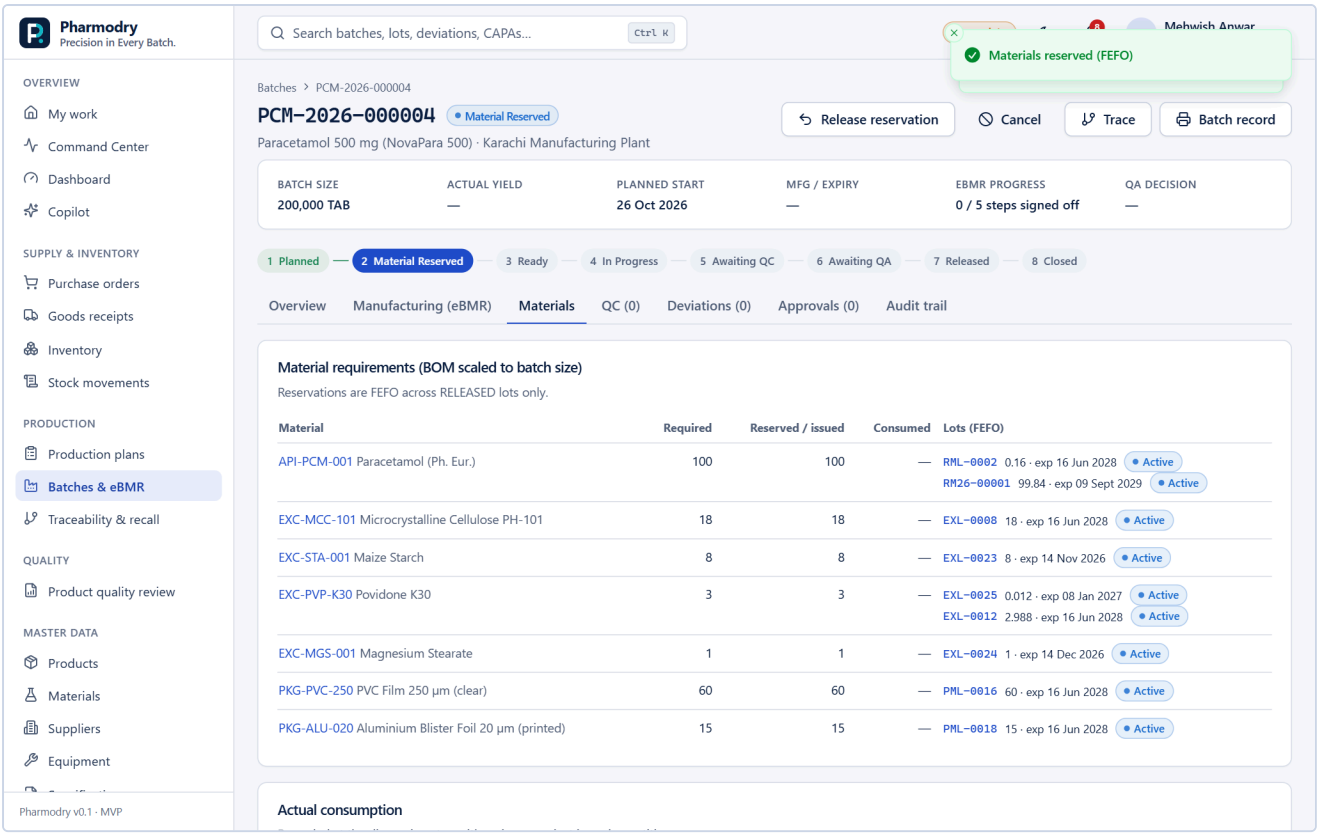


Figure 27 — Materials reserved (FEFO). Reserved lots in first-expiry-first-out order. (screen captured while signed in as Supply Chain Planner)

Start the batch

1. Open the READY batch and select Start batch. The batch is IN PROGRESS and eBMR steps can be executed.

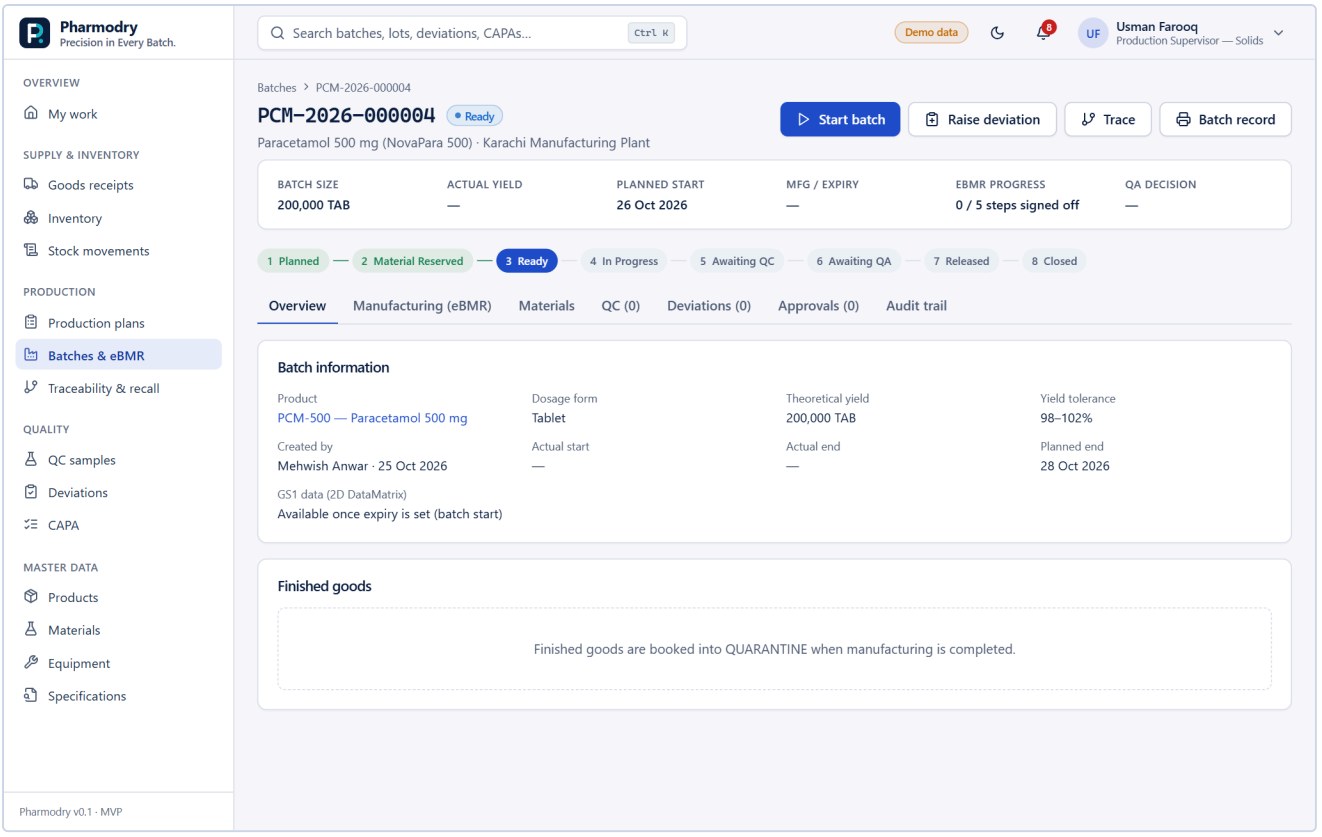


Figure 28 — Batch ready to start. Materials issued; the supervisor starts the batch. (screen captured while signed in as Production Supervisor)

Execute an eBMR step

1. Open the batch and the Manufacturing (eBMR) tab. Steps must be done in order.
2. Select Start on the next step (enabled only when the previous step is done) and choose the equipment. Equipment must be qualified and within calibration; expired calibration is blocked (or, under WARN policy, needs an Override reason).
3. Perform the operation per the master instructions.
4. Select Record & sign, enter every process parameter and in-process control reading. Out-of-limit readings are flagged (“Out-of-limit reading”).
5. Enter Remarks if needed, re-enter your password and sign.

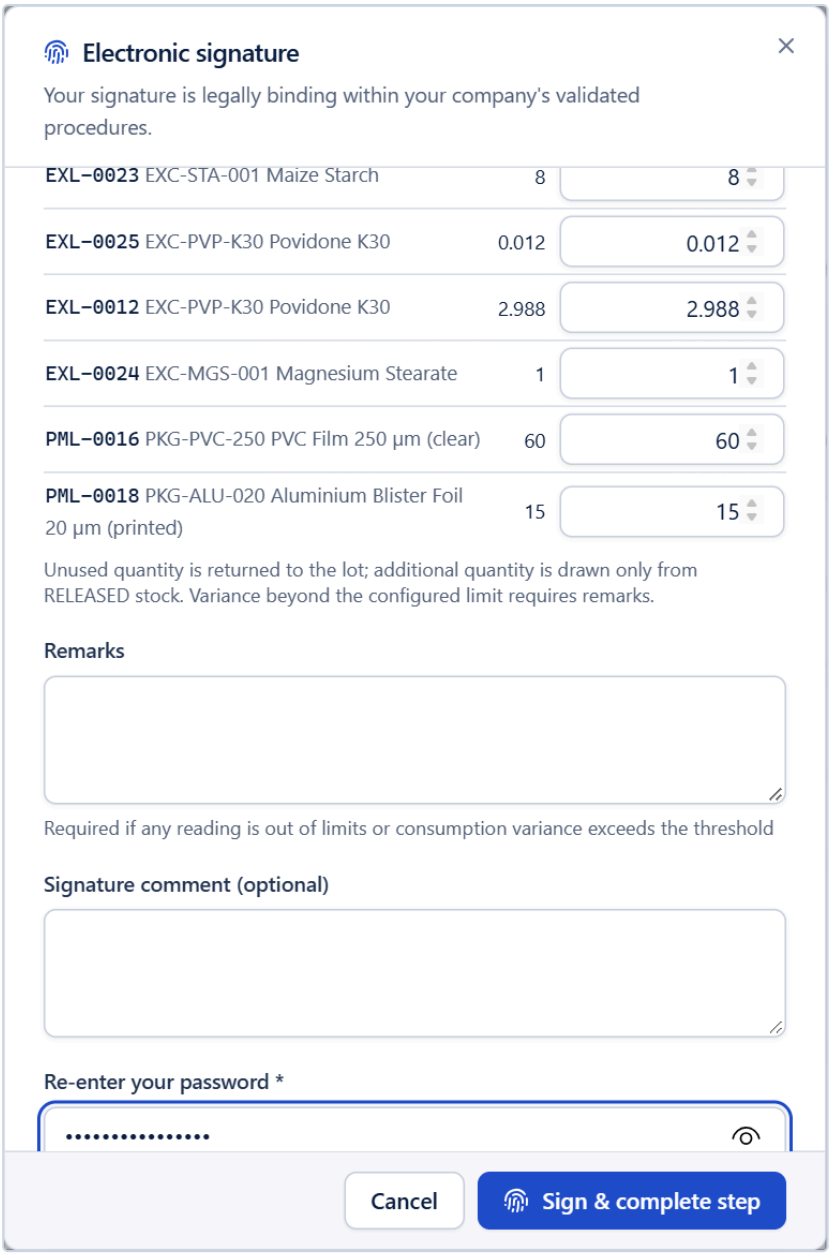


Figure 29 — Complete and sign step DISP. Recording parameters and in-process checks, then signing the step. (screen captured while signed in as Production Operator)

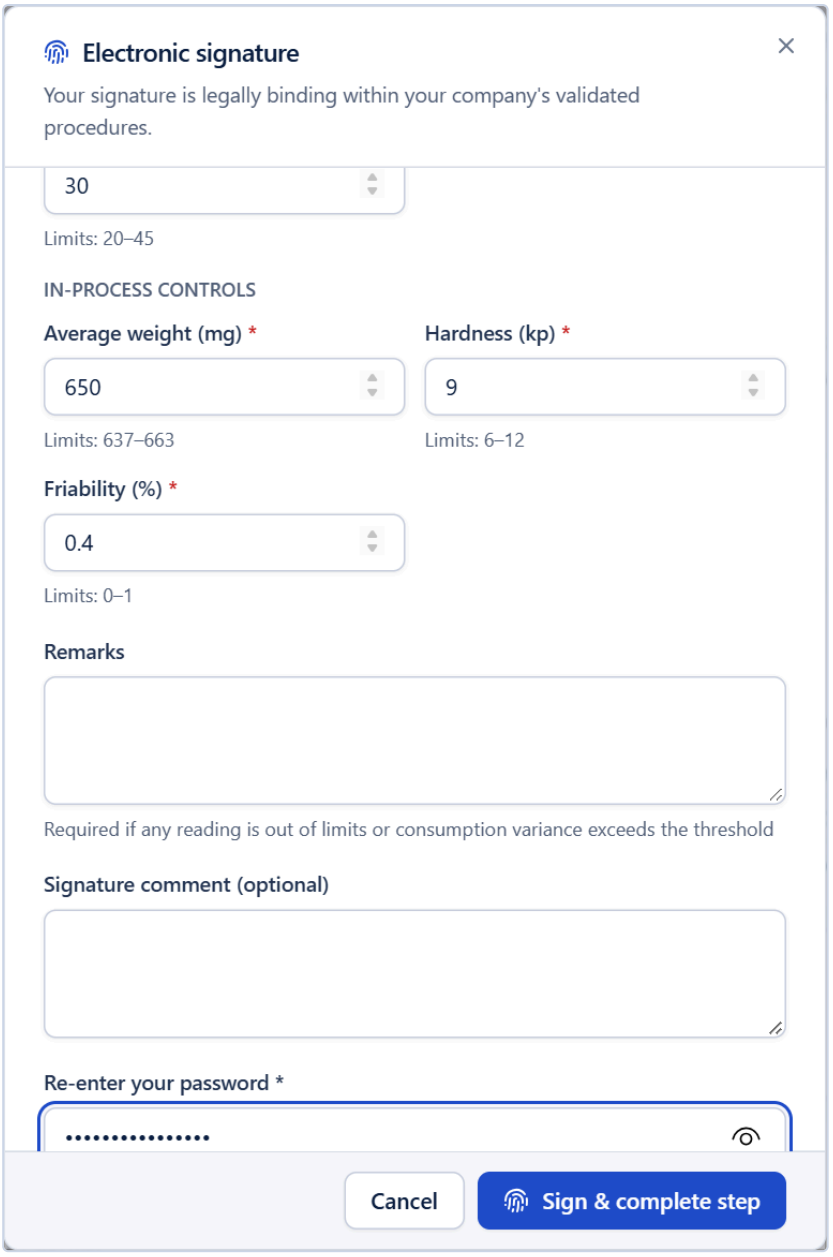


Figure 30 — Complete and sign step COMP. Recording parameters and in-process checks, then signing the step. (screen captured while signed in as Production Operator)

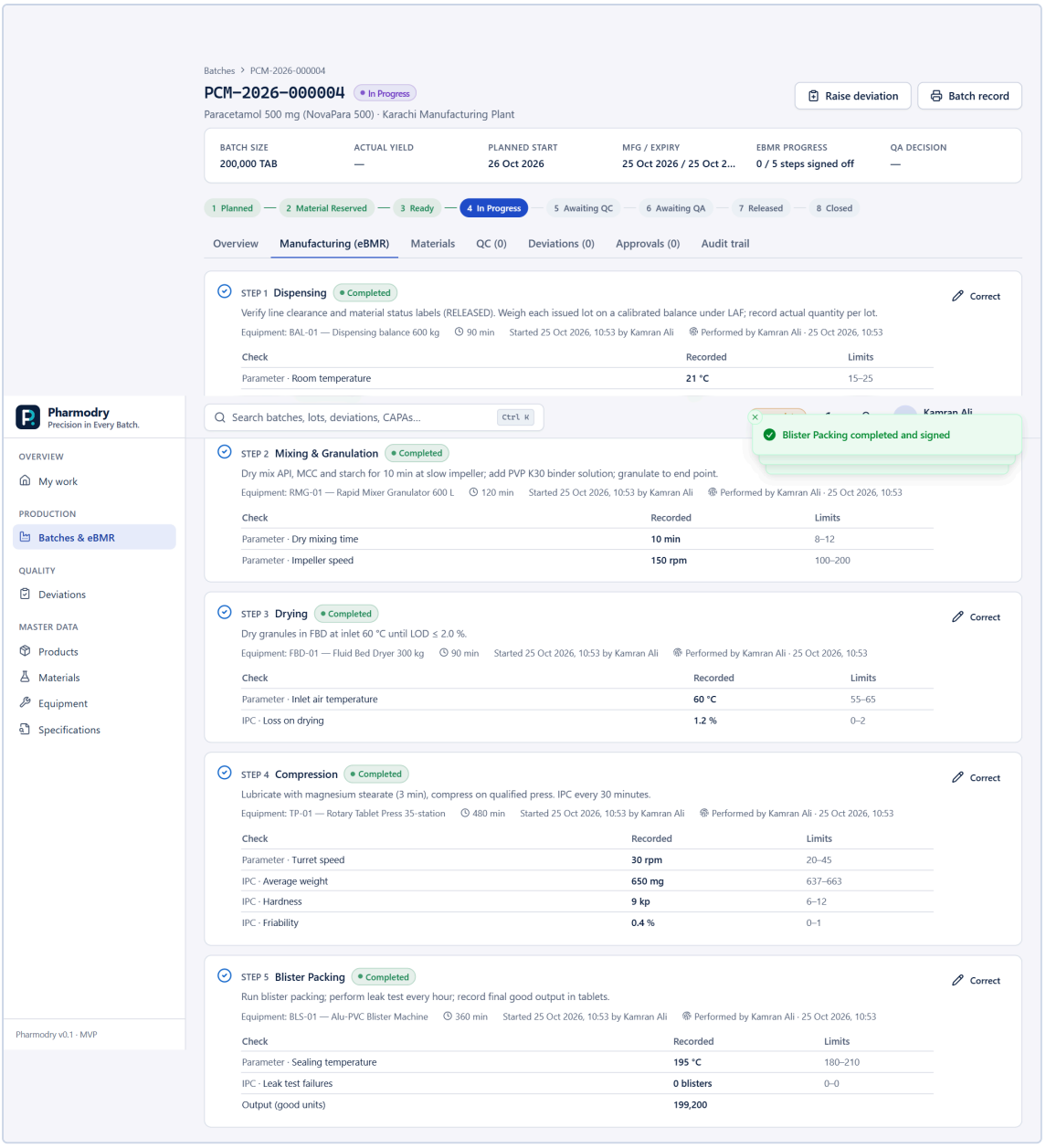


Figure 31 — All steps completed. All eBMR steps completed and signed; awaiting supervisor verification. (screen captured while signed in as Production Operator)

Verify eBMR steps

1. Open the Manufacturing (eBMR) tab.
2. Review each completed step's readings and signature.
3. Select Verify and sign the electronic signature. You cannot verify a step you performed yourself.

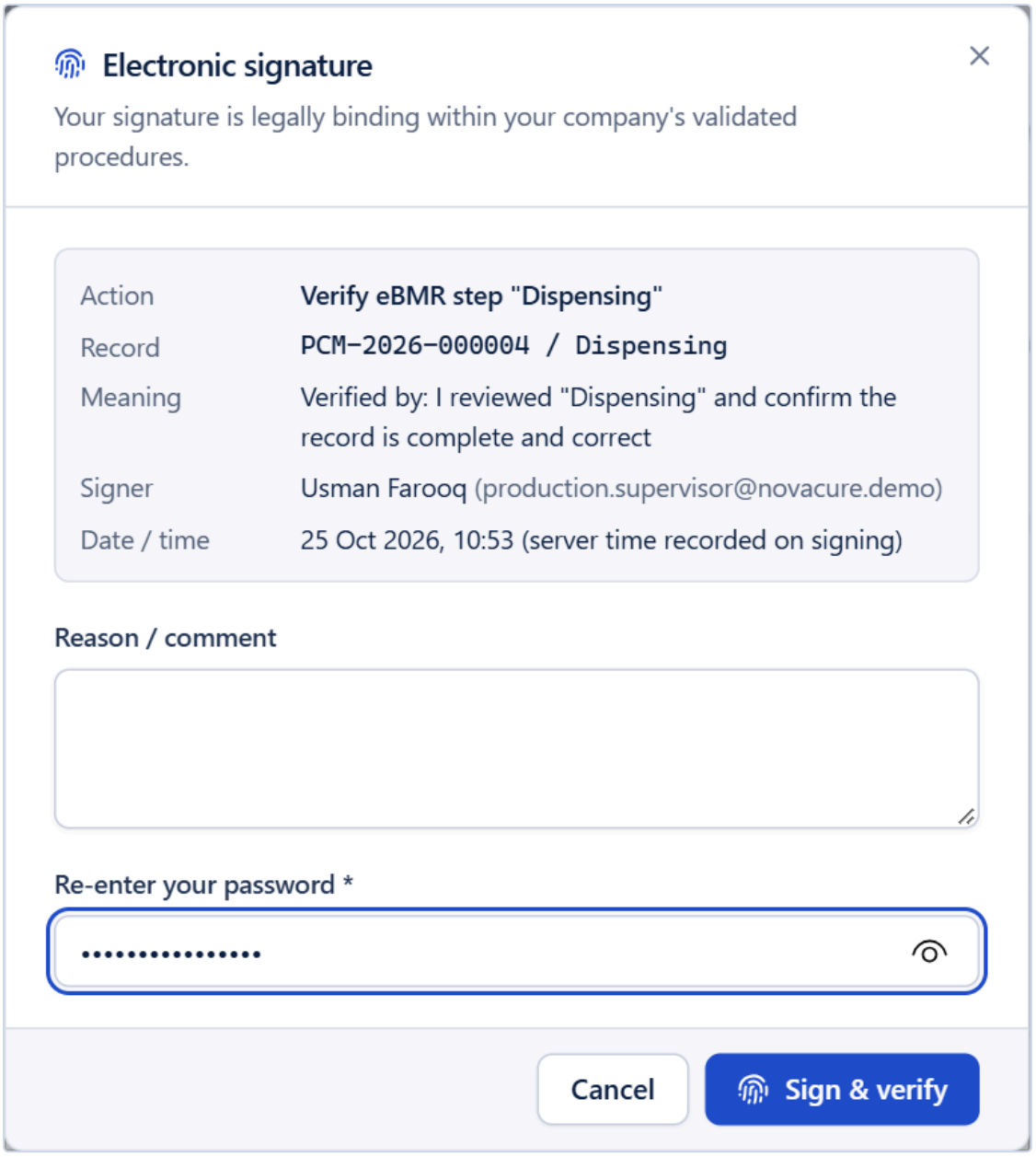


Figure 32 — Verify a step with e-signature. Supervisor verification of an operator step (a different person must verify). (screen captured while signed in as Production Supervisor)

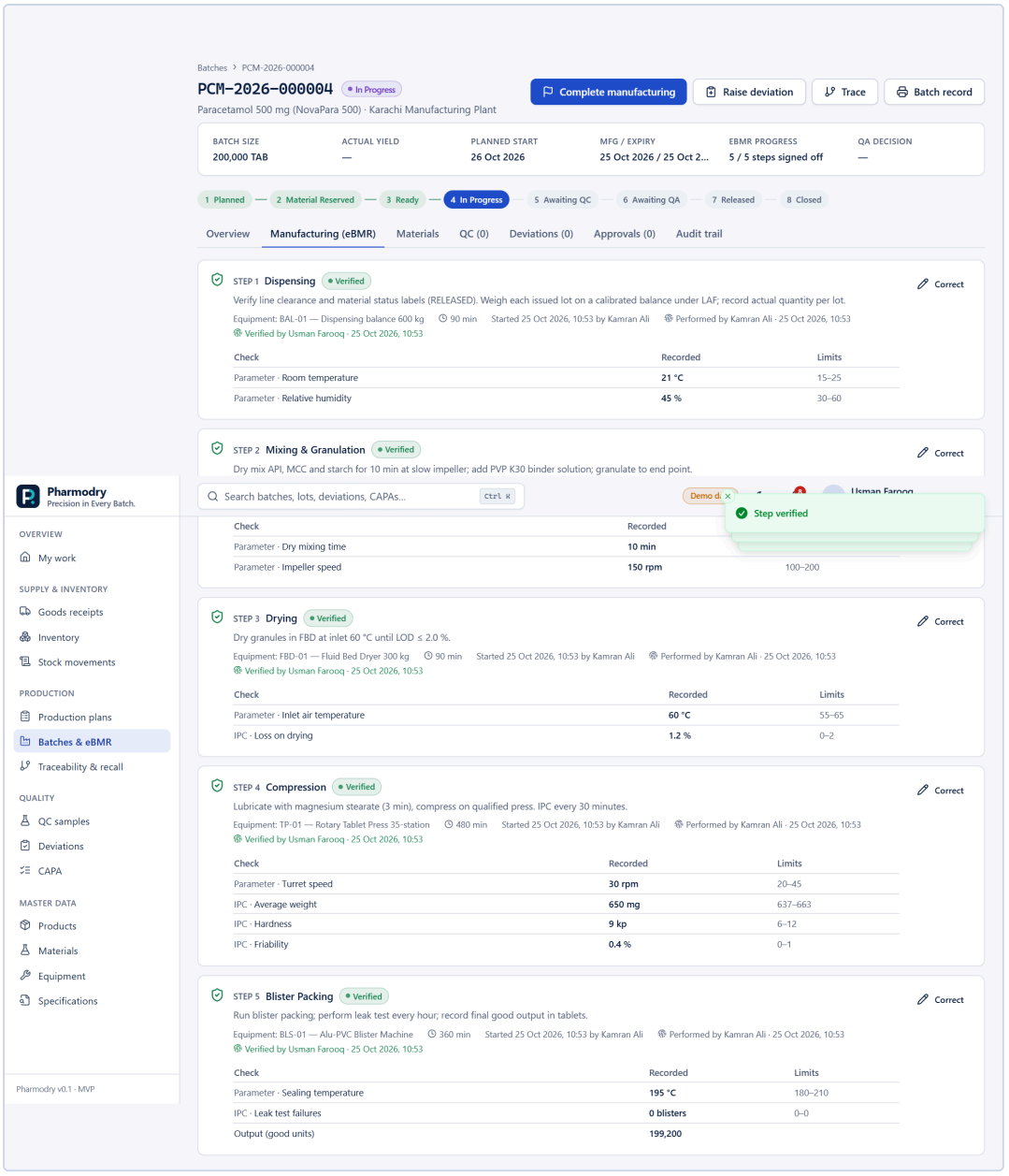


Figure 33 — All steps verified. Every step verified; the batch can be completed. (screen captured while signed in as Production Supervisor)

Record a correction

1. On a completed step select Correct.
2. Choose the Field, enter the New value and the Reason for correction.
3. Select Record correction. The original value, new value, reason and your identity are kept; a verified step returns to COMPLETED and must be re-verified.

Complete manufacturing

1. When all steps are verified select Complete manufacturing.
2. Review the Calculated yield against the tolerance. If it is outside tolerance you must select a Linked deviation (recommended) or enter a documented justification.
3. Confirm. Finished goods are booked into QUARANTINE and a finished-product QC sample is requested; the batch is AWAITING QC.

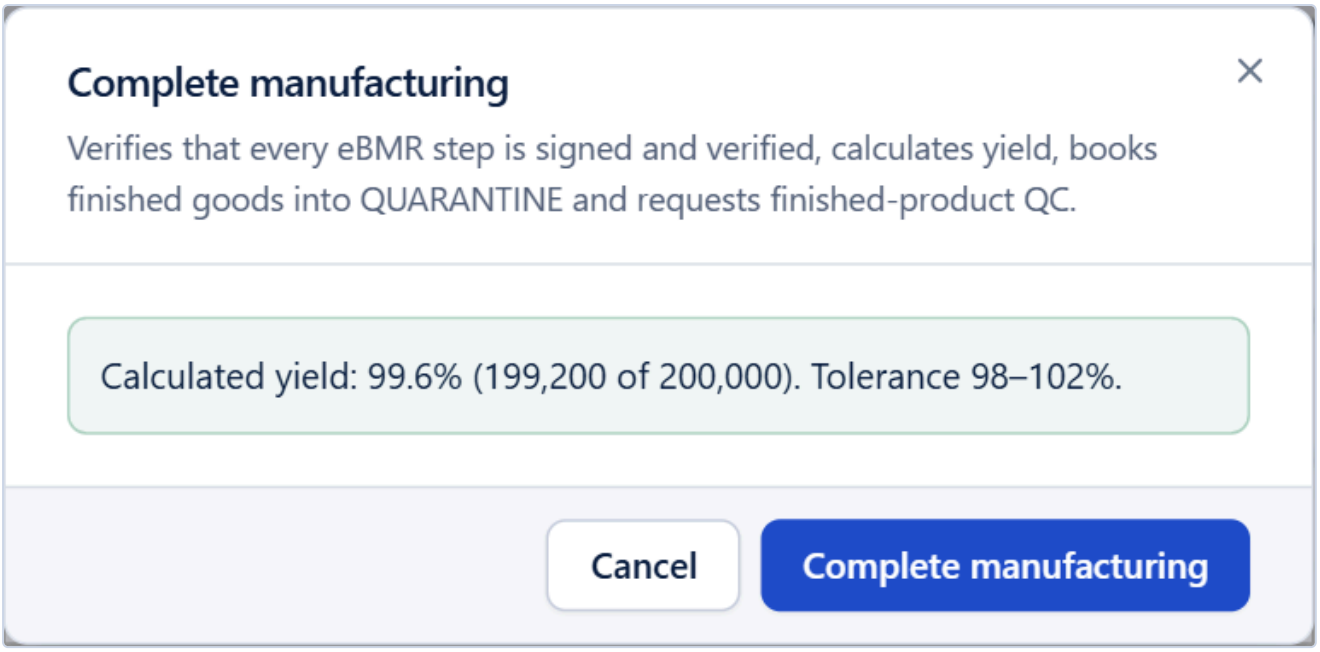


Figure 34 — Complete manufacturing. Completion shows the calculated yield against tolerance.

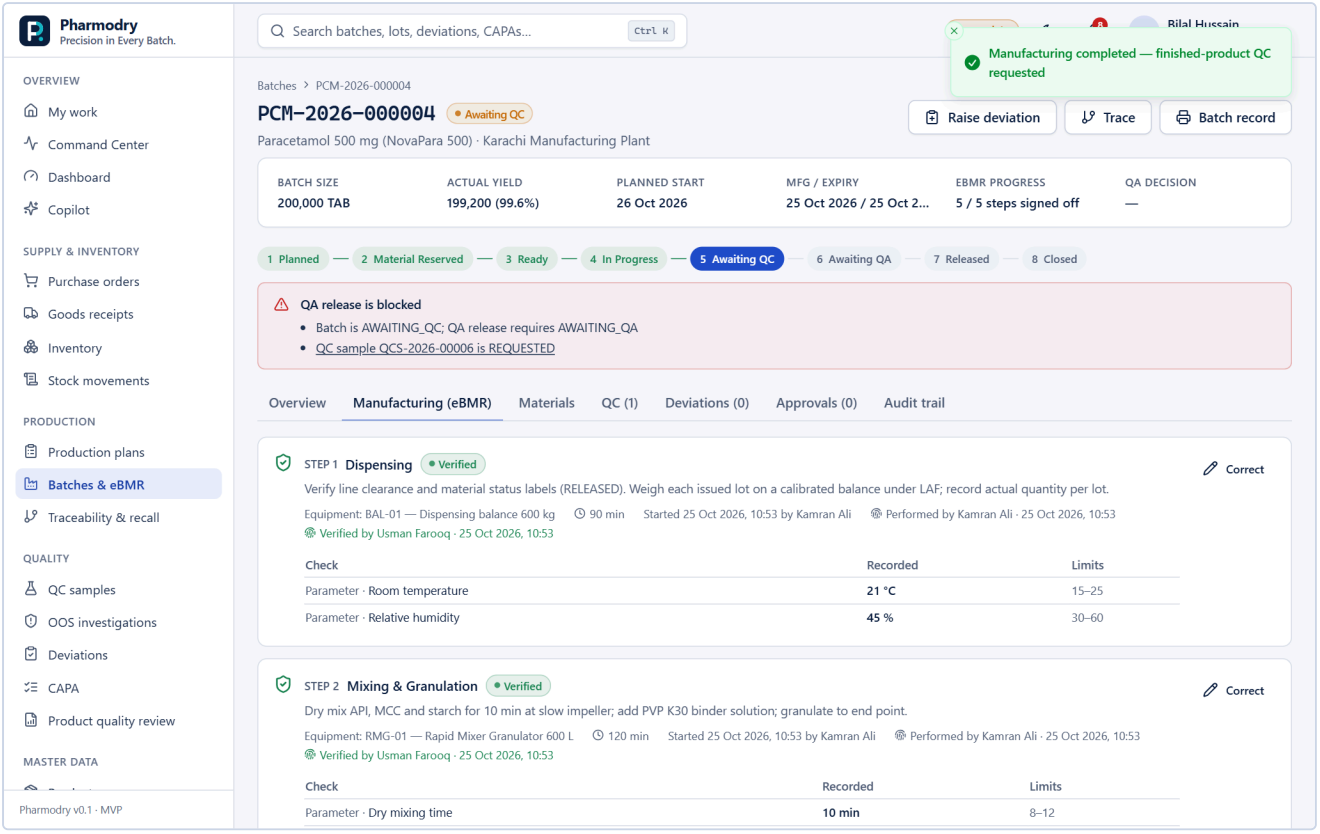


Figure 35 — Batch awaiting QC. Finished goods in QUARANTINE; finished-product QC requested.

Print the batch manufacturing record

1. Open the batch and select Batch record (opens in a new tab).
2. Use Print / the browser's Save as PDF.

Raise a deviation from a batch

1. On the batch page select Raise deviation; the batch is pre-filled.

> **Not available to your role:** Issue materials to production — requires inventory.issue; QA release or rejection of a batch — requires batch.approve.

Traceability and mock recall

Backward genealogy (what went into a batch: material lots, suppliers, supplier batches, GRNs and QC release evidence), forward trace (where a material lot went: batches and finished goods) and the mock-recall reconciliation of a batch (produced vs on hand + dispatched − returned ± adjustments/samples, with any unaccounted quantity highlighted). Customer-level distribution is held in the ERP/distribution system and is not captured in this release.

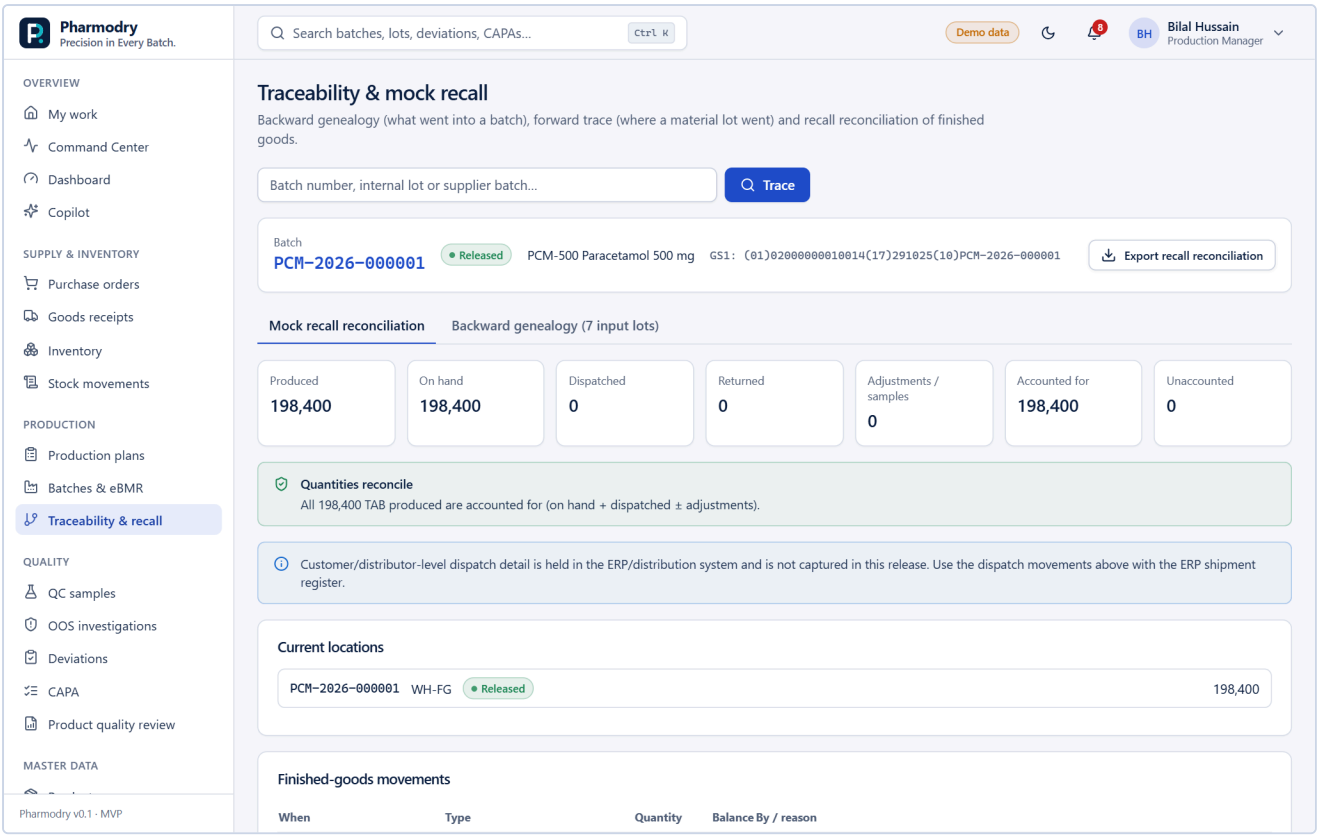


Figure 36 — Traceability & mock recall. Mock recall reconciliation of a released batch.

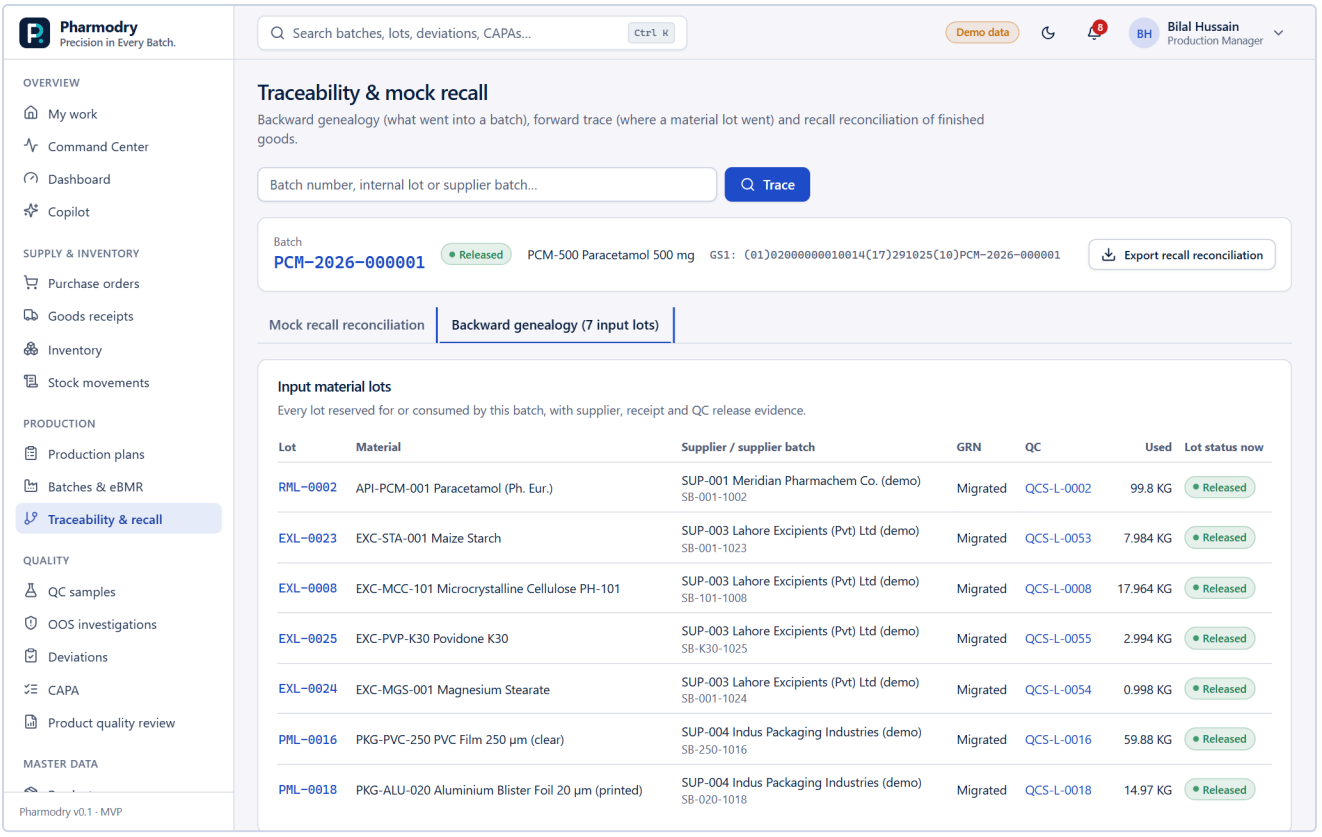


Figure 37 — Backward genealogy. Input material lots with supplier, GRN and QC release evidence.

Run a mock recall for a batch

1. Open Traceability & recall (or Trace on a batch page).
2. Type a batch number, internal lot or supplier batch and select Trace; choose the batch in the results.
3. Read the reconciliation cards. “Quantities reconcile” means every unit is accounted for; “Reconciliation gap” shows the unaccounted quantity to investigate.
4. Review Current locations and Finished-goods movements.
5. Users with report.export can select Export recall reconciliation (CSV, audited).

Trace a supplier lot forward

1. Search the internal lot or supplier batch, or select Trace on a material lot page.
2. The forward trace lists every batch that reserved or consumed the lot and the finished-goods lots produced, with on-hand and dispatched quantities.

QC samples and results

Sample request → collection → testing → submission → review → approval (e-signature). Samples are requested automatically by goods receipts (raw/packaging material) and batch completion (finished product). Each result is evaluated against a snapshot of the specification; results outside specification are flagged as OOS candidates and an OOS investigation opens automatically.

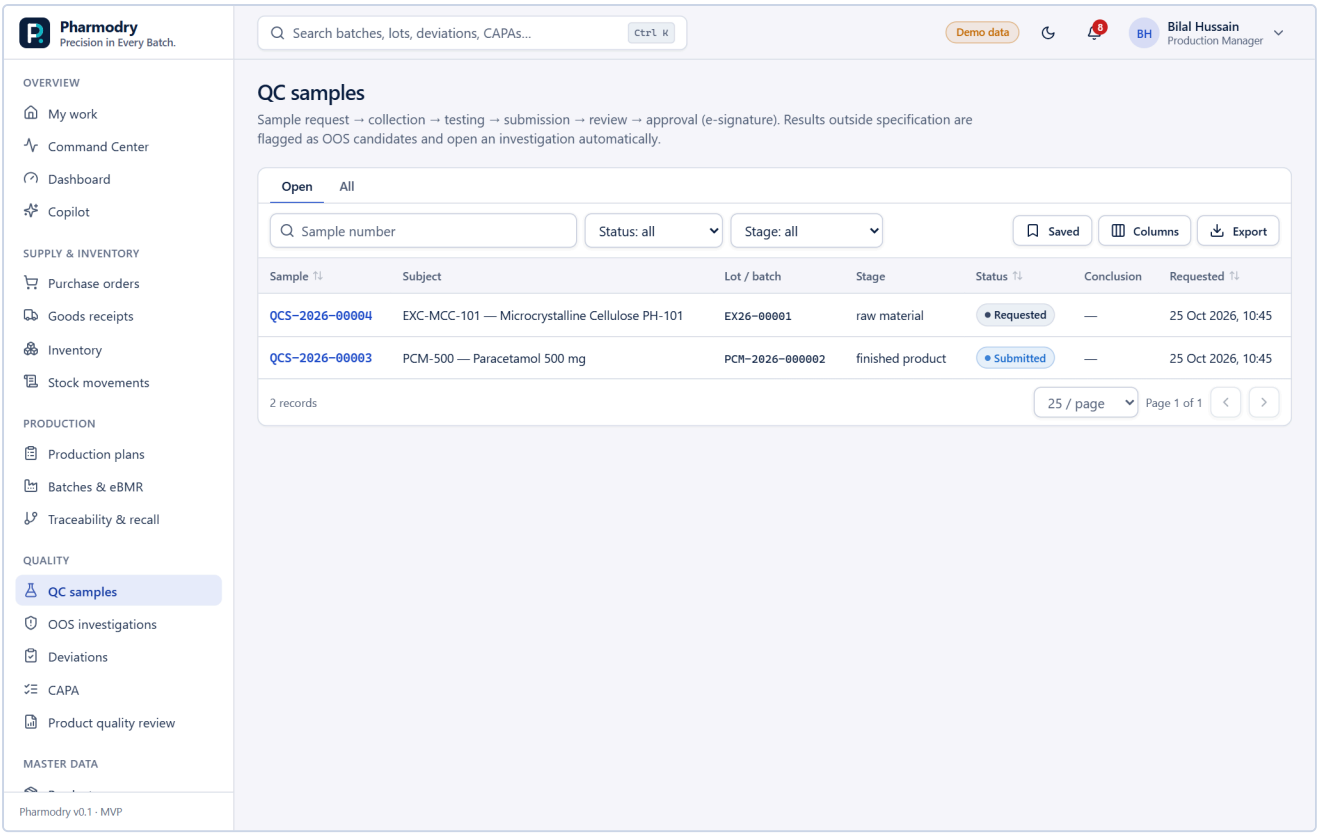


Figure 38 — QC samples. QC sample list by status and stage.

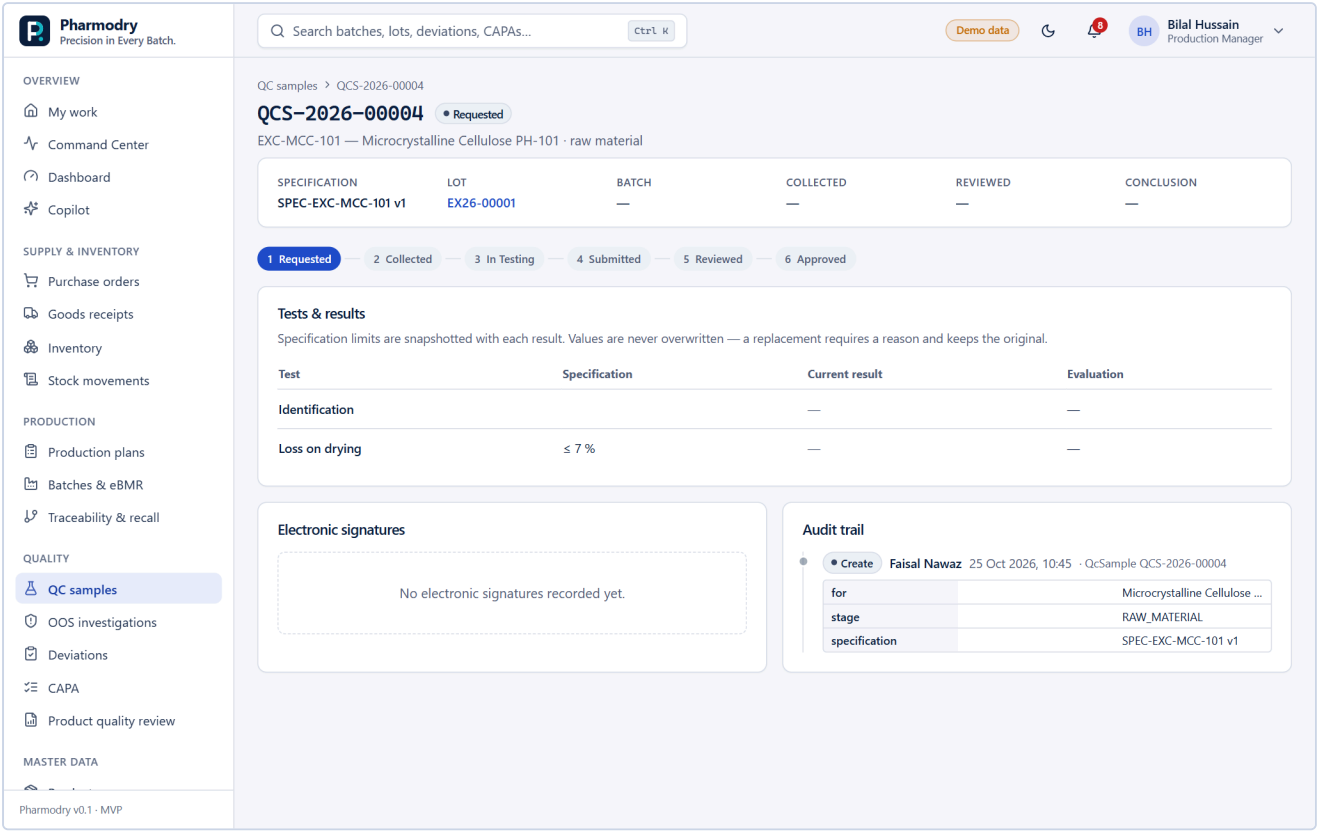


Figure 39 — QC sample detail. A QC sample awaiting collection, with specification tests.

> **Not available to your role:** Record sample collection — requires qc.sample.collect; Enter results — requires qc.result.enter; Review results — requires qc.result.review; Approve a sample — requires qc.result.approve; Replace a result — requires qc.result.enter.

OOS investigations

Out-of-specification results are reported automatically when a QC result falls outside its specification. The investigation follows fixed phases: Reported → Preliminary investigation → Laboratory investigation (Phase I) → Manufacturing investigation (Phase II) → Root cause → Impact assessment → CAPA → QA approval → Closed. Phases cannot be skipped. The conclusion — CONFIRMED OOS or INVALIDATED (lab error) — is a QA decision with electronic signature; invalidation requires an assignable laboratory error.

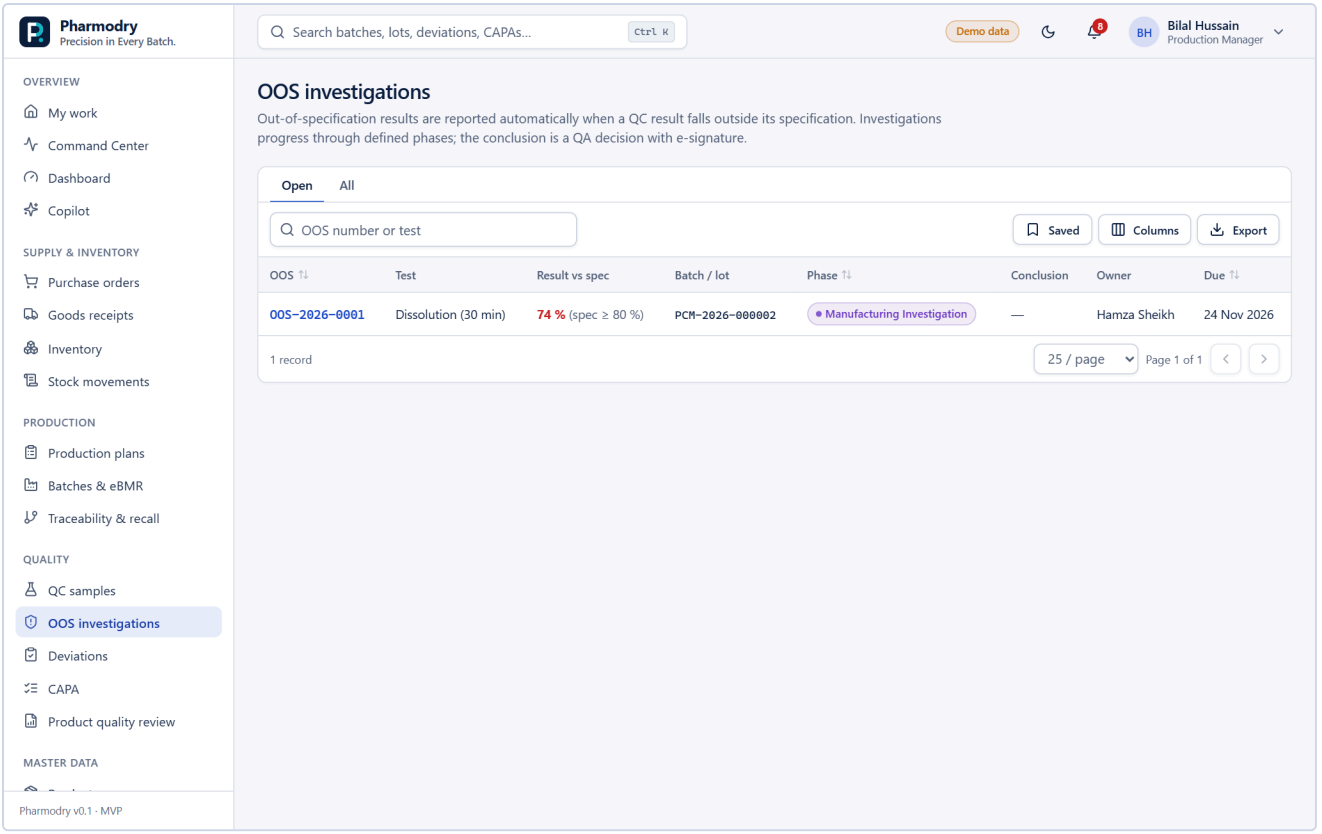


Figure 40 — OOS investigations. Out-of-specification investigations with phase and owner.

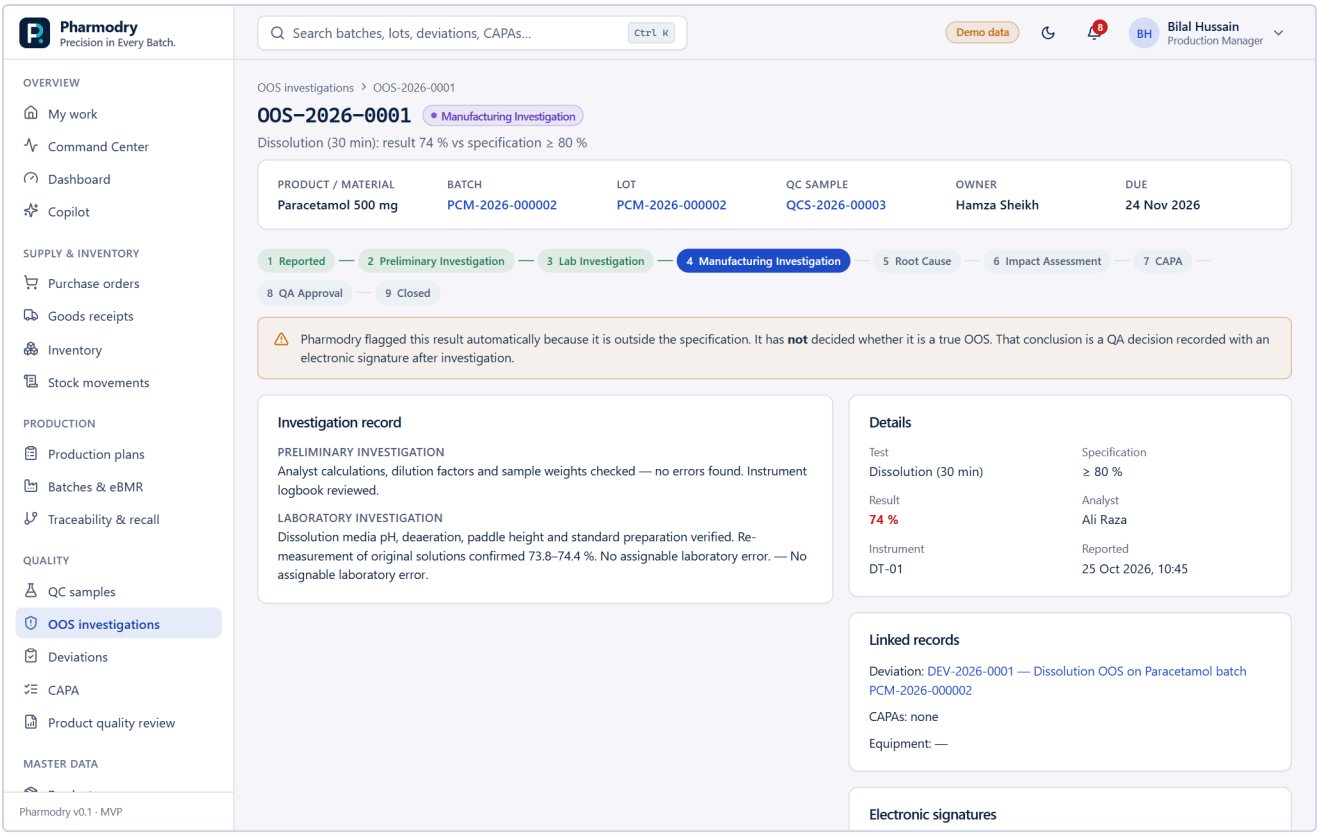


Figure 41 — OOS investigation detail. OOS investigation with phase stepper, linked records and QA conclusion area.

> **Not available to your role:** Record an investigation phase — requires oos.investigate; Conclude the investigation (QA) — requires oos.approve.

Deviations

Draft → Submitted → QA review → Investigation → Root cause → CAPA → Effectiveness check → QA closure. Aging: 0–15 days normal, 16–30 attention, more than 30 overdue (configurable). Root-cause analysis uses structured 5-Whys and a fishbone (6M) category; each save is versioned and QA approves the final root cause.

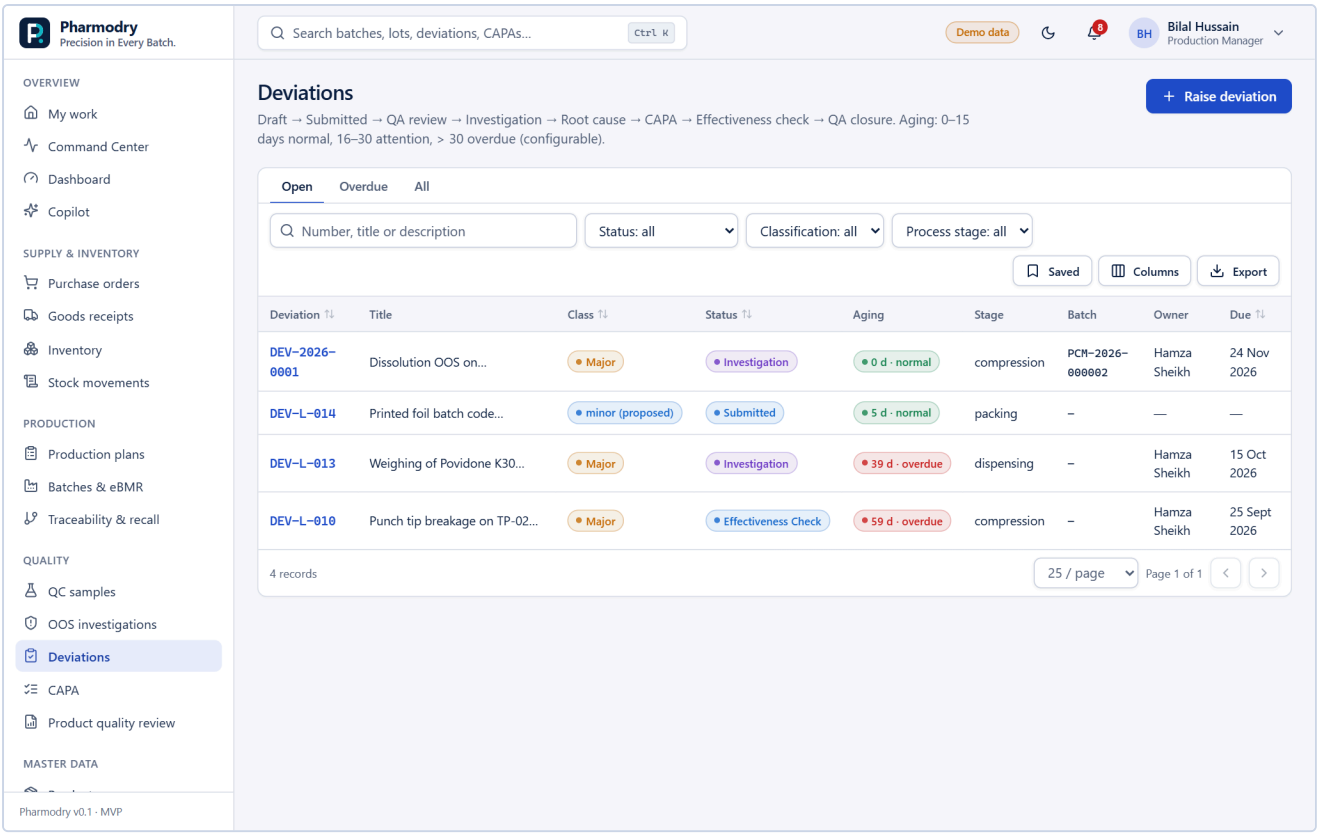


Figure 42 — Deviations. Deviation register with classification, aging and owner.

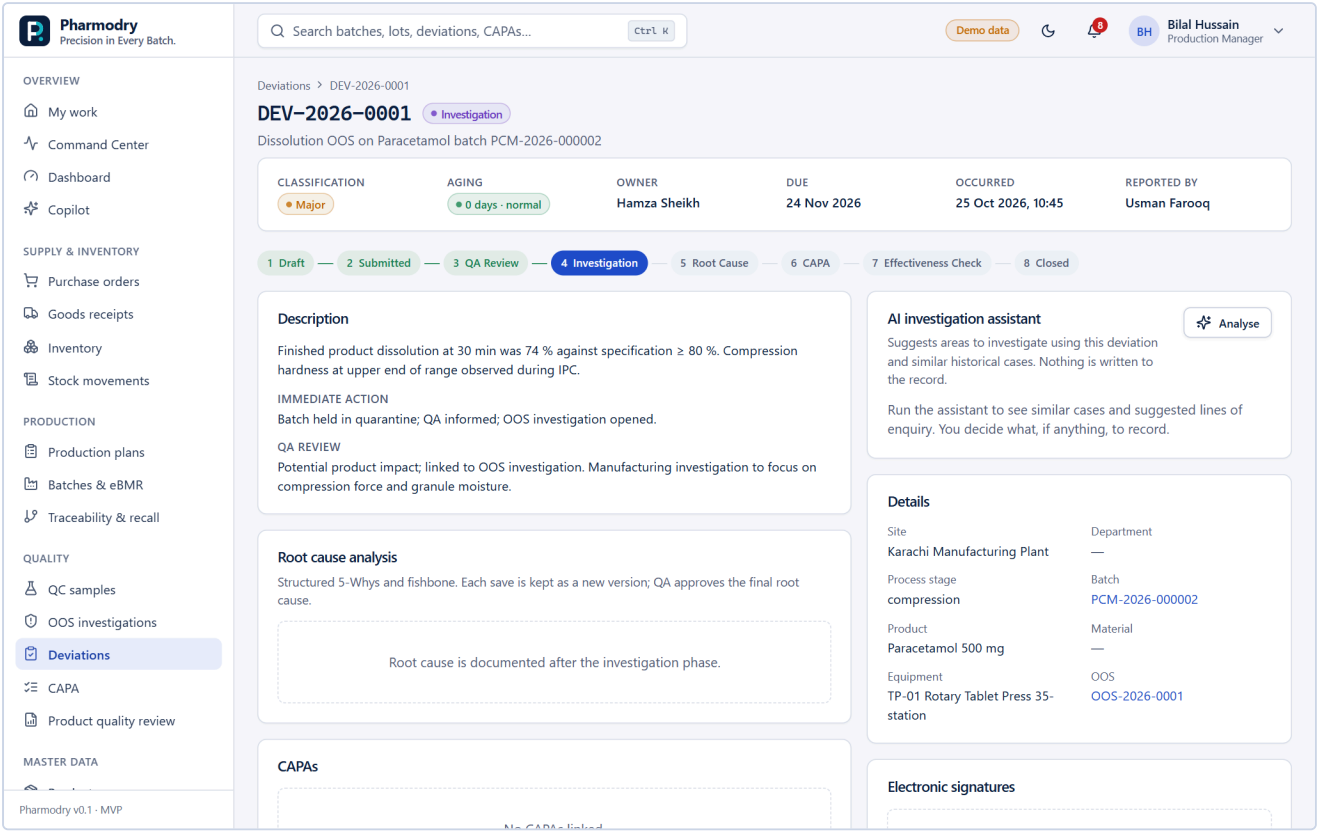


Figure 43 — Deviation detail. Deviation record with workflow actions, RCA, CAPAs and evidence.

Raise a deviation

1. Open Deviations → New (or Raise deviation on a batch, OOS or lot).

2. Under What happened enter Title, Description, Immediate action taken, Occurred at and optionally a Proposed severity.
3. Under Where & what is affected choose Site, Process stage and any Batch, Product, Material or Equipment.
4. Save as draft (then Submit to QA from the deviation page) or submit directly. Record what happened as soon as possible — QA classifies severity during review.

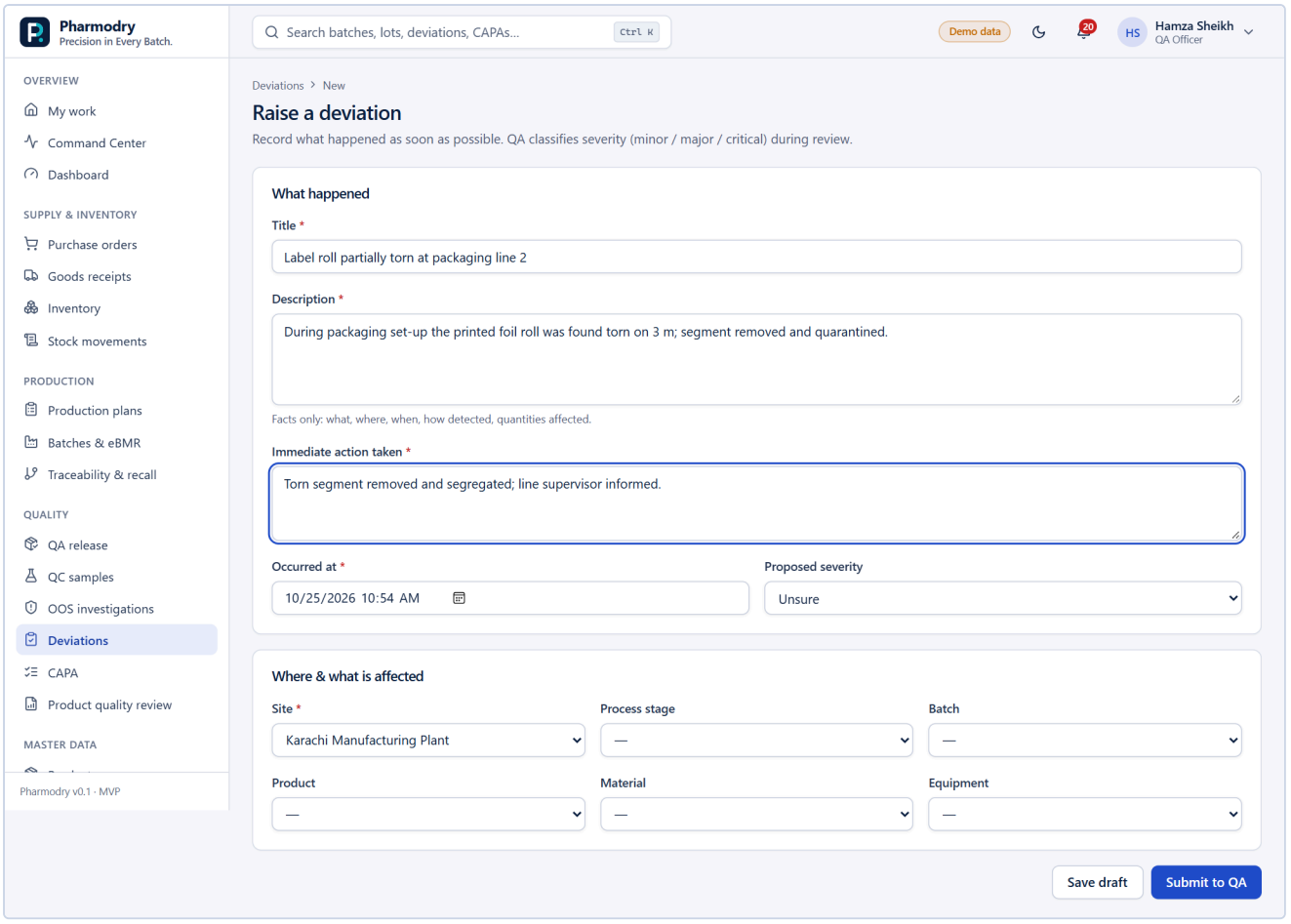


Figure 44 — Raise a deviation. Deviation form: what happened, immediate action and affected records. (screen captured while signed in as QA Officer)

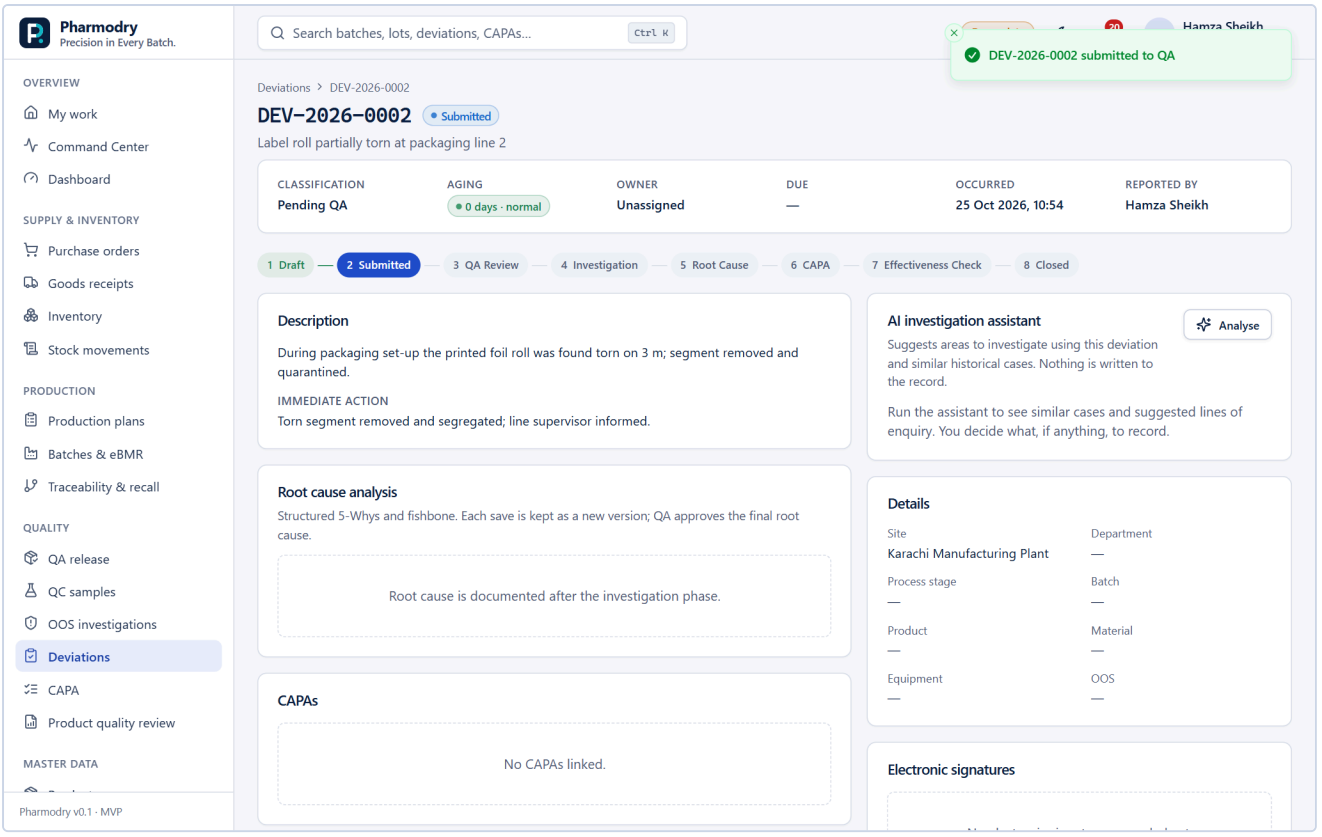


Figure 45 — Deviation submitted. Submitted deviation awaiting QA review. (screen captured while signed in as QA Officer)

Use the AI investigation assistant

1. On the deviation page open AI investigation assistant.
2. Review similar historical cases (computed by Pharmodry) and any suggested areas to investigate.
3. Nothing is written to the record; copy only what you agree with into your own investigation.

> **Not available to your role:** QA review and classification — requires deviation.approve; Investigate and record the root cause — requires deviation.investigate; CAPA decision and closure — requires deviation.approve.

CAPA

Corrective and preventive actions from deviations, OOS, OOT, audits, complaints, supplier issues and recalls. Each CAPA has actions with owners and due dates, effectiveness criteria and an effectiveness check. Closure requires a successful effectiveness check and a QA electronic signature; a failed check returns the CAPA to In Progress and is counted in the effectiveness-failure KPI.

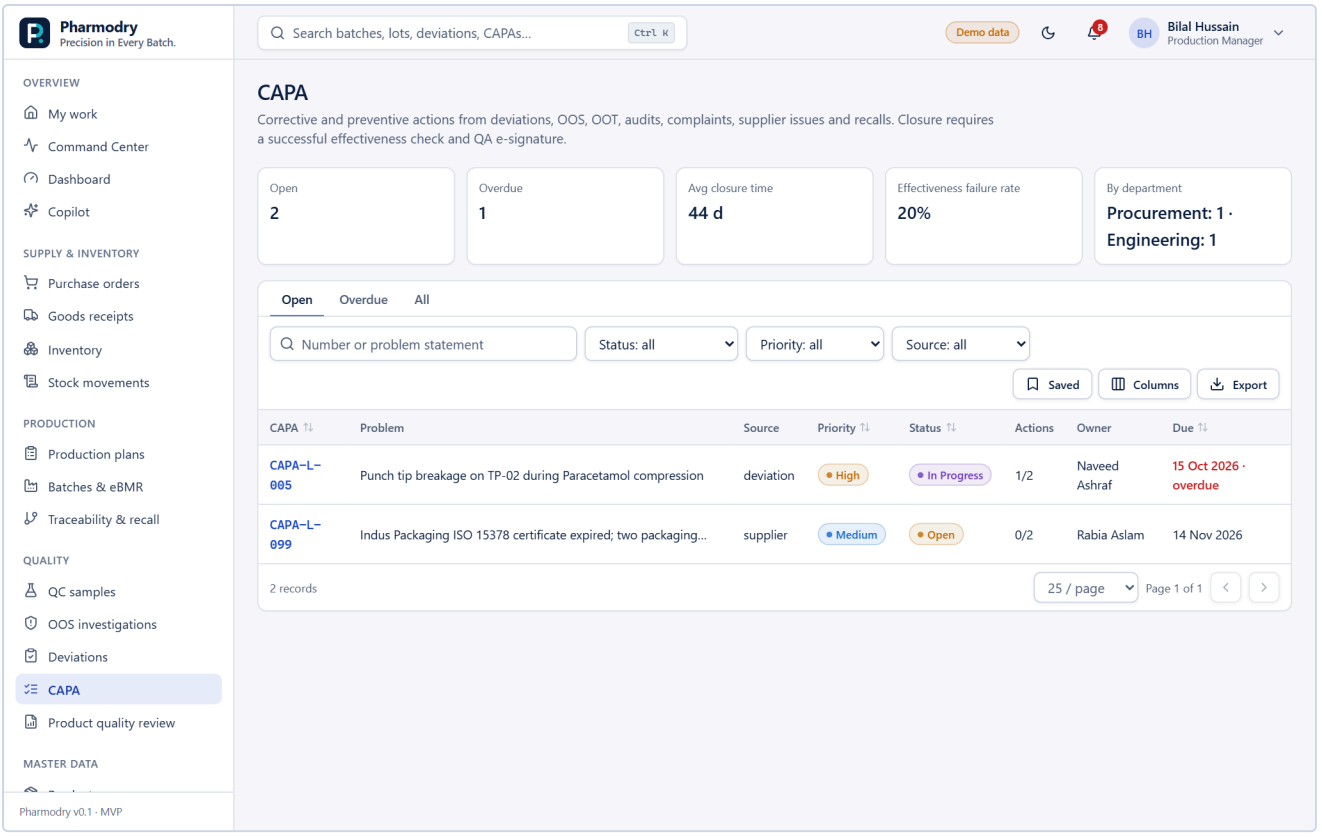


Figure 46 — CAPA. CAPA register with priority, owner and due date.

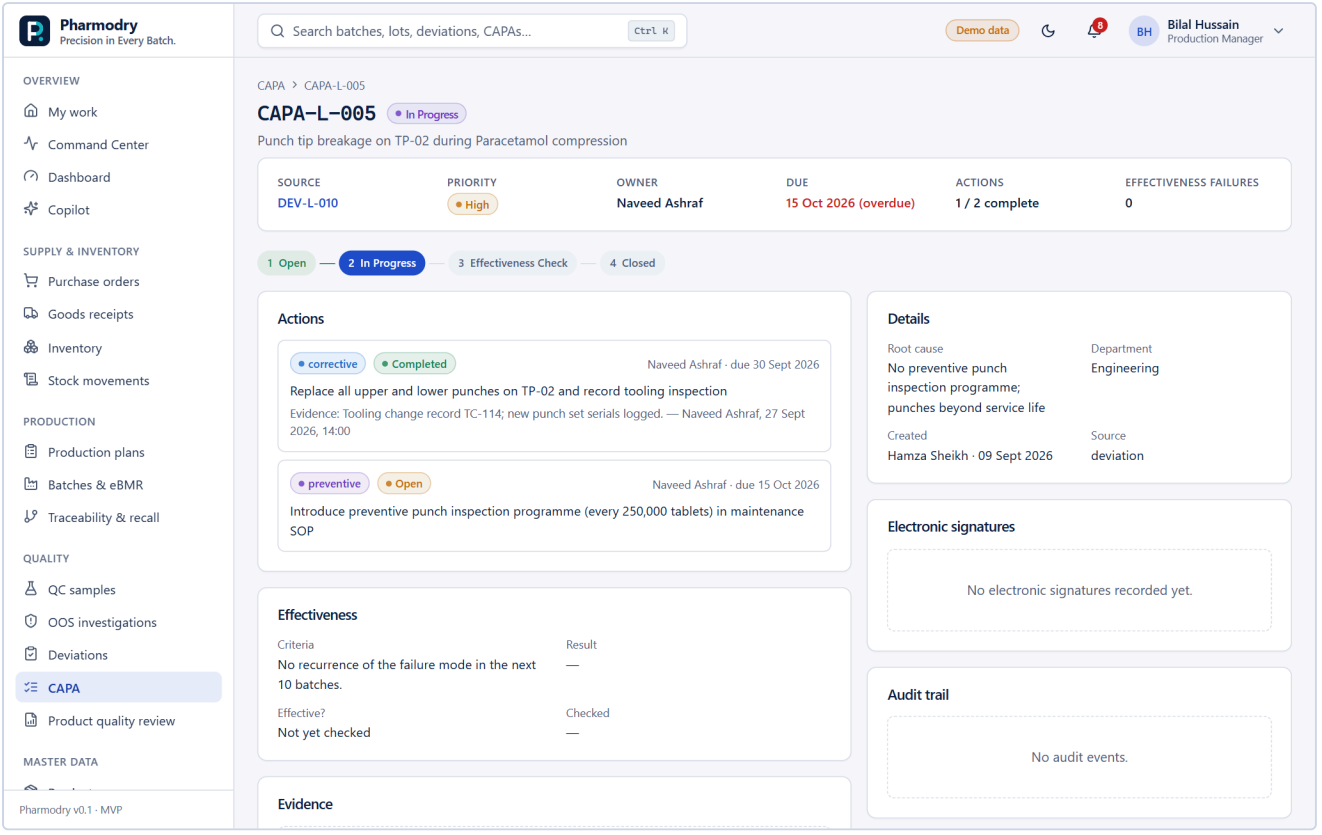


Figure 47 — CAPA detail. CAPA with actions, effectiveness check and closure.

> **Not available to your role:** Create a CAPA — requires capa.create; Complete actions — requires capa.manage; Effectiveness check — requires capa.verify; Close a CAPA — requires capa.close.

Product Quality Review (PQR) data pack

Figures computed for the PIC/S GMP Chapter 1 product quality review: batches manufactured with yield statistics, finished-product QC results (approved samples), deviations, OOS investigations, CAPAs and starting materials with suppliers. QA authors the evaluation and conclusions. Sections not captured in this release (for example stability, complaints, returns, change control) are listed so they can be added from source systems.

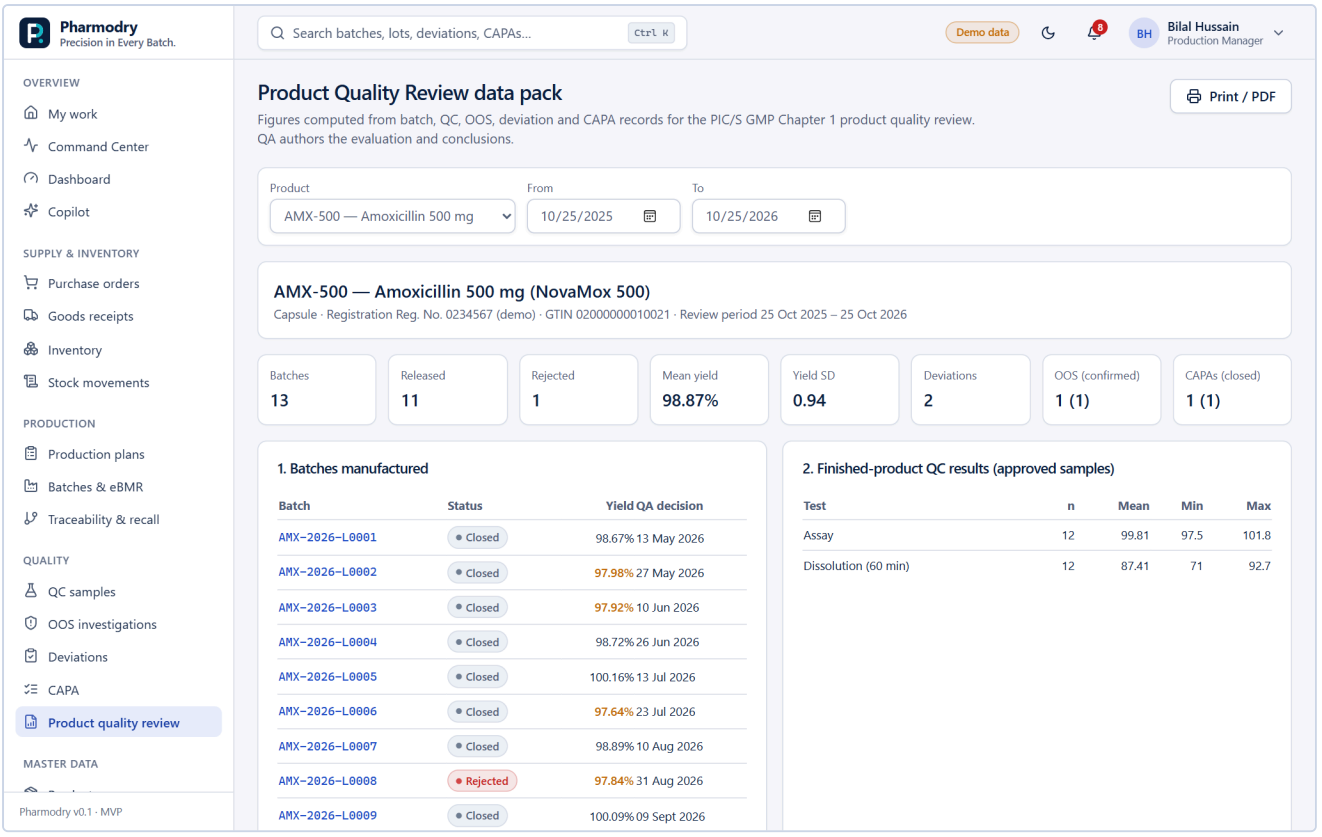


Figure 48 — Product Quality Review. Product Quality Review data pack for a product and period.

Prepare a PQR data pack

1. Open Product quality review.
2. Choose Product, From and To dates.
3. Review sections 1–6 and the “Sections not captured in this release” notice.
4. Select Print / PDF to save the pack.

Master data: products, materials, suppliers, equipment and specifications

Regulated master data is versioned: every change requires a reason and creates a new immutable version (Version history). Products carry registration, shelf life, storage, GTIN (GS1, check digit verified), BOM and manufacturing template; materials carry type, retest period, safety stock and lead time; suppliers carry qualification status, approved materials and certifications; equipment carries calibration frequency and history; specifications hold versioned acceptance criteria.

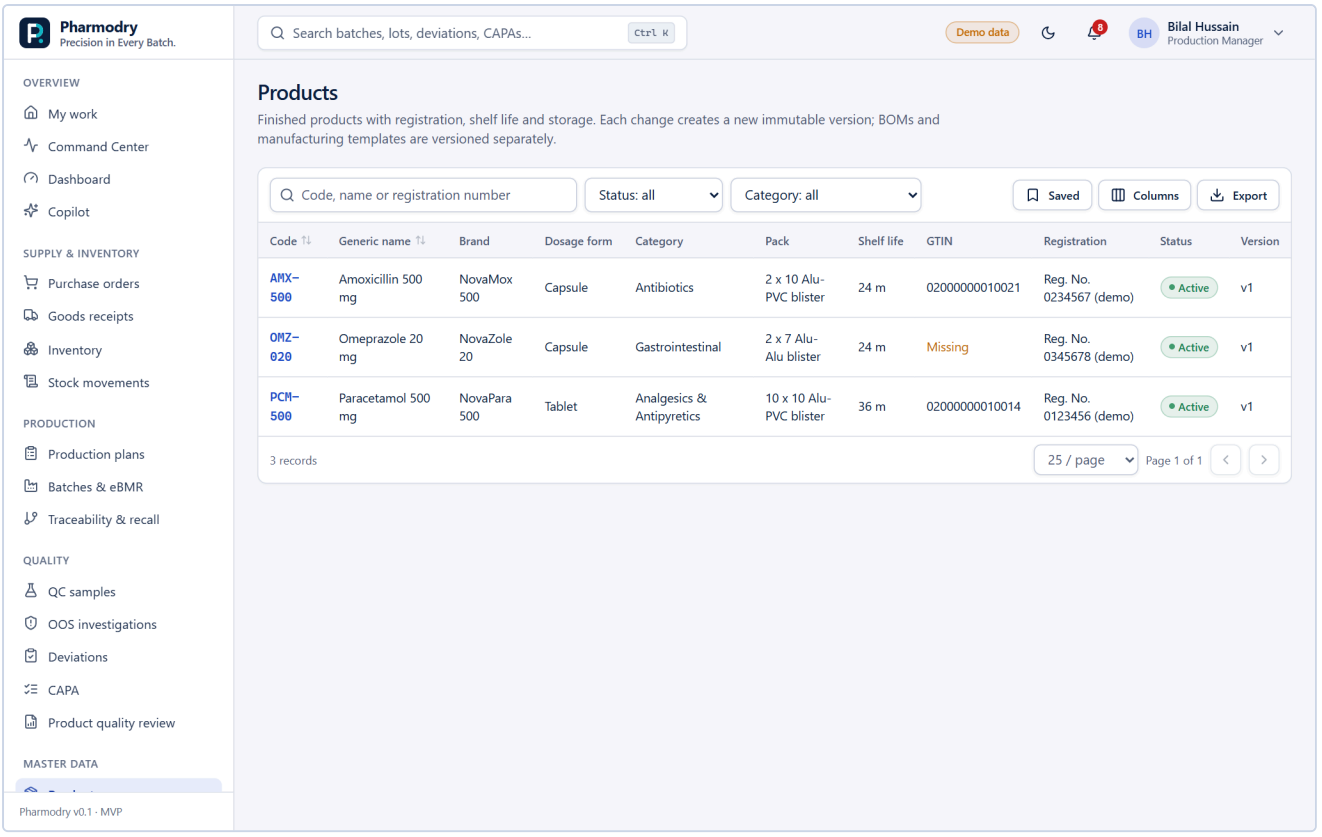


Figure 49 — Products. Product master list with GTIN and registration.

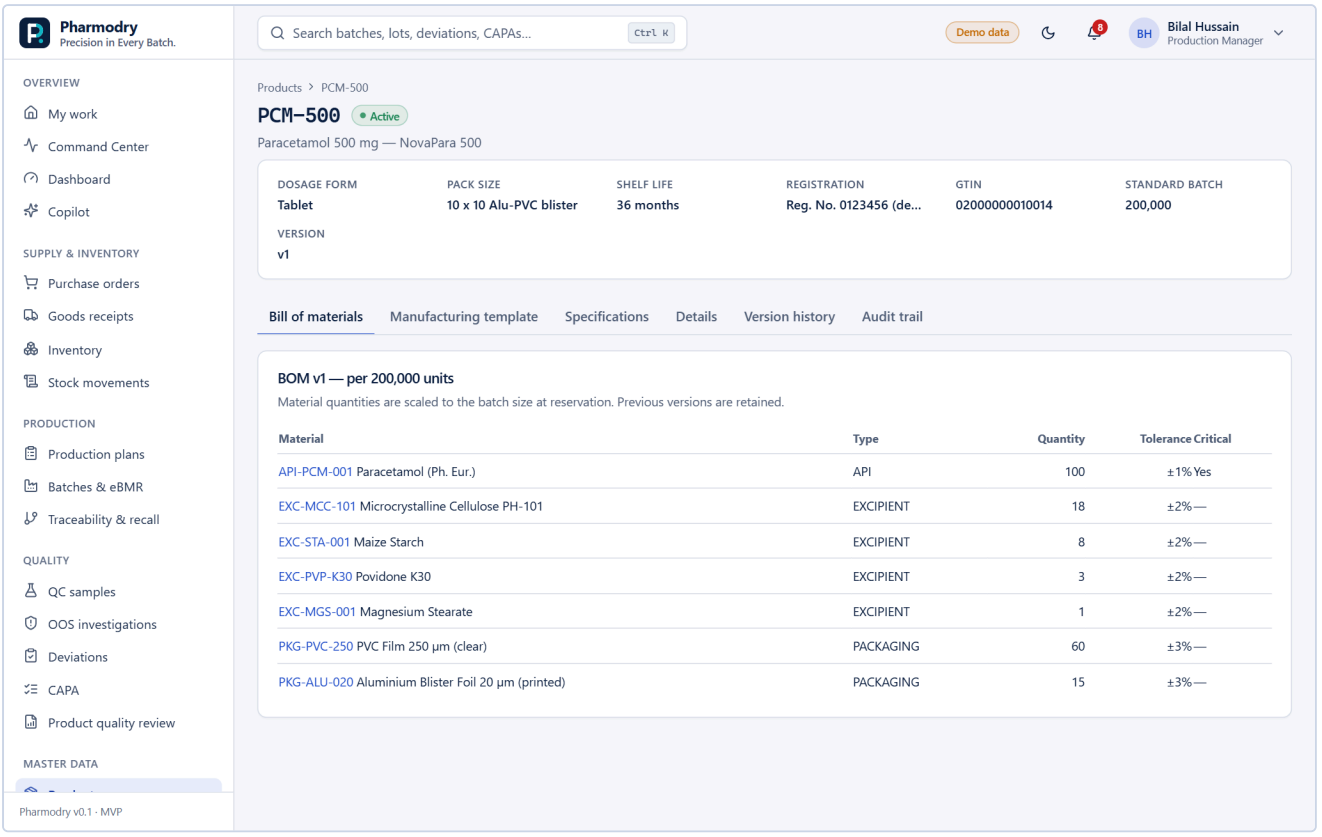


Figure 50 — Product detail. Product detail: BOM, manufacturing template, specifications and versions.

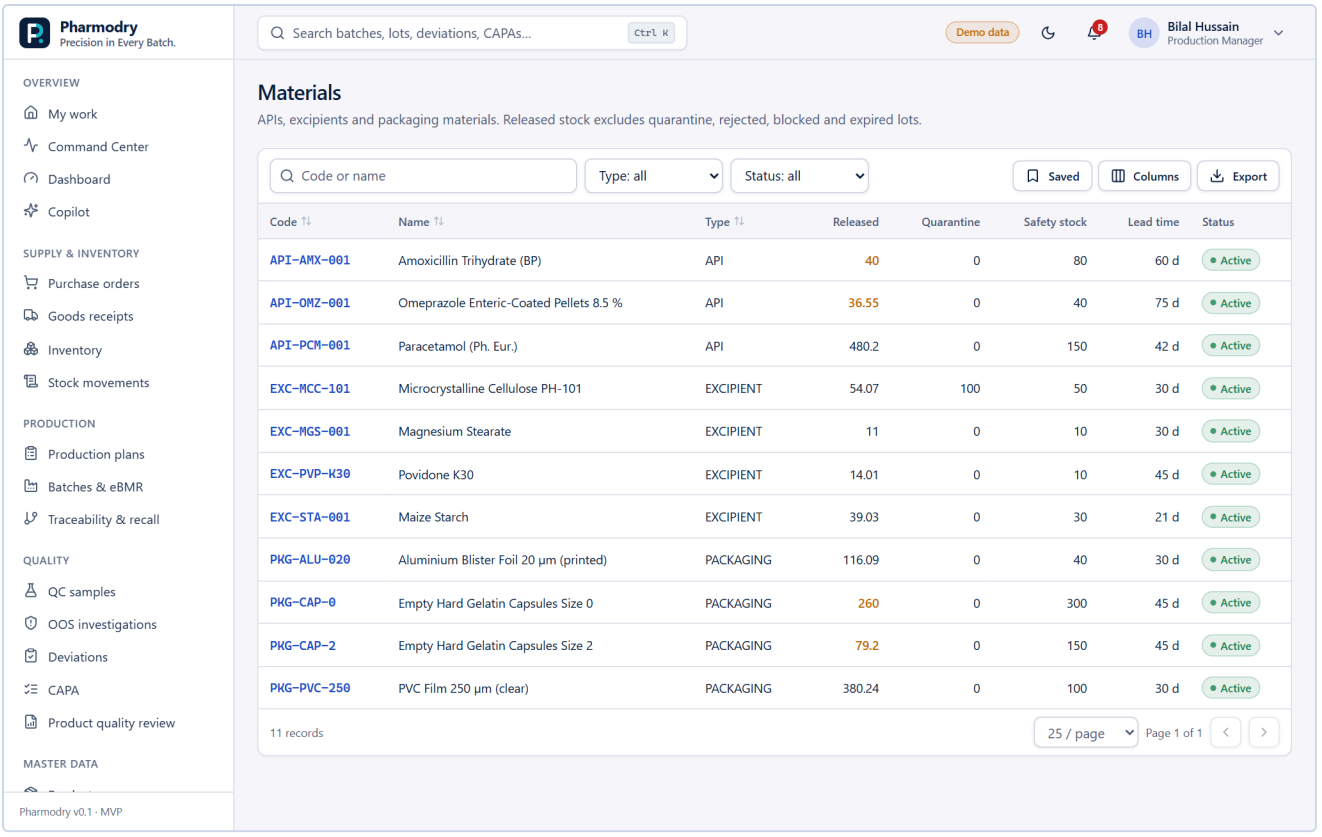


Figure 51 — Materials. Material master with released and quarantined stock.

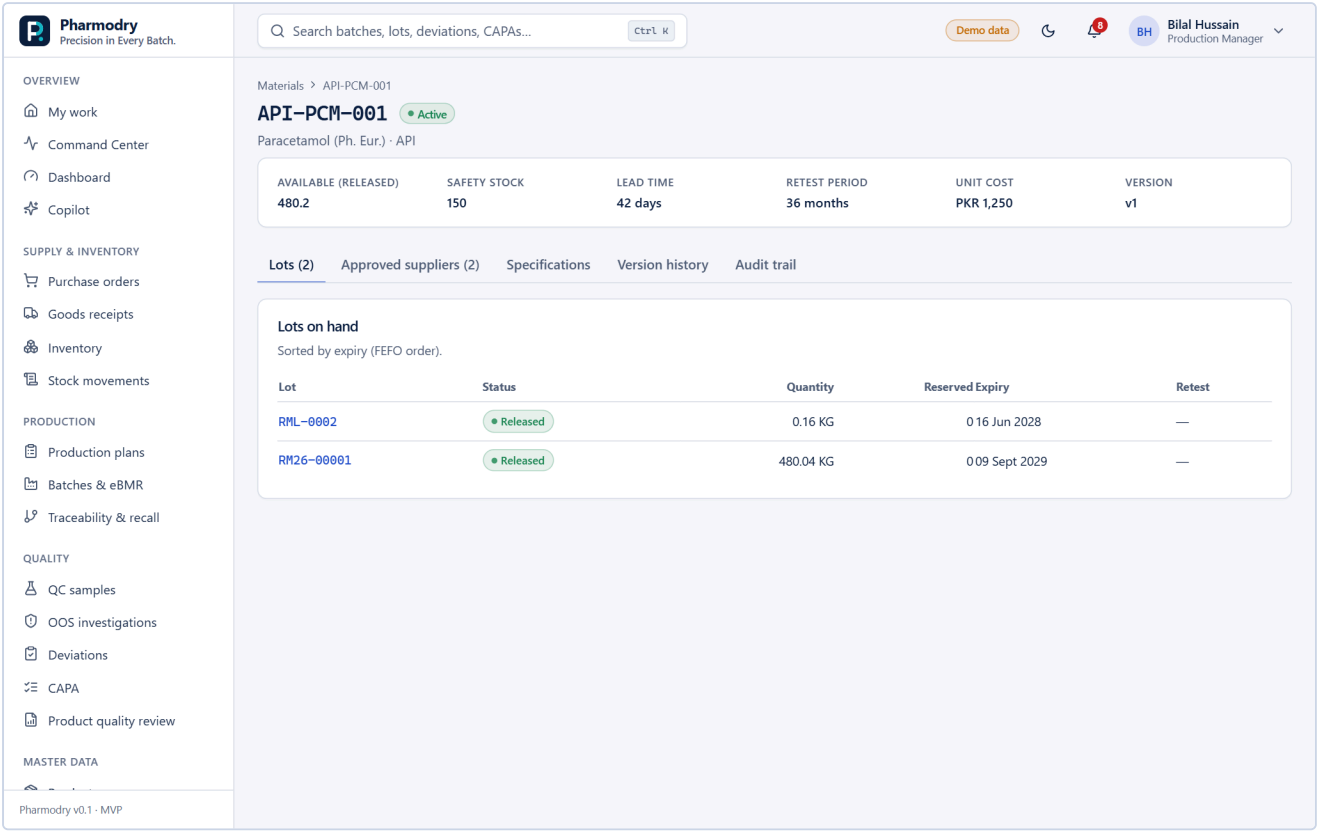


Figure 52 — Material detail. Material detail: lots in FEFO order, approved suppliers, specifications.

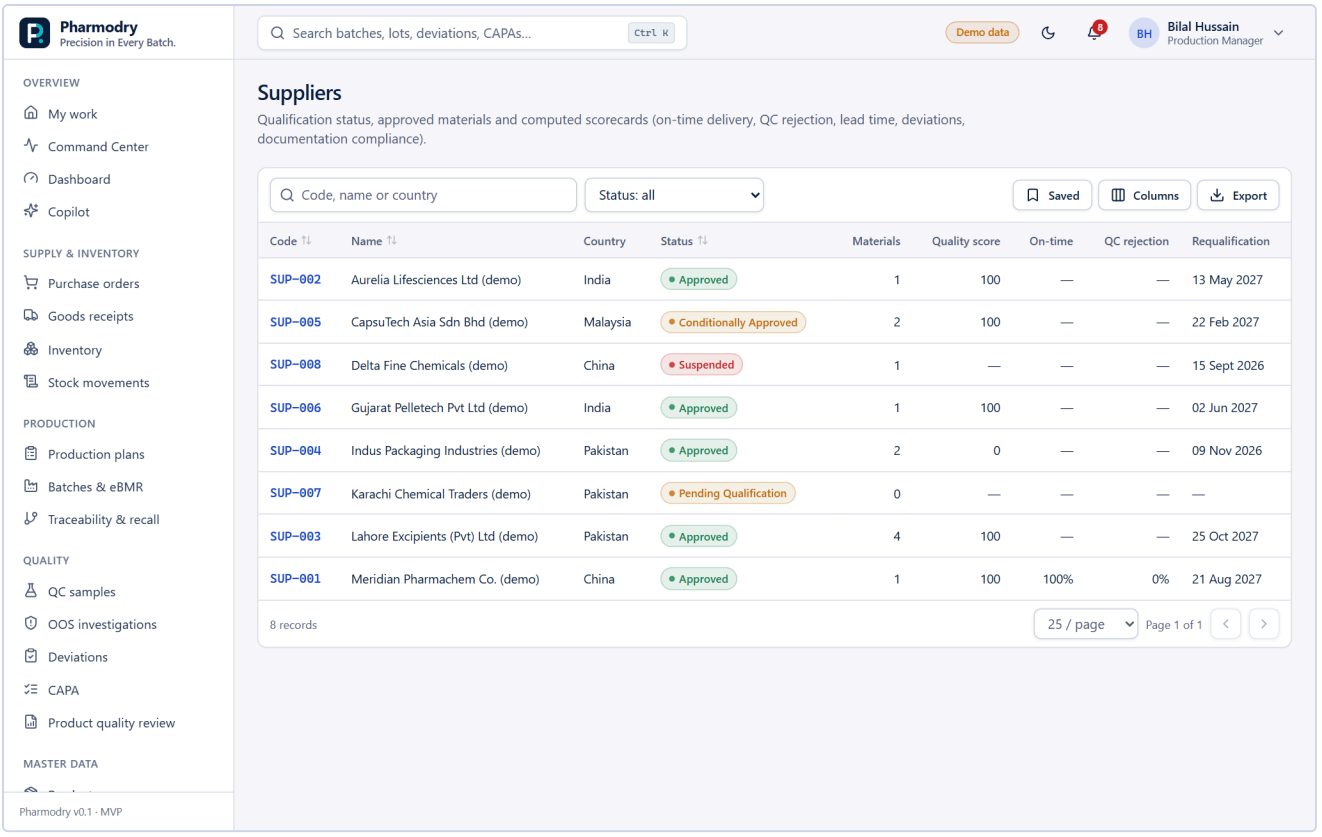


Figure 53 — Suppliers. Supplier list with qualification status and scorecards.

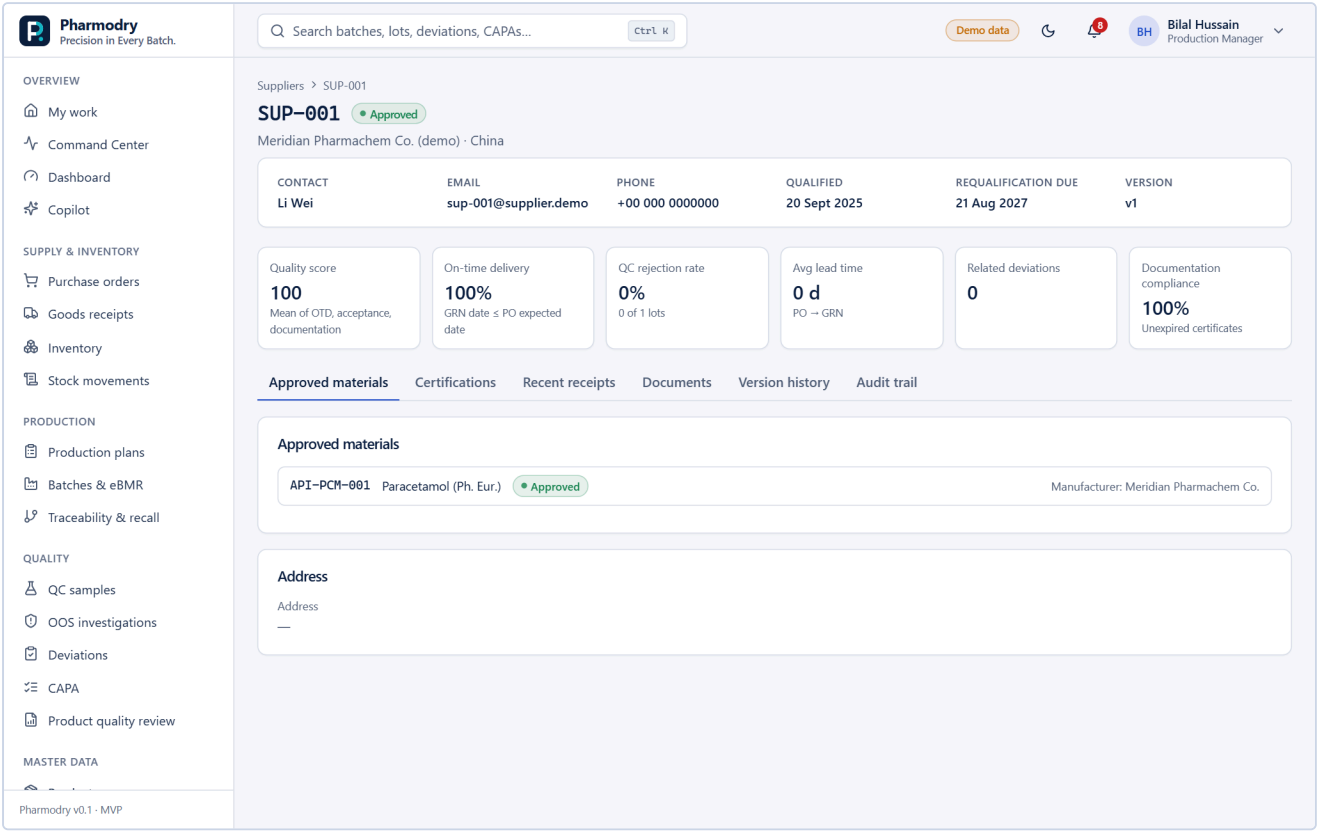


Figure 54 — Supplier detail. Supplier qualification, approved materials, certifications and receipts.

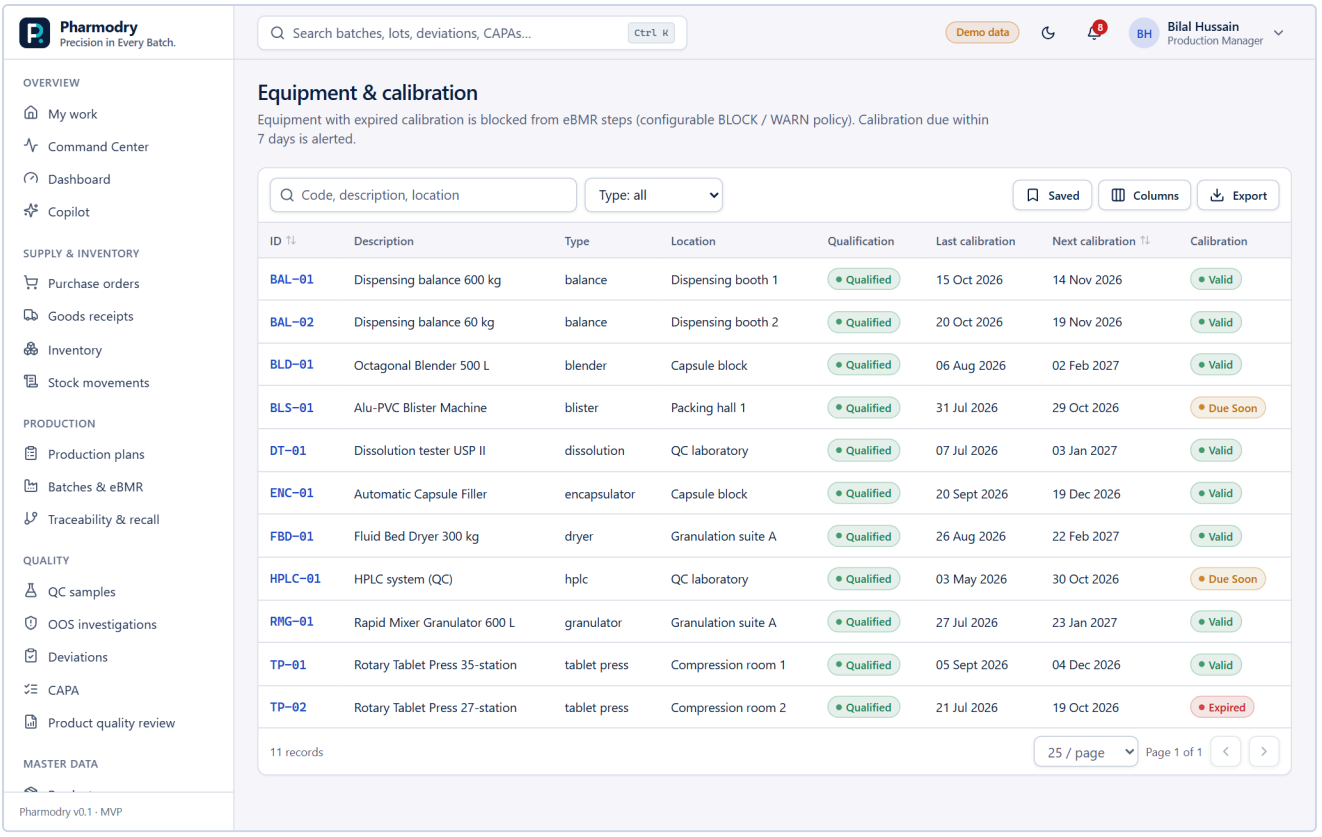


Figure 55 — Equipment & calibration. Equipment list with calibration status.

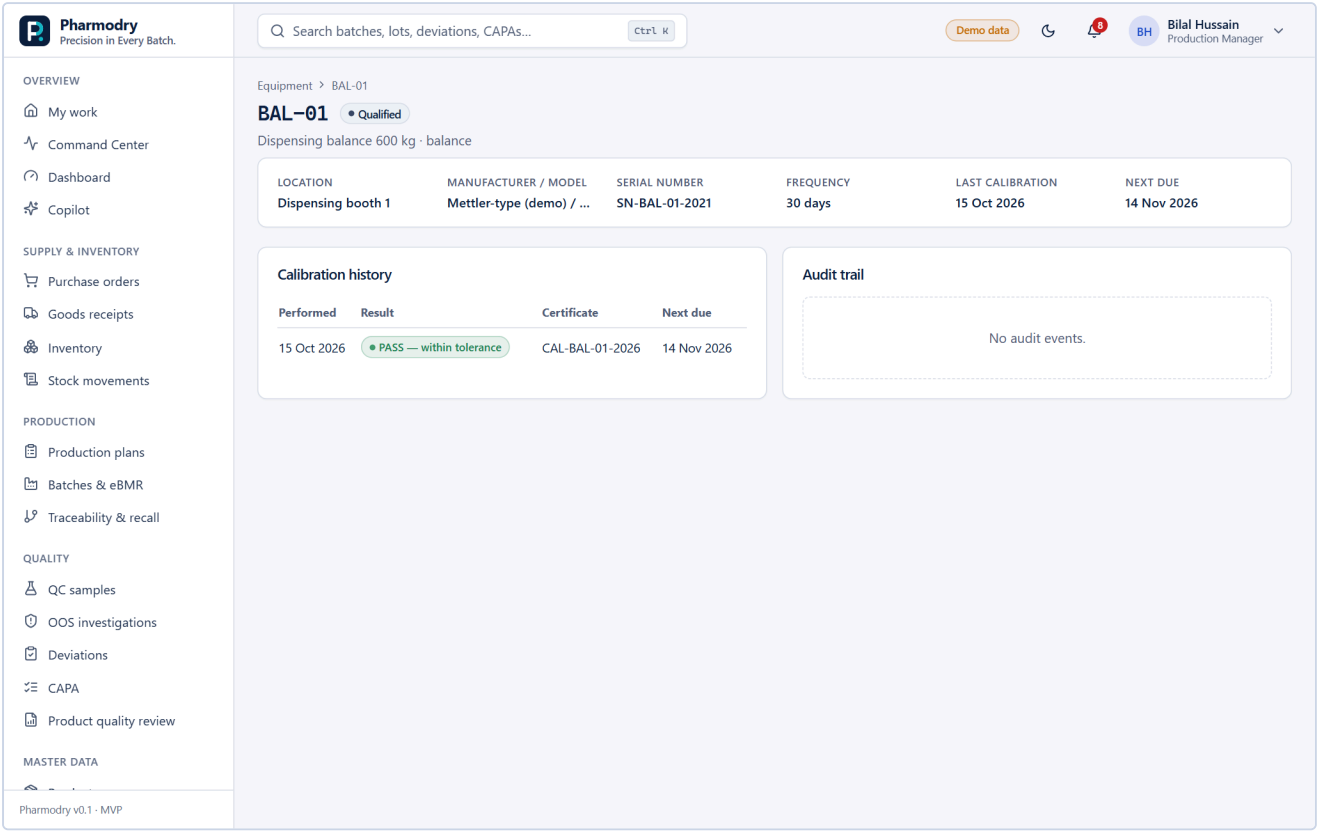


Figure 56 — Equipment detail. Equipment detail with calibration history.

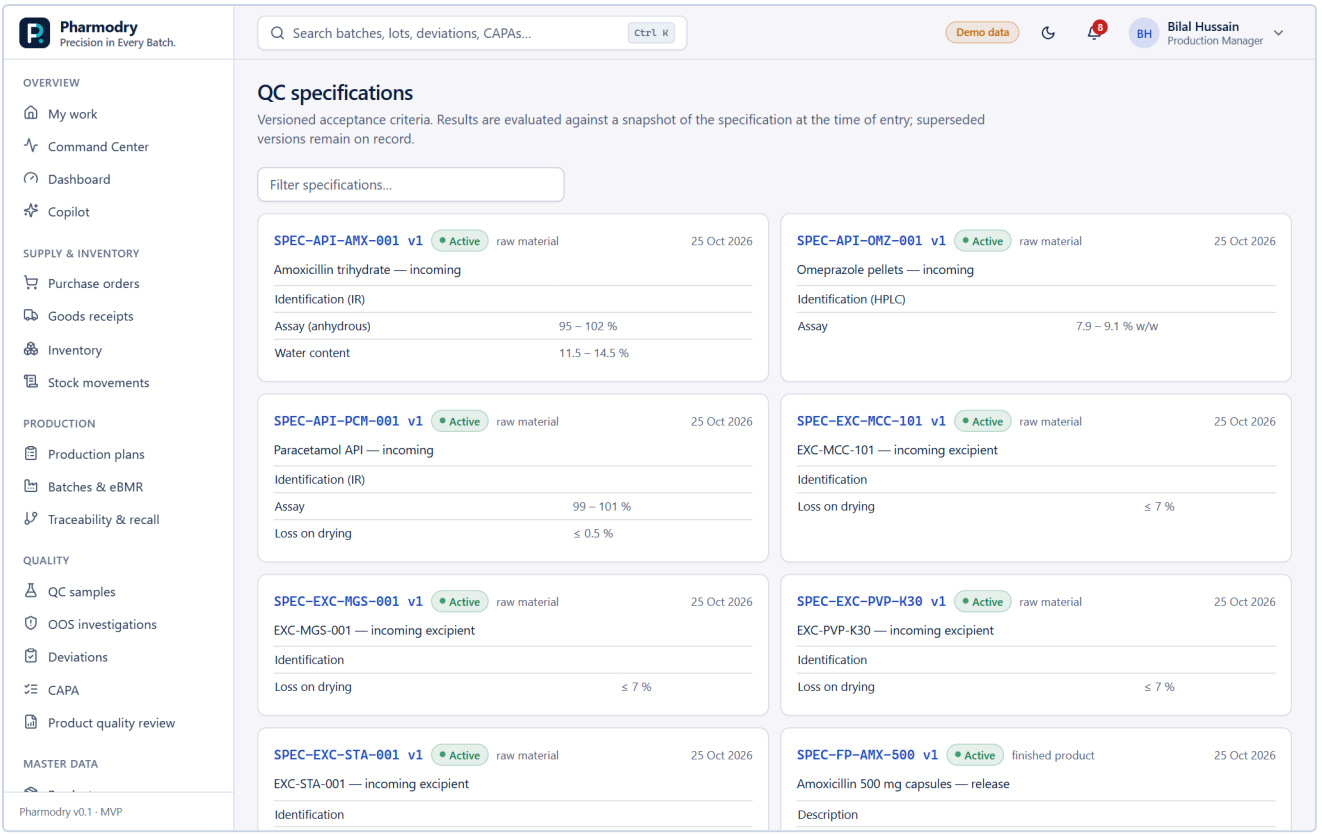


Figure 57 — QC specifications. Versioned QC specifications.

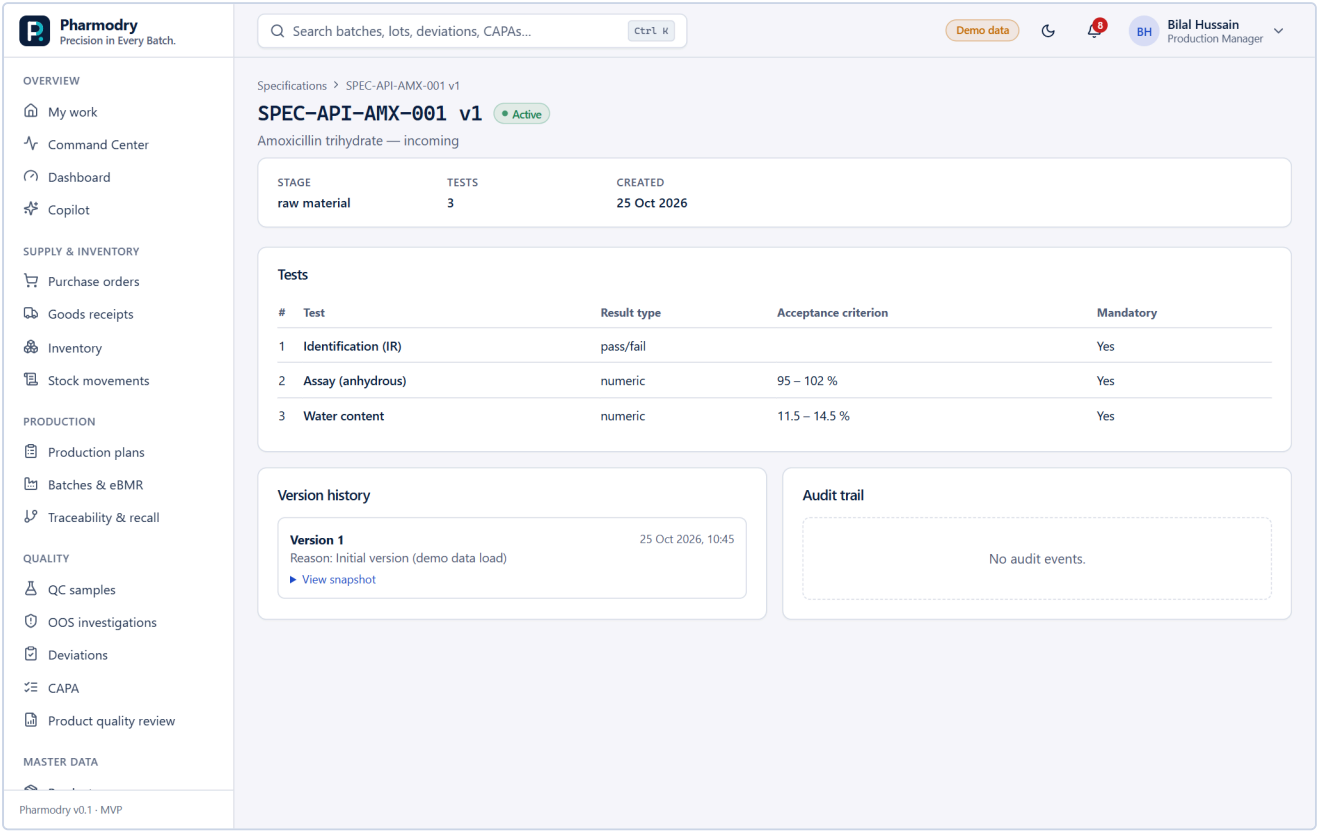


Figure 58 — Specification detail. Specification tests and acceptance criteria.

> **Not available to your role:** Create or edit a product — requires product.manage; Create or edit a material — requires material.manage; Create a supplier — requires supplier.manage; Change supplier qualification / approve a material — requires supplier.approve or supplier.manage; Record equipment calibration — requires equipment.manage.

Audit trail

Centralised, append-only record of who did what, when, with previous and new values, reasons and e-signatures. Entries cannot be edited or deleted from the application, and the database rejects any attempt to modify them. Each record page also has its own Audit trail tab.

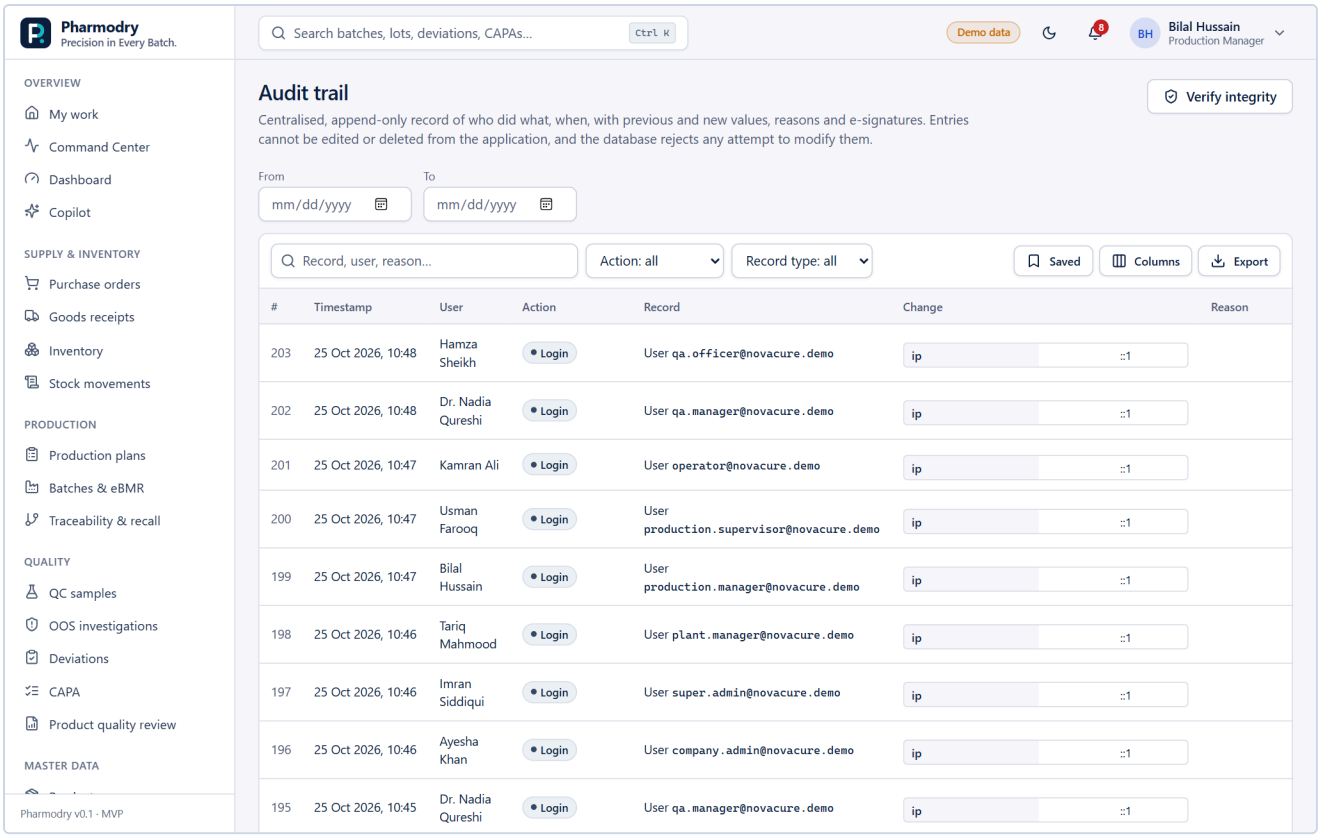


Figure 59 — Audit trail. Central audit trail with hash-chain verification.

Review the audit trail

1. Open Audit trail; filter by Action, Record type, user or date.
2. Expand an entry to see previous and new values and the reason.
3. Select Verify integrity to recompute the hash chain: “Hash chain verified” confirms integrity; “Integrity failure” must be escalated immediately.

> **Not available to your role:** Export the audit trail — requires audit.export.

Reports and exports

CSV exports open in Excel (UTF-8, formula-injection safe): Batch summary, Inventory status, Near-expiry inventory, Expired stock, Audit trail; links to the Deviation, CAPA, OOS, QC and Supplier performance registers (use Export on each list); and the print-ready Batch manufacturing record. Exports are recorded in the audit trail.

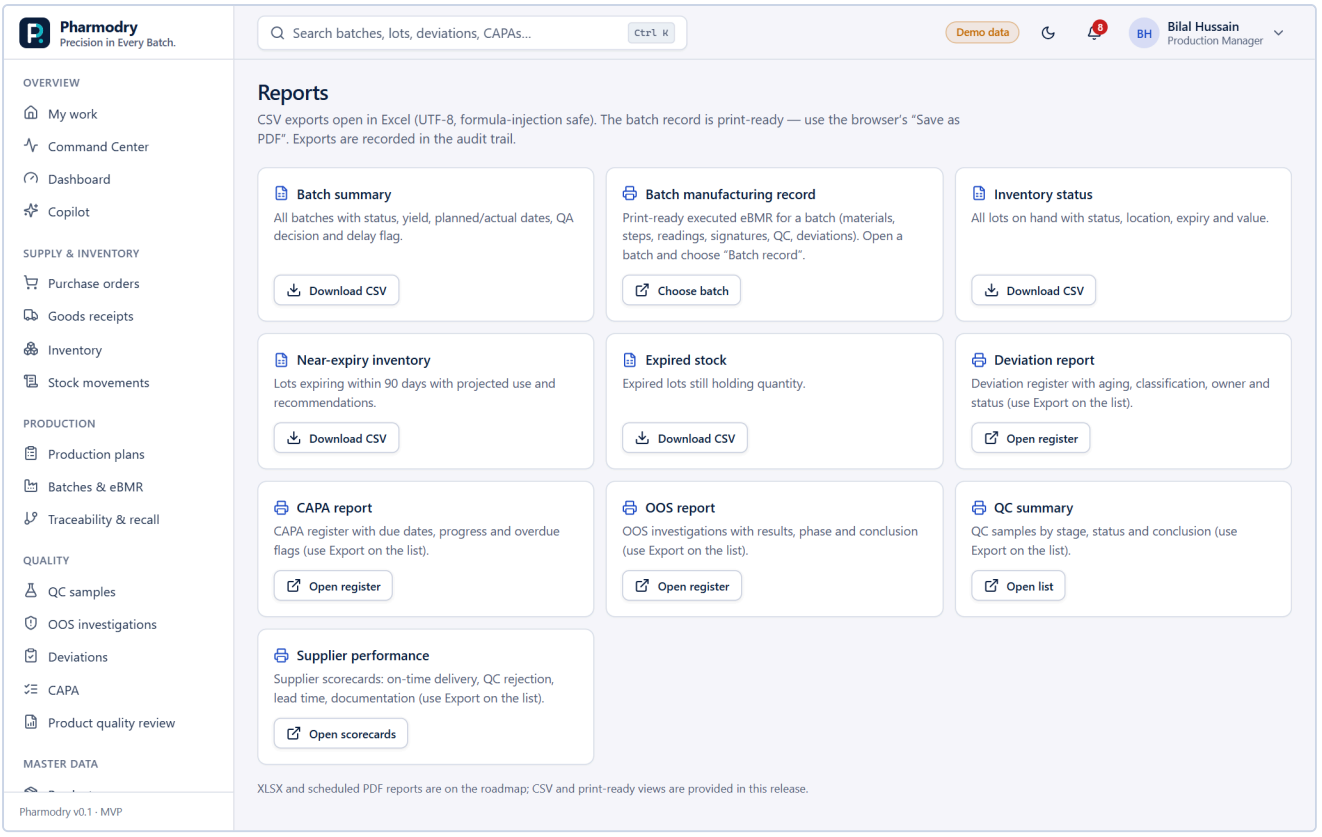


Figure 60 — Reports. Report and export catalogue.

7 Field-by-field explanations, required inputs and validation rules

General rules: required fields are marked; dates use the date picker; numbers must be positive unless stated; reasons must be at least 5 characters; unknown or malformed inputs are rejected by the server. Forms that you cannot submit are still shown for reference where you can view the module.

Inventory, lots and stock movements

Field	Required	Rules and meaning
Destination warehouse / Bin (transfer)	Yes	Target location.
Quantity (transfer)	Yes	Cannot exceed available (unreserved) quantity.
Quantity change (signed) (adjustment)	Yes	Negative reduces stock; the result can never be negative.
Reason	Yes	Recorded in the audit trail (minimum 5 characters).

Batches and electronic batch records (eBMR)

Field	Required	Rules and meaning
Product	Yes	Active product with an active BOM and manufacturing template.
Production plan line	No	Plan must be APPROVED.
Batch size	Yes	Material quantities are scaled from the BOM.
Planned start	Yes	Date/time.
Equipment (step start)	If required	Qualified and calibrated equipment only.
Process parameters / IPC readings	Yes	Checked against limits; out-of-limit flagged.
Yield justification	If out of tolerance	At least 10 characters, or link a deviation.

Deviations

Field	Required	Rules and meaning
Title	Yes	Short summary.
Description	Yes	At least 10 characters.
Immediate action taken	Recommended	Containment actions.

Field	Required	Rules and meaning
Occurred at	Yes	Date and time (local).
Proposed severity	No	QA decides the classification.
Site / Process stage	Yes	Where it happened.
Batch / Product / Material / Equipment	No	Links affected records.

8 Buttons and their functions

Module	Button	What it does	Available to you
Purchase orders	New purchase order	Opens the order form.	No (needs procurement.po.create)
Purchase orders	Add line / Remove line	Adds or removes order lines.	No (needs procurement.po.create)
Purchase orders	Create purchase order	Saves the order as DRAFT.	No (needs procurement.po.create)
Purchase orders	Approve	Approves a draft (not your own).	No (needs procurement.po.approve)
Purchase orders	Cancel order	Cancels with a reason.	No (needs procurement.po.create)
Goods receipts (GRN)	New	Opens the receipt form.	No (needs inventory.receive)
Goods receipts (GRN)	Load lines	Copies lines from the selected PO.	No (needs inventory.receive)
Goods receipts (GRN)	Add line / Remove line	Manages receipt lines.	No (needs inventory.receive)
Goods receipts (GRN)	Create GRN	Posts the receipt; lots go to QUARANTINE.	No (needs inventory.receive)
Inventory, lots and stock movements	QA release / Reject	QA disposition with electronic signature.	No (needs inventory.release)
Inventory, lots and stock movements	Transfer	Moves stock between locations.	No (needs inventory.transfer)
Inventory, lots and stock movements	Adjust	Posts a signed quantity change with reason.	No (needs inventory.adjust)
Inventory, lots and stock movements	Block / Unblock	Temporarily prevents use of a released lot.	No (needs inventory.release)
Inventory, lots and stock movements	Trace	Opens forward traceability for a material lot.	Yes
Batches and electronic batch records (eBMR)	Reserve materials	FEFO reservation of released lots.	Yes
Batches and electronic batch records (eBMR)	Release reservation	Returns reservations (reason required).	Yes
Batches and electronic batch records (eBMR)	Issue materials	Dispenses reserved lots to production.	No (needs inventory.issue)
Batches and electronic batch records (eBMR)	Start batch	Moves READY → IN PROGRESS.	Yes
Batches and electronic batch records (eBMR)	Start / Record & sign	Executes and signs eBMR steps.	Yes
Batches and electronic batch records (eBMR)	Verify	Second-person verification with signature.	Yes
Batches and electronic batch records (eBMR)	Correct	Corrects a recorded value with reason.	Yes

Module	Button	What it does	Available to you
Batches and electronic batch records (eBMR)	Complete manufacturing	Calculates yield, books FG into quarantine.	Yes
Batches and electronic batch records (eBMR)	QA release / Reject	Final disposition with electronic signature.	No (needs batch.approve)
Batches and electronic batch records (eBMR)	Close batch	Closes a released or rejected batch.	No (needs batch.approve)
Batches and electronic batch records (eBMR)	Cancel	Cancels a batch before it starts (reason required).	Yes
Batches and electronic batch records (eBMR)	Trace	Opens traceability & mock recall for the batch (needs inventory.view too).	Yes
Batches and electronic batch records (eBMR)	Batch record	Printable executed batch record.	Yes
Traceability and mock recall	Trace	Searches batches and material lots (needs batch.view and inventory.view).	Yes
Traceability and mock recall	Export recall reconciliation	CSV export, recorded in the audit trail.	Yes
QC samples and results	Record collection	Records collection and sampled quantity.	No (needs qc.sample.collect)
QC samples and results	Save results	Stores and evaluates results.	No (needs qc.result.enter)
QC samples and results	Submit for review	Sends results for review.	No (needs qc.result.enter)
QC samples and results	Accept review / Return to analyst	Second-person review or return to analyst.	No (needs qc.result.review)
QC samples and results	Approve (e-sign)	Approval with electronic signature.	No (needs qc.result.approve)

Buttons you cannot use are hidden or disabled in the application; the server refuses the action even if attempted another way.

9 Workflow, approvals and electronic signatures

The table shows the main GMP workflow and where your role takes part. Segregation of duties is enforced by the server: the person who performs a step cannot review, verify or approve it.

Workflow step	Permission	Your role
Raise purchase order	procurement.po.create	—
Approve purchase order (different person)	procurement.po.approve	—
Receive goods (GRN) → lots in QUARANTINE, QC sample requested	inventory.receive	—
Collect sample and enter results	qc.result.enter	—
Review results (different person)	qc.result.review	—
Approve QC sample (e-signature)	qc.result.approve	—
QA disposition of material lot (e-signature)	inventory.release	—
Create / approve production plan	production.plan.manage	You perform this
Create batch	batch.create	You perform this
Reserve materials (FEFO, released lots only)	batch.reserve	You perform this
Issue materials to production	inventory.issue	—
Start batch	batch.start	You perform this
Execute and sign eBMR steps	batch.execute	You perform this
Verify eBMR steps (different person, e-signature)	batch.verify	You perform this
Complete manufacturing (yield, FG to QUARANTINE)	batch.complete	You perform this
Finished-product QC (enter / review / approve)	qc.result.enter	—
QA batch release or rejection (e-signature)	batch.approve	—
Raise deviation	deviation.create	You perform this
Investigate deviation / OOS	deviation.investigate	—
Classify, approve and close deviations	deviation.approve	—
Conclude OOS (e-signature)	oos.approve	—
Create and manage CAPA	capa.create	—

Workflow step	Permission	Your role
Close CAPA (e-signature)	capa.close	—

Electronic signatures

- Critical actions (approvals, releases, rejections, verifications, closures) open an Electronic signature dialog.
- Read the meaning of the signature shown in the dialog, enter a reason where requested and re-enter your password.
- “Your signature is legally binding within your company’s validated procedures.” A wrong password changes nothing and the failed attempt is recorded.
- The signature (your name, meaning, reason, time) is stored with the record and in the audit trail.

10 Status meanings

Status	Meaning
DRAFT	Draft, not yet submitted/approved.
APPROVED	Approved (sample, supplier, PO or plan).
PARTIALLY RECEIVED	Part of the PO quantity received.
RECEIVED	Fully received (PO) / receipt posted (GRN).
CLOSED	Completed and closed.
CANCELLED	Cancelled with a reason; kept on record.
COMPLETED	eBMR step completed and signed by the operator (or CAPA action completed).
QUARANTINE	Received or produced, not yet approved for use.
UNDER TEST	Sampled; QC testing in progress.
RELEASED	Approved by QA (lot) or released for sale (batch).
REJECTED	Rejected by QA; may not be used or sold.
BLOCKED	Temporarily blocked by QA.
EXPIRED	Past expiry date; cannot be used.
RETURNED	Returned stock.
RECALLED	Subject to a recall.
PLANNED	Batch created; materials not yet reserved.
MATERIAL RESERVED	Released lots reserved (FEFO).
READY	Materials issued; batch can start.
IN PROGRESS	Manufacturing in progress (batch, step or CAPA).
AWAITING QC	Manufacturing complete; finished-product QC pending.
AWAITING QA	QC complete; QA decision pending.
PENDING	eBMR step not started.
VERIFIED	eBMR step verified by a second person.
REQUESTED	QC sample requested.
COLLECTED	Sample collected.
IN TESTING	Results being entered.
SUBMITTED	Submitted for review.
REVIEWED	Reviewed; awaiting approval.
PASS	QC conclusion: meets specification.
FAIL	QC conclusion: does not meet specification.

Status	Meaning
WITHIN SPEC	Result within specification.
OOS CANDIDATE	Result outside specification — OOS investigation opened.
REPORTED	OOS reported; investigation not started.
PRELIMINARY INVESTIGATION	OOS preliminary investigation.
LAB INVESTIGATION	OOS Phase I laboratory investigation.
MANUFACTURING INVESTIGATION	OOS Phase II manufacturing investigation.
ROOT CAUSE	Root cause being documented.
IMPACT ASSESSMENT	Impact on other batches/products assessed.
CAPA	CAPA decision stage.
QA APPROVAL	Awaiting QA conclusion.
CONFIRMED OOS	QA concluded the result is a true OOS.
INVALIDATED LAB ERROR	QA invalidated the result due to an assignable laboratory error.
QA REVIEW	Deviation under QA review.
INVESTIGATION	Deviation under investigation.
EFFECTIVENESS CHECK	Awaiting effectiveness check.
MINOR	Minor deviation.
MAJOR	Major deviation.
CRITICAL	Critical deviation / priority.
OPEN	Open CAPA or action.
LOW	Low priority.
MEDIUM	Medium priority.
HIGH	High priority.
ACTIVE	Active master data.
INACTIVE	Inactive; cannot be used for new transactions.
PENDING QUALIFICATION	Supplier not yet qualified.
CONDITIONALLY APPROVED	Supplier approved with conditions.
SUSPENDED	Supplier suspended.
BLACKLISTED	Supplier barred.
QUALIFIED	Equipment qualified.
NOT QUALIFIED	Equipment not qualified.
RETIRED	Equipment retired.

11 Search, filters, sorting and pagination

- Search table: type to filter rows in the current list.
- Quick views and filters above the table (for example Open / Overdue / All, status, type, warehouse) narrow the list; Saved filters stores your combination.
- Choose columns shows or hides columns; select a column header to sort.
- Rows per page and Previous / Next page control pagination (up to 200 rows per page).
- Export CSV downloads the current list (requires the export permission; exports are audited).

12 Reports and exports

Report	Where	Export allowed
Batch summary (CSV)	Reports → Batch summary	Yes
Batch manufacturing record (print / PDF)	Batch page → Batch record	Yes (print)
Inventory status (CSV)	Reports or Inventory → Export CSV	Yes
Near-expiry inventory (CSV)	Reports or Inventory → Expiry intelligence	Yes
Expired stock (CSV)	Reports	Yes
Deviation register (CSV)	Deviations → Export CSV	Yes
CAPA register (CSV)	CAPA → Export CSV	Yes
OOS register (CSV)	OOS investigations → Export CSV	Yes
QC summary (CSV)	QC samples → Export CSV	Yes
Supplier performance (CSV)	Suppliers → Export CSV	Yes
Mock-recall reconciliation (CSV)	Traceability & recall → Export recall reconciliation	Yes
Product Quality Review data pack (print / PDF)	Product quality review → Print / PDF	Yes (print)

Every export is recorded in the audit trail. CSV files are UTF-8 and protected against spreadsheet formula injection.

13 Notifications and alerts

- The bell shows unread notifications; Notifications → Inbox lists them and Mark all read clears the counter.
- Alerts come from workflows (for example QC sample requested, OOS reported, batch awaiting QA, assignment) and scheduled compliance scans (expiry, low stock, overdue CAPAs/deviations, calibration due, QA backlog).
- My preferences lets you choose which notifications you receive. In-app delivery is on by default; e-mail delivery requires SMTP configuration by an administrator and is not active in this release.

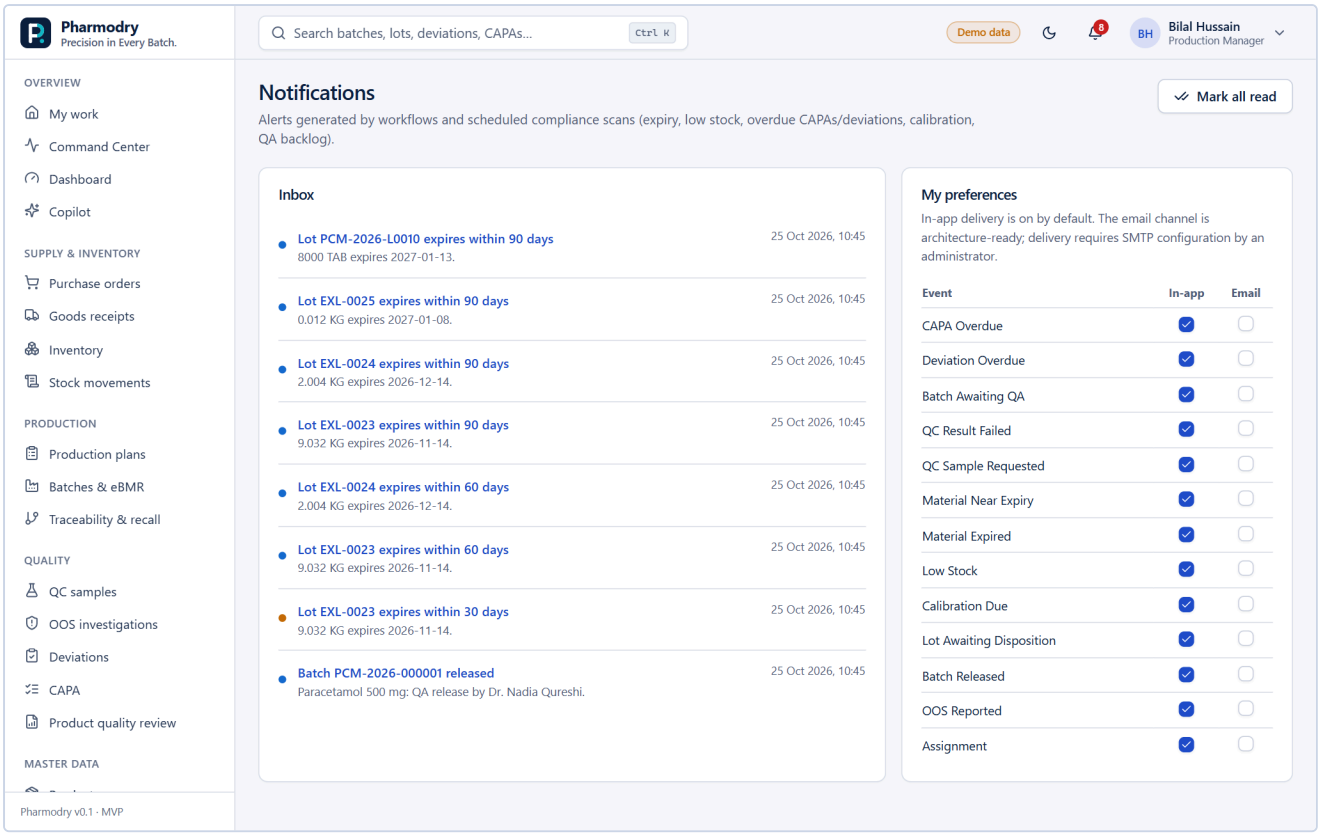


Figure 61 — Notifications. Notification inbox and preferences.

14 Error messages and troubleshooting

Message or symptom	Cause	What to do
403 / “Missing permission …”	Your role does not include the permission, or it is limited to another site.	Ask your Company Admin to review your role assignment.
“Action blocked by business rules” with a list of reasons	A GMP rule prevents the action (for example QC not approved).	Open each reason link and resolve it; the rule cannot be overridden.
“Account locked after repeated failed attempts”	Five wrong passwords.	Wait 15 minutes or ask an administrator to unlock.
Session expired / returned to sign-in	Inactive session or revoked session.	Sign in again.
Electronic signature failed	Wrong password.	Re-enter the password; nothing was saved.
“… was modified concurrently; reload and retry”	Another user changed the record.	Reload the page and repeat your action.
Unknown field / validation messages on forms	Missing or invalid input.	Correct the highlighted fields.
Order rejected by business rules	Supplier not qualified, material not approved for the supplier, or invalid quantities.	Read the listed reasons; ask QA to qualify the supplier/material if needed.
Segregation of duties: the creator of a purchase order cannot approve it	You created the order.	Ask another user with approval rights.
Receipt rejected by business rules	Expired material, unqualified supplier, unapproved material or invalid dates.	Correct the data, or quarantine the delivery physically and inform QA/procurement.
Supplier not qualified	The supplier is pending, suspended or blacklisted.	Do not receive; contact QA (supplier qualification).
This lot cannot be released yet	QC sample not approved, failed, or other blockers.	Complete QC; follow up the listed blocker links.
Insufficient released material to reserve this batch	Released, unexpired stock is lower than required (quarantined stock never counts).	Expedite QC/QA of quarantined lots or procure more.
Insufficient released material to reserve this batch	Shortage of released stock.	Escalate to planning/QA; do not use quarantined stock.
Calibration expired	Selected equipment is past its calibration due date.	Use other equipment or contact Engineering; under BLOCK policy the step cannot start.
Segregation of duties: the operator who performed a step cannot verify it	Same person.	Ask another supervisor.
QA release is blocked	Release blockers exist.	Resolve each listed reason first.
Batch … was modified concurrently; reload and retry	Someone else changed the batch.	Reload the page and repeat.
Enter at least one result	Save without values.	Enter results first.
Segregation of duties: an analyst cannot approve their own results	Same person.	Another authorised person approves.

Message or symptom	Cause	What to do
OOS investigation open — approval and release are blocked	A result is an OOS candidate.	Complete the OOS investigation first.
Phase cannot be completed	Required findings missing.	Fill in the phase findings.
Segregation of duties: the investigation owner cannot give the QA conclusion	Same person.	Another QA approver concludes.
Action blocked by business rules	Required information missing for the next step.	Read the listed reasons and complete them.
Action blocked	For example closing without a successful effectiveness check.	Complete the effectiveness check first.
GTIN … is not a valid GS1 GTIN (length or check digit)	Wrong length or check digit.	Enter the 8, 12, 13 or 14-digit GTIN exactly as allocated by GS1.

15 Common mistakes and how to prevent them

- Entering supplier batch, expiry or quantities that differ from the container label or CoA — always copy exactly.
- Trying to use quarantined or under-test material — only RELEASED lots can be reserved; wait for QA release.
- Signing for someone else or sharing passwords — each signature must be your own.
- Delaying deviation reporting — raise the deviation as soon as the event is known; QA classifies it.
- Replacing results to “fix” an OOS — OOS results cannot be replaced during an investigation; follow the OOS procedure.
- Closing CAPAs without objective evidence — the effectiveness check must be documented.

16 Security and access limitations

Permissions granted to your role (as enforced by the server):

Permission	Meaning
ai.assist.use	Use AI investigation and CAPA assistants
ai.copilot.use	Use the management copilot
analytics.view	View dashboards and analytics
audit.view	View audit trail
batch.complete	Complete batches and record yield
batch.create	Create batches
batch.execute	Execute eBMR steps as operator
batch.reserve	Reserve and dispense materials for batches
batch.start	Start batches
batch.verify	Verify eBMR steps as supervisor
batch.view	View batches and eBMR
capa.view	View CAPAs
commandcenter.view	View Pharma Command Center
deviation.create	Raise deviations
deviation.view	View deviations
inventory.view	View inventory lots and movements
masterdata.view	View master data
oos.view	View OOS investigations
procurement.po.view	View purchase orders
production.plan.approve	Approve production plans
production.plan.manage	Create and change production plans
production.plan.view	View production plans
qc.view	View QC samples and results
report.export	Export reports
supplier.view	View suppliers

Not granted (34 permissions): admin.tenant.manage, admin.site.manage, admin.user.manage, admin.role.manage, product.manage, material.manage, supplier.manage, supplier.approve, equipment.manage, spec.manage, procurement.po.create, procurement.po.approve, inventory.receive, inventory.transfer, inventory.adjust, inventory.issue, inventory.release, batch.approve, qc.sample.create, qc.sample.collect, qc.result.enter, qc.result.review, qc.result.approve, oos.create, oos.investigate, oos.approve, deviation.investigate, deviation.approve, capa.create, capa.manage, capa.verify, capa.close, audit.export, ai.admin.

Procedures you cannot perform

- Purchase orders: Create a purchase order — requires procurement.po.create

- Purchase orders: Approve a purchase order — requires procurement.po.approve
- Purchase orders: Cancel a purchase order — requires procurement.po.create or procurement.po.approve
- Goods receipts (GRN): Receive material — requires inventory.receive
- Inventory, lots and stock movements: Release or reject a material lot (QA disposition) — requires inventory.release
- Inventory, lots and stock movements: Transfer stock — requires inventory.transfer
- Inventory, lots and stock movements: Post a stock adjustment — requires inventory.adjust
- Inventory, lots and stock movements: Block or unblock a released lot — requires inventory.release
- Batches and electronic batch records (eBMR): Issue materials to production — requires inventory.issue
- Batches and electronic batch records (eBMR): QA release or rejection of a batch — requires batch.approve
- QC samples and results: Record sample collection — requires qc.sample.collect
- QC samples and results: Enter results — requires qc.result.enter
- QC samples and results: Review results — requires qc.result.review
- QC samples and results: Approve a sample — requires qc.result.approve
- QC samples and results: Replace a result — requires qc.result.enter
- OOS investigations: Record an investigation phase — requires oos.investigate
- OOS investigations: Conclude the investigation (QA) — requires oos.approve
- Deviations: QA review and classification — requires deviation.approve
- Deviations: Investigate and record the root cause — requires deviation.investigate
- Deviations: CAPA decision and closure — requires deviation.approve
- CAPA: Create a CAPA — requires capa.create
- CAPA: Complete actions — requires capa.manage
- CAPA: Effectiveness check — requires capa.verify
- CAPA: Close a CAPA — requires capa.close
- Master data: products, materials, suppliers, equipment and specifications: Create or edit a product — requires product.manage
- Master data: products, materials, suppliers, equipment and specifications: Create or edit a material — requires material.manage
- Master data: products, materials, suppliers, equipment and specifications: Create a supplier — requires supplier.manage
- Master data: products, materials, suppliers, equipment and specifications: Change supplier qualification / approve a material — requires supplier.approve or supplier.manage
- Master data: products, materials, suppliers, equipment and specifications: Record equipment calibration — requires equipment.manage
- Audit trail: Export the audit trail — requires audit.export

- Your role may additionally be limited to specific sites; records of other sites are then hidden or read-only.
- Never share your password: it is also your electronic-signature credential.
- Sign out or lock your screen when leaving the workstation.

17 Practical end-to-end example

Based on the fictional NovaCure demonstration data:

1. All eBMR steps of a Paracetamol batch are verified.
2. Open the batch → Complete manufacturing; the dialog shows the calculated yield (e.g. 99.6 %) within the 98–102 % tolerance.
3. Confirm. Finished goods go to QUARANTINE and QC receives a finished-product sample request.

18 Frequently asked questions

Why is the release button disabled?
The server has found release blockers; they are listed on the page with links.

Can I delete a record I created by mistake?
No. Regulated records are cancelled or corrected with a reason, never deleted.

Why can I see a module but not its action buttons?
View and action permissions are separate; your role may be read-only for that module.

Does the AI make decisions?
No. The Copilot and assistants only explain computed facts or draft suggestions; people decide and sign.

What time zone are dates shown in?
Times are stored in UTC and displayed in Pakistan time (Asia/Karachi) in this release.

Is this the official batch record?
The executed eBMR in Pharmodry is the electronic record; the printable view is a copy for review. Your SOPs define the official record.

19 Glossary

Term	Meaning
ALCOA+	Data-integrity principles: Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring, Available.
API	Active Pharmaceutical Ingredient.
BOM	Bill of materials — quantities of materials per standard batch.
CAPA	Corrective and Preventive Action.
CoA	Certificate of Analysis from the supplier.
Deviation	Departure from an approved instruction or standard.
DRAP	Drug Regulatory Authority of Pakistan.
eBMR	Electronic Batch Manufacturing Record.
E-signature	Electronic signature: password re-entry linked to a record, meaning and reason.
FEFO	First Expiry, First Out — earliest-expiring released lots are used first.
FG	Finished goods.
GMP	Good Manufacturing Practice.
GRN	Goods Receipt Note.
GS1 / GTIN	Global standards body / Global Trade Item Number used in the 2D DataMatrix.
IPC	In-process control.
Lot	A quantity of one material or product with one internal lot number at one location.
OOS	Out of specification — a result outside its acceptance criteria.
OOT	Out of trend.
PQR	Product Quality Review (annual review of a product's quality data).
Quarantine	Status of material that may not be used until released by QA.
Retest date	Date after which a material must be re-tested before use.
SoD	Segregation of duties — the same person may not perform and approve/verify.
SOP	Standard Operating Procedure.
Yield	Actual output as a percentage of theoretical output.

20 Escalation

Escalate through the roles configured in your company. This release has no built-in external support desk and no vendor contact details; use your company's documented support and incident procedures.

- Yield out of tolerance or process failure: raise a deviation and inform QA. Equipment out of calibration: Engineering. Material shortages: Supply Chain Planner / Warehouse Manager.
- Access problems (locked account, missing permission): Company Admin.
- Any suspected data-integrity problem (for example "Integrity failure" on audit verification): QA Manager immediately.